

Mach-Zehnder Modulator Based Electronic-Photonic Integrated Circuits for Optical Interconnects Applications

by

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Dedicated to my wife and family for their unconditional love and support.

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Biographical Sketch

Jing Gao was born in Jiayu, Hubei Province, China in 1985. He attended Nanyang Technological University, Singapore, and graduated with a Bachelor of Science degree in 2009. He came to the University of Rochester in the Fall of 2011 and began graduate studies in Electrical Engineering. He completed his Master of Science degree in Electrical Engineering at the University of Rochester in 2013. He pursued his research in high performance optical transceiver design under the direction of Professor Hui Wu. He worked at Cisco System between May 2014 and September 2014. In August 2016, he joined Infinera as Senior Hardware Development Engineer.

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Abstract

Silicon photonics promises low cost and high volume production of highly integrated electronic-photonic systems by taking advantage of the well characterized and maintained complementary metal-oxide semiconductor (CMOS) manufacturing infrastructures in the microelectronic industry. It is considered as a promising solution to meet the future demand for high speed optical interconnects which continue replacing its electrical counterpart. High performance silicon photonic devices fabricated on standard CMOS or CMOS compatible processes have been studied intensively. Thanks to the tremendous research effort, the performance of both passive and active silicon photonic components is advancing in a phenomenal pace. The convergence of photonics and electronics into a single platform catalyzes the development of electronic-photonic integrate circuits (EPIC). Complete electro-optical systems including both the electronic and photonic circuits have been demonstrated through either monolithic or multi-chip integration with good performance. However, challenges still remain in photonic device performance and electronic-photonic system co-design and integration. First of all, silicon does not have a strong electro-optical effect compared to LiNO_3 or III-V group materials. It is challenging to make a modulator that can operate at high speed and at the same time has large modulation efficiency. Large voltage swing is usually required to compensate for the low modulation efficiency, which is however, not practical for advanced CMOS processes. Further, as the performance continues to improve in both electronic and photonic sides, the electro-optical interface becomes the

bottleneck of the whole system. A co-design of these two sub-systems in a single, integrated design flow will become necessary to continue data rate road map. Last but not least, the wide adoption of the silicon photonic technology also requires the exploration of new functional blocks to enrich the component library of silicon photonics.

To address these challenges, we firstly proposed an integrated, multi-physics design flow for EPIC design to seamlessly interfacing electronics and photonics. The design flow integrates various simulation packages to provide fabrication process, photonic device and electronic circuit modeling capabilities. In addition, it provides a good representation of the electro-optical interface through multi-lateral file sharing. Furthermore, we propose a new device model for accurate device modeling and system performance evaluation. The design flow is exemplified through the design of a traveling-wave Mach-Zehnder modulator (TWMZM). The detailed design optimization process is demonstrated through various device and system modeling tools.

Use the EPIC design flow as the design methodology, we further proposed a multi-function EPIC design based on a single Mach-Zehnder modulator (MZM). This design exploits a new operation mode of a MZM in the time domain to realize new functions. Different from the conventional design, the two arms of the modulator are driven by time-shifted signals of tunable amplitude. A single MZM is capable of performing optical pulse shaping in addition to electro-optical (EO) conversion, or pulse generation with tunable pulse width at double clock rate when biased properly. The modulator incorporated with pulse shaping capability helps cancel the inter-symbol interference (ISI) caused by bandwidth limited modulator or detector. Meanwhile, the optical pulse generation circuit can be used as frequency doubler and for on-chip generation of optical return-to-zero (RZ) signal. The system EO performance has been evaluated using the EO interfacing capability provisioned by the EPIC design flow. The simulation results demonstrate

the capabilities of the new EPIC as an optical pulse shaper and optical pulse generator. This prototype is fabricated using IMEC 50G Si photonic process.

As an example of electronic-photonic co-design, a novel high speed TWMZM with built-in feedback equalization is designed using the same EPIC design methodology. The new equalization technique takes advantage of the natural time delay in a travel-wave electrode (TWE) to form an infinite impulse response (IIR) filter. In the time domain, the delayed output signal at the end of the TWE is weighed and added back to the input to pre-distort the signal as in a feed forward equalizer (FFE). In the frequency domain, the transfer function of the TWE is equalized by the negative feedback through low frequency component suppression and high frequency component boosting. Furthermore, the frequency domain analysis based on a mathematical model shows that this technique not only equalizes the magnitude of the transfer function, but also reduces the overall group delay variation (GDV) in the frequency range of interest. A 3-stage cascaded TWMZM with the feedback equalization in each stage is designed, and two types (linear and limiting) of feedback circuits are investigated. The photonic device is designed based on IMEC 50G Si photonic process and the electronic circuits are designed based on GlobalFoundries 0.13 μm SiGe process. Modeling of devices and circuits are carried out based on the EPIC design flow. Simulation results show that the equalized TWE increases the 3-dB bandwidth by 1.4 times and the 6-dB bandwidth is improved by 57%. Transient simulation shows the equalization extends the error-free data rate from 40 Gb/s to 60 Gb/s. The simulated total power consumption is 190 mW for linear feedback and 171 mW for limiting feedback.

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List of Abbreviations

APD		Avalanche Photodetector/Photodiode
BER		Bit Error Rate
BOX		Buried Oxide
CE		Common Emitter
CML		Compact Model Library
CMOS		Complementary-Metal-Oxide-Semiconductor
CW		Continuous Wave
DFE		Decision Feedback Equalization
DNW		Deep N-Well
EDA		Electronic Design Automation
EF		Emitter Follower
EHP		Electron Hole Pair
EM		Electromagnetic
EO		Electro-Optical
EPIC		Electronic Photonic Integrated Circuit
ER		Extinction Ratio
FDM		Finite Difference Method
FDTD		Finite Difference Time Domain
FEM		Finite Element Method
FFE		Feed Forward Equalizer
FIR		Finite Impulse Response
GBP		Gain-Bandwidth Product
GD		Group Delay
GDV		Group Delay Variation

IIR		Infinite Impulse Response
ISI		Inter-Symbol Interference
MM		Multimode
MMI		Multi-Mode Interferometer
MZI		Mach-Zehnder Interferometer
MZM		Mach-Zehnder Modulator
NW		N-Well
NRZ		Non-Return to Zero
PAM		Pulse Amplitude Modulation
PCW		Photonic Crystal Waveguide
PD		Photodetector/Photodiode
PDK		Process Design Kit
PIC		Photonic Integrated Circuit
PW		P-Well
QAM		Quadrature Amplitude modulation
QE		Quantum Efficiency
RF		Radio Frequency
RZ		Return to Zero
SNR		Signal-to-Noise Ratio
SOA		Semiconductor Optical Amplifier
SOI		Silicon-on-insulator
STI		Shallow Trench Isolator
TE		Transverse Electric
TIA		Transimpedance Amplifier
TM		Transverse Magnetic
TWE		Traveling-Wave Electrode
TWMZM		Traveling-Wave Mach-Zehnder Modulator
UI		Unit Interval
VCSEL		Vertical Cavity Surface Emitting Laser
VNA		Vector Network Analyzer
WDM		Wavelength Division Multiplexing

Chapter 1

Introduction

1.1 Optical Interconnects

The exponential increase of Internet traffic over the past decade, fueled by the prevailing of cloud computing, social media and video streaming, has posed a formidable challenge to the conventional way we transport data. The ever-demanding bandwidth requirement has driven aggressive scaling of interconnect capacities. In addition, the power consumption as a result of Internet traffic has also increased dramatically, which consists of almost 10% of the world's total energy usage by 2018 [1]. Traditional electrical interconnects cannot scale in the same pace as the bandwidth and power efficient requirements. On the other hand, optical interconnect emerges as an alternative to its electrical counterpart and has largely replaced copper wires with shorter and shorter distances, from long-haul telecommunications to board-level interconnects, or even inter-chip communications within a multi-chip module [2], due to the advantages of low attenuation, large bandwidth, and immunity to electromagnetic interference.

Long-haul optical communications or metro network typically use single mode fibers at either 1310 or 1550 nm windows where the fiber attenuation is minimal. Single mode optical fiber has been made with very small loss, typically 0.2 dB/km, allowing for long distance transfer of data. Since the cost of laying down

an optical fiber is very high, the number of fibers available in telecommunications is very limited. The scarcity of channels drives the demand for better spectral efficiency which in turn drives the development of expensive equipment at transmitter and receiver ends to allow advanced modulation schemes and wavelength division multiplexing (WDM). As an example, quadrature amplitude modulation (QAM) is used to convey two digital bit streams in digital pulse amplitude modulation (PAM) using the same wavelength and polarization but with 90° out of phase. Depends on the type of the PAM, the QAM can increase the bit/symbol to 4 (QAM16) or even 6 (QAM64). Increasing channel count by transmitting at different wavelengths is another way to increase the spectral efficiency. Today's dense WDM systems use 50 GHz or even 25 GHz channel spacing for up to 160 channels operation [3]. To compensate for fiber loss at a long distance, repeaters made of fiber amplifiers are inserted for every 10 to 500 km, depending on applications. The optical transceivers in long-haul optical communications is dominated by expensive discrete components or III-V group material based photonic integrated circuit (PIC), which usually requires a significant amount of hybrid integration.

Short-haul optical interconnects between and within data centers have gained much attention in recent years because of the development of the data centers and high performance computers (HPCs). Since the distance is much smaller, adding more fiber does not come with a high cost. Short-haul optical interconnects require cost and power efficient transceiver technologies for fast and efficient data communications. Multimode (MM) fibers optimized at 850-nm are widely adopted. Cost- and power- efficient MM laser diodes and light emitting diodes (LEDs) which can be directly modulated are the dominant technology in these networks. However, due to the multi-mode nature of the lasers and fibers, they can't support long distance interconnects because the strong mode-dispersion will deteriorate the signal to noise ratio (SNR). For example, one of the best MM fiber, optical mode-4 fiber, has maximal reach of 400 meters at 10 Gb/s data rate.

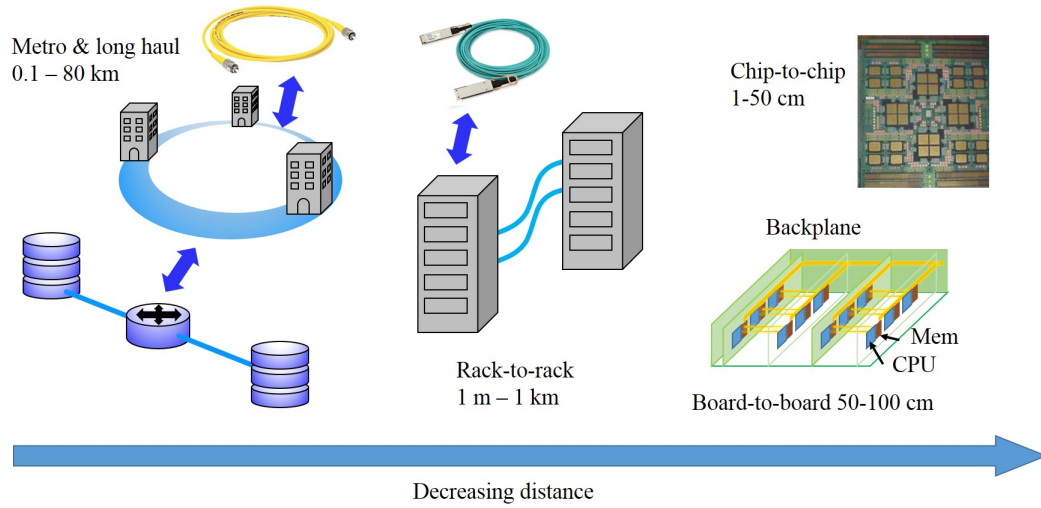


Figure 1.1: Current applications of optical interconnects with respect to interconnect length. Long-haul communication resort to single mode fiber system for long distance transmission, while short-haul optical interconnects usually use MM VCSEL based optical transceiver technologies. Research on board-to-board, or on-board communication are very active.

The continued success of optical interconnects in both long-haul and short-haul communications has inspired the research and application of optical interconnects in shorter distances (10 meters or less), such as the optical backplane for board-to-board communications [4, 5], optical multi-chip module for module-to-module communications on-board [6, 7], or even optical interconnects for intra-chip communications [8, 9]. As cost and power consumption of the optical devices continue to drop, accompanied by emerging of novel packaging technologies, optical interconnects have started to make progress in these short reach interconnect systems.

1.2 Optical Transceivers

Optical transceiver is the core of any optical interconnect systems. The schematic of a 4-channel WDM optical link is shown in Fig. 1.2. The data that goes into the optical transceiver is firstly processed by a serializer which multiplexes many low

speed parallel channels into a high speed stream of serial data. The high speed data stream is then fed into a modulator driver which encodes electrical signal into optical signal. In the optical domain, an optical multiplexer is used to combine all 4 wavelengths into a single fiber or waveguide (for on-chip interconnect) and transmitted over to the receiver. On the receiver side, a demultiplexer separates these four wavelengths and send them to corresponding photodetectors where optical signal is converted back to electrical signal. Electrical amplifier circuit is then used to receive and amplify the electrical signal and then a deserializer will demultiplex this high speed data stream into low speed parallel channels and send them to their designed destinations.

Optical transceivers based on discrete devices have been widely applied in short-haul optical interconnect within HPCs, catalyzed by the development of inexpensive (vertical cavity surface emitting laser) VCSEL and packaging techniques. However, the package becomes increasingly complicated as the number of channel increases. IBM developed a holey CMOS technique to integrate large number of VCSELs and PDs in an optical transceiver [10]. The photonic devices were flip-chip bonded onto the electrical driver chip. To access these photonic devices, the CMOS chip was drilled with many holes for light beams to vertically go in and out. Above these holes, lens array was used to couple light beams in and out of fibers. In one example, Doany *et al.* [6] demonstrated a Tb/s class optical printed circuit board link for module-to-module communications by using flexible polymer waveguides on traditional PCBs. They employed microlens and 45° mirrors in the system to couple light in and out of flex polymer fibers. The optical transceiver was based on a silicon carrier with an array of 24 850-nm vertical surface emitting lasers (VCSELs), an array of 24 PDs, and an electrical circuit chip integrated on. The silicon carrier also had 48 open holes for optical access to the flip-chip bonded photonic devices. Each of these 24 channels operated at 20 Gb/s, which amounts to 480 Gb/s aggregated data rate. Nevertheless, the

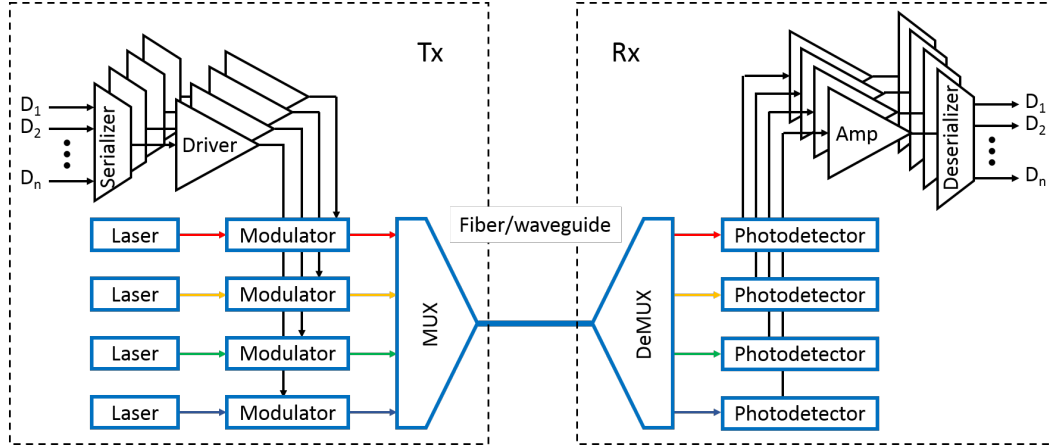


Figure 1.2: The schematic of a 4-channel WDM optical link. On the electrical side, low speed parallel data streams, $D_1, D_2, \dots, D_n$, are multiplexed into a high speed serial data stream by a serializer, which is then converted to optical signal by a modulator. On the optical side, four different wavelengths (represented by four different colors) are combined by an optical MUX and transmitted in a single fiber or waveguide. Thereby, the utilization of the optical channel spectrum is maximized.

holey technique results in tremendous cost overhead and reduces product yield. As the complexity of optical transceiver scales up further, the cost of integration will grow exponentially.

In contrast, PIC has the inherent advantage of large scale integration. Optical transceivers based on PIC are progressing at an unprecedented pace. The early photonic circuits are based on InP platform which is a natural extension of InP based lasers [11]. Later on, the scale of integration grows exponentially as more and more components are demonstrated on InP platform. Today, InP platform based PIC remains an important transceiver technology for all kinds of optical interconnect systems [12]. Recently [13], Infinera demonstrated an InP based PIC integrated with 14 optical coherent transceiver channels. Each channel achieved 33Gbaud/16-QAM operation, enabling multi-Tb/s total PIC capacities. Despite the great success of InP based PIC, the cost remains very high because of the dedicate fabrication process and limited volume. Another promising platform for

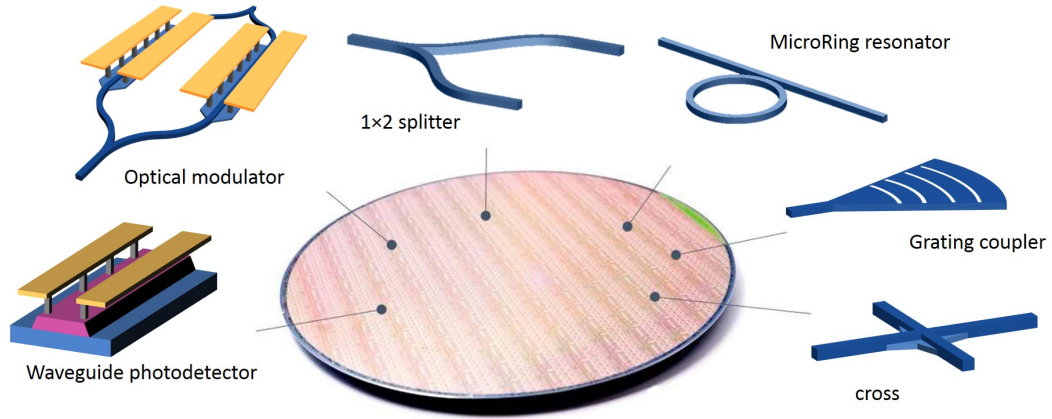


Figure 1.3: Active and passive photonic devices fabricated on a silicon wafer using a CMOS compatible Si photonic process.

optical transceiver integration is silicon. As shown in [14], the scale of integration on silicon grows at a much faster speed than on InP platform and may eventually surpass it in the near future.

1.3 Silicon Photonics

Silicon photonics is receiving attentions recently as a promising technology for optical interconnects, because it promises low cost and high volume production of the highly integrated electronic-photonic system by taking advantages of the existing well-characterized and maintained complementary metal-oxide semiconductor (CMOS) manufacturing infrastructures in the microelectronics industry [15]. In addition, the convergence of both electronics and photonics into a single platform leads to an alluring picture of monolithically integrated electronic-photonic integrated circuit (EPIC). Thanks to the tremendous research effort, the performance of both pass and active silicon photonic components are advancing in a phenomena pace [16] (Fig. 1.3).

Silicon-on-insulator (SOI) wafer with a few micrometers buried oxide (BOX) layer provides an excellent platform for optical devices development. The large

refractive index contrast between silicon and silicon dioxide permits waveguides with sub-micrometer cross-section, and bending with small radius of curvature, suitable for large scale, high density integration. Due to its transparency in the telecommunication wavelengths, low loss silicon waveguide is achievable. The loss of silicon waveguides mainly comes from the interaction between optical mode and waveguide sidewalls. Numerous effects have been made to reduce the roughness of silicon waveguide sidewalls. Ultra low loss of 0.3 dB/cm in SOI waveguides has been achieved using etchless process based on selective oxidation [17]. Nevertheless, the loss in a silicon waveguide is still much higher than that of a single mode fiber. Therefore, for relatively long distance optical interconnect (>10 cm), optical fiber is still needed, which necessitates the waveguide-to-fiber coupler. Coupling the light from a waveguide to a fiber is not trivial given the single mode nature of the silicon photonics technology. A conventional single mode fiber has a mode fiber diameter of about $11\text{ }\mu\text{m}$, which suggests the mismatch between the on-chip optical mode and the fiber optical mode should be within micrometer scale. The coupling loss remains to be the dominant source in the system insert loss. Taking the grating coupler as an example, the Fresnel reflection at the air-silicon interface limits the coupling efficiency of a fully etched grating coupler to a theoretical limit of 49% (3 dB loss) [18]. By using shallow etch and back reflector, the theoretical estimation can be improved to 90% coupling efficiency [19]. Other passive components, such as waveguide cross, 3-dB coupler, ring resonator *etc.*, have all been demonstrated on SOI platform.

Active components, such as modulators and switches fabricated on SOI platform have also been extensively studied and demonstrated with excellent performance. Modulators is one of the most fundamental active components in any optical interconnect systems. Unfortunately, silicon doesn't have linear electro-optical effects (Pockels effect) as in LiNO_3 . Plasma dispersion effect [20] and thermal effect [21] are among the most successfully exploited effects in silicon. The

plasma dispersion effect provides fast enough response for high speed operation. A high speed modulator based on the plasma dispersion effect was first demonstrated by Intel in 2004 [22] and since then, we have seen continued improvement in the performance of modulators. Today, silicon modulators operate at upto 60 Gb/s have been demonstrated with Mach-Zehnder interferometer (MZI) configuration [23]. An optical switch has numerous applications, such as add/drop multiplexing, bypass switching, packet switching. The plasma dispersion effect can be used to make a switch. However, introducing dopant causes additional optical loss. On the other hand, silicon has a very large thermo-optic coefficient. Very compact, low loss, and low power switches are achievable by thermo-optic effect [24]. Photodetector (PD) is another important device in optical interconnect systems. Since silicon is transparent in telecommunication window, other materials have to be introduced to increase the quantum efficiency (QE). Germanium is traditionally a silicon-compatible material due to the small lattice mismatch between silicon and germanium (4.2% [25]). It has much higher absorption coefficient than silicon for wavelength up to 1.6 μm . Epitaxial growth of germanium on silicon is a mature technology and it also has been adopted in commercial foundry, *e.g.* in SiGe process. High speed (30-40 GHz) and high responsivity (~ 1 A/W @1.55 μm) Ge-on-Si PDs have been demonstrated in either butt coupling [26] or evanescent coupling configurations [27].

Meanwhile, electrical driving and receiving circuits are also developing in parallel with the maturing of photonic devices. These electrical circuits have to address issues associated with Si photonic modulators and receivers. For instance, as mentioned previously, silicon photonic doesn't have a strong electro-optical effect, so the modulation efficiency of a silicon phase modulator is very low. To compensate for low modulation efficiency, long device or heavy doping have to be made. However, both methods cause large parasitic capacitance that limits the bandwidth of driving circuits. One solution is to adopt a traveling-wave driv-

ing scheme that is typical applied in traditional LiNO_3 Mach-Zehnder modulator (MZM). A lossless transmission line exhibits infinite bandwidth and linear phase response. The large capacitance can be absorbed into the transmission line and only affects its characteristic impedance and effective mode index if no parasitic resistance is present. Therefore, a traveling-wave electrode (TWE) can achieve high bandwidth and large modulation depth. In addition, it presents a single resistive load to driving circuits that simplifies the driver design. As a result, traveling-wave MZM (TWMZM) is a popular modulator structure and has been studied intensively [28–31]. Another solution is to break the large capacitance into small pieces and one driver for each piece to partition the load [32–34]. However, a few ps analog delay is difficult to implement. LC network is often used, which, however, takes up large chip area.

Equalization is a popular technique used in wireline communication system to compensate for channel loss. Even though the optical channel in optical interconnects has a almost flat frequency response, equalization techniques can still find its application in an optical link to over-drive a bandwidth limited photonic devices [35] or to realize new receiver structure [36]. The design methodology in wireline communication can be easily ported over to the optical transceiver circuit. However, electrical equalization methods increase the circuit design complexity and add extra noise. Equalization in optical domain has also been exploited. Applying the same principle as in electrical equalization, optical equalization utilizing photonic elements to shape the optical pulse has been demonstrated in ref. [37].

The promise of the silicon photonic technology lays on the fact that it can be fabricated using the heavily invested infrastructures in microelectronics world. It may offer the ultimate integration capability to produce highly integrated electronic-photonic systems with most of the elements fabricated directly in CMOS. As data rate continues to increase, it will be desirable to move optical transceivers close to processors or memory chips to eliminate any parasitics in between. Suc-

successful demonstration of monolithic integration of photonic and electronic were reported by Sun *et al.* from MIT [38]. In [38], a 9-wavelength WDM optical transceiver system based on ring modulator was demonstrated by using $0.18\ \mu\text{m}$ bulk CMOS process. Photonic devices were based on amorphous polysilicon layer deposited on $1.2\ \mu\text{m}$ silicon dioxide filled inside a deep trench. Ring resonators were used as modulators, filters and PDs and on-chip wavelength locking circuits were used to stabilize the resonance frequencies of the ring resonators. Driver and receiver circuits including serializer and deserializer were all integrated on the same chip. The transceiver achieved error free (bit error rate (BER) = 10^{-12}) transmission at 5 Gb/s for a single channel and consumed 15 pJ/bit overall energy efficiency. Even though making all the electronic and photonic components together on the same chip in a single process may sound exciting, it may be technologically too complicated and economically too expensive to integrate photonic devices with cutting edge CMOS processes because the yield of advanced CMOS processes are so sensitive to process modification and the real-estate on CMOS chips is so expensive that it can't afford bulky photonic components. Many of the benefits of monolithic integration can also be achieved through multi-chip integration thanks to the advancement in 3D chip integration. Further, it has the additional benefits of allowing separate optimization of both photonic and electronic chips thereby optimization of integrated systems. Luxtera [39] commercialized the first optical transceiver system based on the silicon photonic technology using multi-chip integration. The electrical chip and the laser chip were flip-chip bonded to the silicon photonic chip which hosts a variety of active and passive photonic devices. The optical transceiver had 4 channels and achieved single channel data rate of 25 Gb/s.

Electronic design automation (EDA) tools and standardized design flow are another aspects of the microelectronic industry that silicon photonics can take advantages of. As silicon photonic devices maturing and the library of functioning

devices growing, there is a strong interest in developing dedicate design tools and standardizing design flow for silicon photonic integrated systems. A fabless photonic company will be possible which focus solely on design and integration of systems rather than optimizing individual photonic devices. As one of such effort, Pond *et al.* [40] presented a unified design flow of silicon photonic circuits with iteration between photonic devices simulation and photonic circuit simulation, layout, verification. Moreover, with all process data, layout, design rules included in a process design kit (PDK) provided by a foundry, the design flow can be greatly simplified. In 2015 [41], IME partnered with Lumerical developed a calibrated compact model library (CML) for IME's silicon photonic platform and PDK. The CML includes 15 active and passive components with process data, layout table, design rules to help silicon photonic circuit designers to improve the accuracy and reliability of their design.

Silicon photonics technology is a single mode platform operating at telecommunication wavelengths, thus it has much longer reach than VCSEL and MM fiber based optical interconnect technology. Therefore, silicon photonics will has advantages in ware-house scale data centers and multi-data center interconnects over extended distance. In addition, silicon photonics technology can utilize the WDM system well-developed for telecommunications to increase the number of channels per fiber [38]. This will substantially reduce the number of fibers which already takes up the available back of drawer area and reduce cost. The ability to pack more channels with a given spectrum will be important to address 400 Gb/s and even 1 Tb/s transceiver data rate that will become necessary for future optical interconnect systems [42].

Nevertheless, the wide adoption of silicon photonics technology in optical interconnects still requires further development of this technology from the improvement of the individual components to the exploration of new functional blocks. As an example, pulse shaping, particularly equalization, is well-developed in elec-

trical domain for inter-symbol interference (ISI) mitigation. Optical equalization techniques have been developed to cancel the ISI arises from chromatic dispersion or transmitter bandwidth limitation, by using very dedicate and expensive optical components, which can't be integrated on a silicon photonic chip. Implementation of equalization techniques in optical domain by using silicon photonic integrated circuits will be an extremely interesting topic. The silicon photonic implementation of optical pulse shaping will provide a cost and power efficient solution for future high speed optical interconnects. Therefore, the growing silicon photonic device library and additional functionalities that can be implemented by silicon photonic integrated circuits will be an enabler for future application of silicon photonic technology in optical interconnects and beyond.

1.4 Dissertation Organization

This thesis is organized as in the following. Chapter 2 is a detailed background review about the silicon optical modulators and equalization techniques in optical interconnects. The first part is about the working principles and categorization of silicon optical modulator. The advantages and disadvantages are discussed, and the state of art devices performance is summarized as well. The second part is about the mechanism of equalization and different equalization techniques. Both electrical domain and optical domain equalizations are discussed.

In Chapter 3, the design methodology: an multi-physics EPIC design flow is presented. An example design of a TWMZM is discussed in detail. Optimization process of each electronic and photonic components is discussed in the framework of this design flow.

In Chapter 4, a multi-function EPIC based on a single MZM is presented. The first part of this chapter introduces the idea based on a generic MZM. The second part demonstrates the pre-emphasis and pulse generation functions of the

EPIC utilizing the EO response modeling capability of the EPIC design flow. A monolithic implementation of the EPIC is also proposed. In the last part of the chapter, prototype fabrication using IMEC 50G Si photonic process is discussed.

In Chapter 5, the design and implementation of a high speed TWMZM with built-in feedback equalization is presented. The design started with an analysis based on a mathematical model. Then a TWE design using IMEC 50G Si photonic process is presented, followed by the system optimization. The design of feedback circuits using GlobalFoundries 0.13 μm SiGe process is discussed subsequently. Finally, system performance of the 3-stage TWMZM is presented.

In Chapter 6, the thesis is summarized and concluded.

Chapter 2

Background and Review

2.1 Silicon Optical Modulators

The workhorse of silicon photonics based optical interconnects is silicon optical modulator. There has been great research interests on silicon optical modulators since long time ago, driven by the strong demand of fabricating photonic devices using the mature CMOS technology to reduce cost. In recent years, the performance of modulators have been improved dramatically in terms of bandwidth, modulation efficiency, and power consumption. However, there are still many challenges remain. An optimal design entails trade-off between many performance merits and the co-design of optics and electronics. In this chapter, a review about the operation mechanism of silicon optical modulators is presented and the challenges in modulator design is discussed.

2.1.1 Electro-Optical Effects in Silicon

Silicon doesn't have many properties that can be used to modulate optical signals. The primary EO effects exploited in other semiconductors, such as Pockels effect, Kerr effect, and the Franz-Keldysh effect are absent in bulk, unstrained silicon at telecommunication wavelengths ($1.3\ \mu\text{m}$ and $1.6\ \mu\text{m}$) [20, 43]. So far,

the most successfully exploited EO effect in silicon is the plasma dispersion effect, which was quantified by Soref and Bennett [20]. They found that the refractive index (both real and imaginary parts) of silicon (both real and imaginary parts) can be manipulated by free carrier concentration (Fig. 2.1). The change of refractive index at $1.5 \mu\text{m}$ can be expressed as [29]:

$$\Delta n = \Delta n_e + \Delta n_h = -[8.8 \times 10^{-22} \times \Delta N + 8.5 \times 10^{-18} \times (\Delta P)^{0.8}] \quad (2.1)$$

$$\Delta \alpha = \Delta \alpha_e + \Delta \alpha_h = 8.5 \times 10^{-18} \times \Delta N + 6.0 \times 10^{-18} \times \Delta P \quad (2.2)$$

where $\Delta n_{e(h)}$, $\Delta \alpha_e(h)$ are changes of refractive index and absorption coefficient respectively, due to the change of free electron (hole) concentrations. The ΔN and ΔP are the changes of free electron and free hole concentrations respectively. The change of refractive index at $1.3 \mu\text{m}$ can be expressed in a similar way as [29]:

$$\Delta n = \Delta n_e + \Delta n_h = -[6.2 \times 10^{-22} \times \Delta N + 6.0 \times 10^{-18} \times (\Delta P)^{0.8}] \quad (2.3)$$

$$\Delta \alpha = \Delta \alpha_e + \Delta \alpha_h = 6.0 \times 10^{-18} \times \Delta N + 4.0 \times 10^{-18} \times \Delta P \quad (2.4)$$

These equations quantify how refractive index changes with free carrier concentrations and it also implies one of the most fundamental trade-off in phase modulator design: modulation efficiency (Δn) and optical loss ($\Delta \alpha$). More free carrier results in large refractive index change but leads to more optical loss. It is worthwhile to elaborate a little more about the modulation efficiency. In a phase modulator employing the EO effect, the modulation efficiency is represented by the voltage which generates π phase shift in a phase modulator of unit length. This quantity is commonly quoted as $V_\pi L_\pi$ and has the unit of $V \cdot \text{cm}$ or $V \cdot \text{mm}$, meaning the voltage for π phase shift in a 1-cm or 1-mm long phase modulator. To measure this quantity, you will need to increase the bias of phase modulator

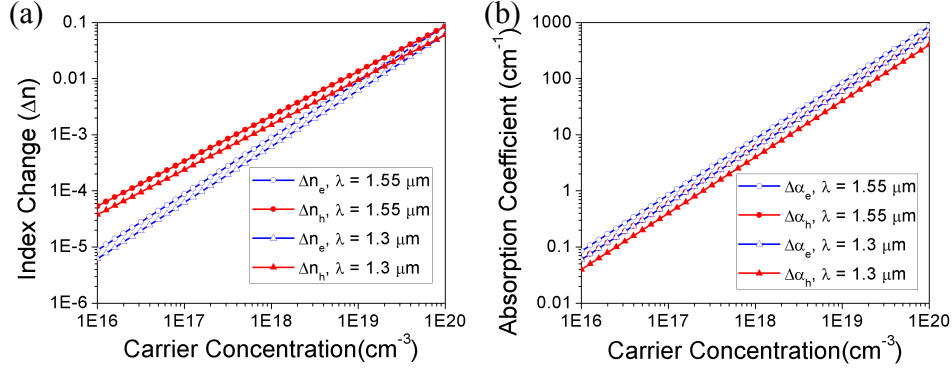


Figure 2.1: Change in refractive index (a) and absorption coefficient (b) as a function of change in carrier concentration for both electrons and holes at a wavelength of $\lambda = 1.55 \mu\text{m}$ and $\lambda = 1.3 \mu\text{m}$. Notice that the red line with solid circle marker and blue line with empty triangle maker overlap in (b).

until π phase shift is achieved. However, in reality, the device size varies from design to design and the π phase shift may be difficult to obtain for some devices. Another way of obtaining the $V_\pi L_\pi$ is to measure the phase shift $\Delta\phi$ versus applied reverse bias V_R on a phase modulator of certain length L_{dut} . The $V_\pi L_\pi$ is calculated by [44]: $V_\pi L_\pi = V_R L_{dut} \pi / \Delta\phi$. $\Delta\phi$ is related to the change of refractive index by $\Delta\phi = \Delta n_{eff} k_0 L_{dut}$, where Δn_{eff} is the change of effective index of optical mode, k_0 is the propagation constant of optical wave in free space. Lower $V_\pi L_\pi$ means lower voltage or shorter device is needed to achieve π phase shift, hence higher efficiency. Another quantity k is used in some literature to represent the modulation efficiency, which is defined as: $\Delta\phi = k \Delta V$. k is given by: $k = \pi / V_\pi$ where V_π is the voltage for π phase shift in a phase modulator. As we can see, this quantity is length dependent, so it is not as common as $V_\pi L_\pi$.

Manipulating the free carrier concentration electrically can be realized in many different ways: carrier accumulation [22, 45, 46], carrier injection [47, 48] and carrier depletion [23] (Fig. 2.2). Carrier accumulation method works in the same principle as a capacitor: voltage applied across a dielectric layer causes the accumulation of the charges at two conducting layers separated by a dielectric

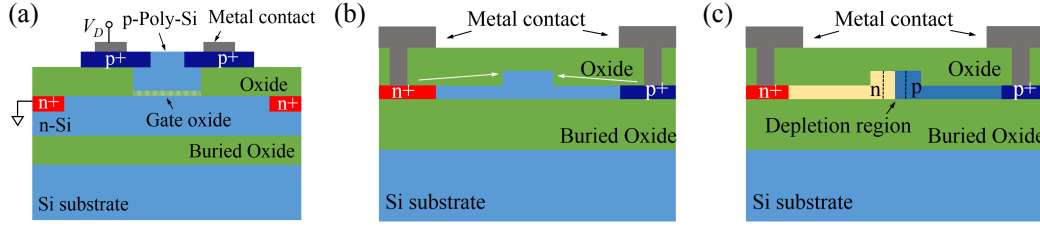


Figure 2.2: Cross-sectional view of silicon optical modulators based on the three common mechanisms for carrier concentration manipulation. (a) Carrier accumulation; (b) carrier injection; (c) carrier depletion.

layer. The first device realized by this method in silicon was demonstrated by Intel in 2004 [22]. In this work, a MOS type capacitor structure was embedded in a silicon waveguide as shown in Fig. 2.2-a. The voltage-induced charge carrier concentration change is governed by:

$$\Delta N = \Delta P = \frac{\epsilon_0 \epsilon_r}{q t_{ox} t} [V_{RF} - V_{FB}] \quad (2.5)$$

where ϵ_0 and ϵ_r are the vacuum permittivity and relative permittivity of silicon dioxide, q is the electron charge, t_{ox} is the gate oxide thickness, t is the effective charge layer thickness, and V_{FB} is the flat band voltage of the MOS capacitor. A bandwidth exceeding 1 GHz was achieved for the first time for a silicon optical modulator by using this device. In [45], a MZM using improved MOS cap device was reported and the bandwidth was increased by 10 times (10 GHz) compared to their first generation. A similar modulator using accumulation method was also reported in [46]. It was implemented by adding a poly-Si layer onto the SOI wafer and the waveguide is formed at the overlap region between the poly-Si and the silicon layer of the SOI wafer. A $V_\pi L_\pi$ of $0.13 \text{ V} \cdot \text{cm}$ was reported and 20 Gb/s data rate was demonstrated. The carrier accumulation devices can achieve very high efficiency given large capacitance as shown in Eq. 2.5. However, the waveguide structure is normally not optimized to minimize loss and the poly-Si layer might introduce excess loss.

Carrier injection is usually implemented as a forward biased PN junction or PIN junction. The excessive minority carrier accumulated in the junction due to the forward biasing changes the refractive index of silicon. The carrier injection method in general has high modulation efficiency because of high excessive minority carrier in PN junction under forward bias [49]. In 2005, Xu *et al.* demonstrated a silicon microring modulator based on a forward biased PIN junction [47]. As shown in Fig. 2.2-b, a lateral PIN junction was embedded in the rib waveguide with intrinsic rib region and p+/n+ doping on the slab layer to minimize doping induced optical loss. In another example, a carrier injection device based MZM was also demonstrated [48]. A $V_\pi L_\pi$ as low as $0.29 \text{ V} \cdot \text{cm}$ was achieved for this device.

Silicon optical modulators using carrier depletion consists of a reversely biased PN or PIN junction inside a rib optical waveguide (Fig. 2.2-c). Free charge carrier was removed from the center of the waveguide by applying a reverse bias. For an ideal PN junction, the width of the depletion region in p-type side is given by:

$$w_p = \frac{N_D}{N_A + N_D} \sqrt{\frac{2\epsilon_0\epsilon_r}{q} \frac{N_A + N_D}{N_A N_D} (V + V_B)} \quad (2.6)$$

where N_A and N_D are the donor and acceptor concentrations, and V_B is the built-in potential of the junction. Similarly, the width of the depletion region in n-type side is given by:

$$w_n = \frac{N_A}{N_A + N_D} \sqrt{\frac{2\epsilon_0\epsilon_r}{q} \frac{N_A + N_D}{N_A N_D} (V + V_B)} \quad (2.7)$$

The change in the free electron and hole concentrations are given as: $\Delta N = \Delta w_n \times N_D$, $\Delta P = \Delta w_p \times N_A$. They are directly proportional to the square root of the applied reverse bias voltage. The performance of the devices depend on the all kinds of design parameters. In ref [31], a comprehensive review of the depletion type silicon optical modulator was presented. Junctions of all kinds of

Ref	$V_\pi L_\pi$ ($V \cdot cm$)	Optical Loss (dB/mm)	Phase Shifter Length (mm)	Data rate (Gb/s)
[48]	0.29	3.9	1	12.5
[49]	0.24		0.5	2*
[50]	0.36		0.1-0.2	10
[51]	0.0025	1	0.5	0.1

Table 2.1: Performance comparison of MZM modulators based on carrier injection mode PN junction. *: 3-dB EO bandwidth

Ref	$V_\pi L_\pi$ ($V \cdot cm$)	Optical Loss (dB/mm)	Capacitance ($fF/\mu m$)	Phase Shifter Length (mm)	Data rate (Gb/s)
[52]	2.8	0.5		1	50
[44]	2.7	1.1	0.22-0.28	3	30*
[23]	1.8	1.6	0.12-0.35	0.75	60
[53]	3.15	1	0.17-0.23	4.2	35*
[30]	2.67	1.04		4	50
[54]	4	1.8		1	40
[55]	3.5	1		4.7	40
[56]	1.5-2.0	1	0.2-0.29	0.75	40

Table 2.2: Performance comparison of MZM modulators based on depletion mode PN junction. *: 3-dB EO bandwidth

orientation, lateral, vertical, interleave, wrap, seesaw, or combination of any of them, were summarized and compared.

The carrier injection type of devices usually exhibit high modulation efficiency because of the large amount of excessive minority carriers under the forward bias voltage. Table 2.1 lists some examples of the MZM based on a forward biased PN junction. All devices show high modulation efficiency and relatively small size. However, due to long life time of the excessive minority carriers in the silicon, the carrier injection devices exhibit relative low operation speed compared with other devices based on other method. Pre-emphasis driving signal is universally applied to compensate for the long minority carrier life time [48, 57]. In contrast,

the overall performance of the modulator based on depletion devices exhibits high operation speed because free carriers can be moved quickly in depletion mode (Table. 2.2). Xiao *et al.* [23] demonstrated 60 Gb/s operation of the depletion type MZM which represents the fastest operation speed of a silicon optical modulator reported as far. However, due to the low free carrier concentration in the PN junction under reverse bias, the modulation efficiency of the modulator based on carrier depletion devices is much lower than that of carrier injection devices and the size is much larger in general.

Other mechanisms have also been investigated to enhance or replace the plasma dispersion effect. The slow light effect in photonic crystal waveguide (PCW) could enhance the optical wave and free-carrier interaction thereby increase the modulation efficiency, resulting significant reduction of the modulator size (Fig. 2.3-a). For example, in [58], by using extraordinary dispersion in a PCW, a MZM of 80 μm length was demonstrated with 1 GHz operation bandwidth. An improved MZM [59] later on demonstrated 2 GHz operation, with a extremely low $V_\pi L_\pi$ of 0.0464 $V \cdot \text{mm}$. However, slow light effect in photonic crystal originates from the resonance, which inherently has narrow optical bandwidth. Alternatively, the electro-absorption based on Franz-Keldysh effect attracts a lot of attention recently. Since the pure silicon shows very small Franz-Keldysh effect, germanium was added to improve the absorbance at telecommunication wavelength (Fig. 2.3-b). SiGe electro-absorption modulators have the advantages of small device footprint, ultra-low energy consumption and CMOS compatibility [60,61]. However, the optical loss due to the intrinsic SiGe absorption even without any applied voltage leads to high insert loss compared with other devices. In addition, the absorption is very sensitive to wavelength, which may limit optical bandwidth [61].

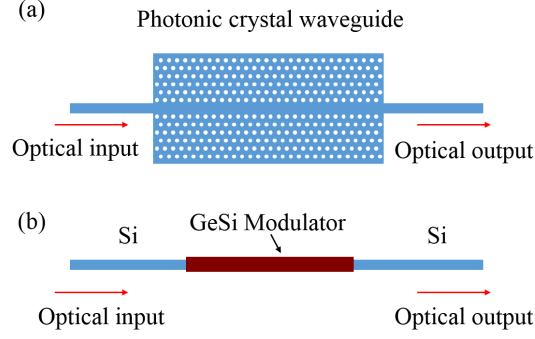


Figure 2.3: Schematic of (a) PCW; (b) electro-absorption modulator.

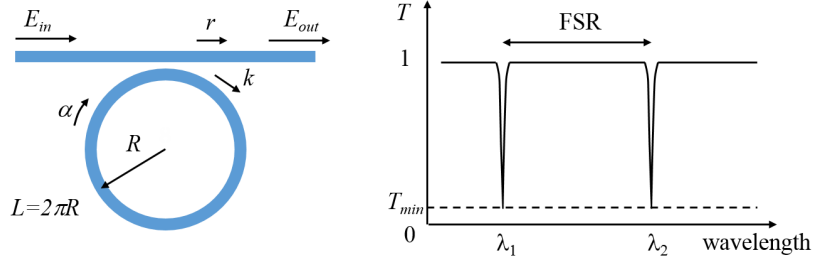


Figure 2.4: Schematic of a microring resonator coupled to an input waveguide and its transmission spectrum.

2.1.2 Optical Intensity Modulators

Most of the silicon optical intensity modulators have one of the three forms: electro-absorption modulator, microring modulator, and MZM. The first one is just a straight waveguide with voltage controlled absorbance (Fig. 2.3-b). The modulation depth is given by: $\eta = 1 - \exp(-\Delta\alpha(E)L)$, where $\Delta\alpha(E) = \alpha(E) - \alpha(0)$ is the absolute change in absorption coefficient when electrical field E is applied the SiGe and L is the length of the device.

The silicon microring resonator is one of the most important component in silicon photonics. It has been researched intensively in recent years because of its ultra-compact size which make it promising for high density integration. As shown in Fig. 2.4, a microring coupled to a straight waveguide formed the simplest

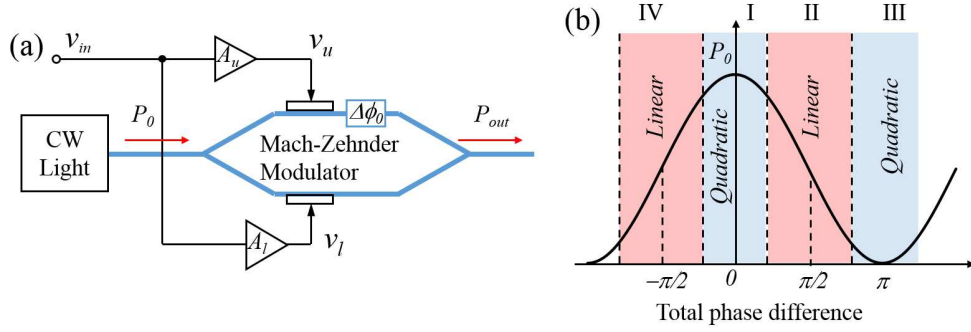


Figure 2.5: (a) Schematic of a MZM and driving circuits. (b) Relationship between optical output and total phase difference.

microring resonator. The electric field at the input and the output is related by:

$$\frac{E_{out}}{E_{in}} = e^{i(\pi+\phi)} \frac{a - re^{-i\phi}}{1 - are^{-i\phi}} \quad (2.8)$$

where $\phi = \beta L$ is the round-trip phase shift and β , L are the propagation constant of the circulating mode and length of the microring, respectively. $a^2 = \exp(-\alpha L)$ is the round trip power loss in the ring and α is the attenuation coefficient. r is the self-coupling coefficient of the microring and the waveguide. Another parameter k is defined as the cross-coupling coefficient. Assuming lossless coupling, they are related by: $r^2 + k^2 = 1$. The ring resonates when the phase shift $\phi = 2m\pi$, where we have transmission minimum.

The disadvantages of silicon microring modulator are obvious: narrowband. The optical bandwidth of the microring modulator can be derived from the width of the transmission dips, which is normally less than 1 nm, even with high order rings [62]. Further, the effective mode index of silicon is also very sensitive to temperature. A temperature change of 5°C would results in a change in refractive index in the order of 10^{-3} . This will lead to change in resonance wavelength in the order of 1 nm, which means a 5°C change in temperature will move the operating wavelength out of the bandwidth of the microring modulator [63].

The MZM is an optical intensity modulator in MZI configuration. As shown

in Fig. 3.2-a, a MZM consists of two Y-junctions at both input and output and two balanced or unbalance arms. The Y-junction at the input splits the input power equally to two arms and the phase of the light traveling in one or both arms are modulated through the plasma dispersion effect. The output Y-junction recombines the modulated light in two arms. Depends on the phase difference between the light in upper and lower arms, the intensity of the light will be modulated.

It is worthwhile to start from the basic equations of the generic MZI. The driving signals generate the phase shift $\Delta\phi_u$ and $\Delta\phi_l$ on the two arms, respectively. A fixed phase difference $\Delta\phi_0$ between the upper and lower arms is added to set the operating point of the MZM. For the output optical signal, the electrical field can be expressed as:

$$\begin{aligned} E_{out} &= \frac{1}{2}E_0[e^{j(\Delta\phi_u+\Delta\phi_0)} + e^{j\Delta\phi_l}] \\ &= E_0\cos\left(\frac{\Delta\phi_u + \Delta\phi_0 - \Delta\phi_l}{2}\right)e^{j\frac{\Delta\phi_l+\Delta\phi_u+\Delta\phi_0}{2}} \end{aligned} \quad (2.9)$$

where E_0 is the electrical field of the input continuous wave (CW) light. The output optical power

$$P_{out} = \frac{1}{2}P_0[1 + \cos(\Delta\phi_0 + \Delta\phi_u - \Delta\phi_l)] \quad (2.10)$$

where P_0 is the input CW light power, and $\Delta\phi_0 + \Delta\phi_u - \Delta\phi_l$ is the total phase difference between the two arms.

To understand operation principle, the MZM output power vs. the total phase difference is plotted in Fig. 3.2-b. Taking the total phase difference as the input, there are four regions of operation depending on the fixed phase shift $\Delta\phi_0$. Among them, II or IV are linear regions and can be used for EO modulation because of their linearity. I and III are quadratic regions and can be used for pulse generation.

The size of the MZM tends to be very large because of the weak EO effect in

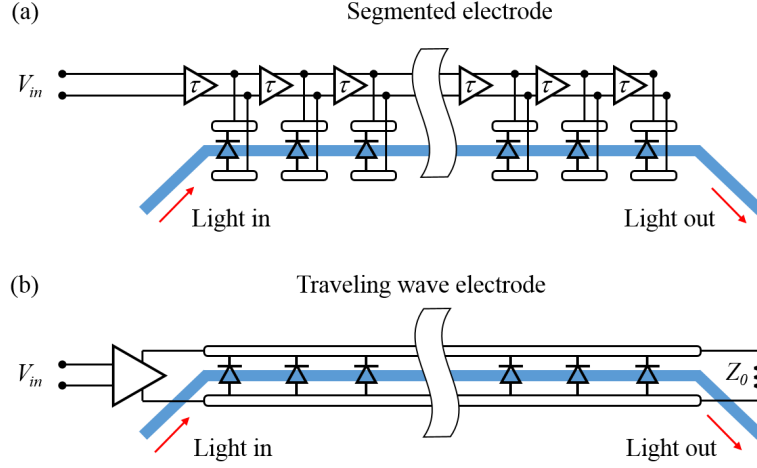


Figure 2.6: (a) Segmented electrode and driver for MZMs. (b) TWE and driver for MZMs. Notice that the transmission line has to be terminated with a matched impedance Z_0 .

silicon [20]. The use of carrier depletion devices can make it even worse because of the scarcity of the charges that can be manipulated. In IME's silicon photonic process [64], a carrier depletion based MZM can be 3-mm long. The large size of the devices translate into large capacitance loading for the modulator driver. Although through doping optimization [23] or replacing the lateral PN junction with interleave PN junction [65], device size can be reduced (0.5 mm), it doesn't directly translate into small load capacitance.

To solve this problem, one can either segment the arms into small pieces and drive each segment sequentially or absorb the capacitance into a transmission line structure as in distributed system (Fig. 2.6). Intel [45] demonstrated a segmented MZM driver in 2005 which was used to driver their 3.5-mm long MZM based on MOS devices (carrier accumulation). Ding *et al.* [32] demonstrated a segmented MZM driver design which achieved 100 Gb/s non-return to zero (NRZ) operation. The velocity matching is very challenging in these segmented driver because delay τ in the *ps* accuracy is required (Fig. 2.6-a). Analog passive delay, like LC combination has to be used which however suffers from PVT and consumes

a lot of chip area. Cignoli *et al.* [33] presented another segmented MZM driver. By utilizing a meandering MZM layout, the author saved a lot of chip area by employing combination of active and passive delay elements.

In contrast, the design of the driver can be eased significantly by using transmission line as the driving electrode for MZMs, which is called TWMZM (Fig. 2.6-b). As in a distributed amplifier [66], the large loading capacitance doesn't translate into RC time constant. Rather, it only changed the characteristics of transmission line. Therefore, the bandwidth of the TWE can be very high. In ideal situation (no intrinsic loss and no series resistance), a TWE has infinite bandwidth. Fig. 2.6-b shows a typical phase modulator with a transmission line as electrode. During operation, the electrical signal will in the transmission line at the same speed as the optical signal propagates in the optical waveguide. Therefore, the optical signal is continuously modulated by the same voltage as it propagates. Most of the high speed silicon MZM demonstrated so far employs the TWE [31,52,64].

Thanks to the advantages of large optical bandwidth, high tolerance to temperature and process variation, silicon MZM is widely used in silicon photonic transceiver chips. The first commercialized silicon photonic transceiver system by Luxtera uses the MZM with high speed phase modulator [39]. High speed MZMs using either carrier accumulation [46], carrier injection [67], or carrier depletion [30,44,52] have been reported. Among all these reported MZM, carrier depletion based MZM operates at much faster speed than the MZM based on other devices (Tab. 2.1 and Tab. 2.2).

2.1.3 Design Challenges

The MZM design involves the trades-off between all the design parameters: footprint, power consumption, optical loss, modulation efficiency, parasitics, and bandwidth. For instance, the doping concentration is one of the most fundamental design parameters. As we discussed previously, higher doping concentration

lends us more free charge carriers which in turn leads to higher modulation efficiency. The effective mode index change Δn_{eff} is approximately proportional to the product of depletion width w and doping concentration. Assuming the depletion region is mainly in p-type side ($N_D \gg N_A$), from Eq. 2.1, $\Delta n_{eff} \propto \Delta w_p N_A$. From Eq. 2.6, we have Δw_p is inversely proportional to the doping concentration, $\Delta w_p \propto \frac{1}{\sqrt{N_A}}$. Therefore, under certain condition, the effective mode index change is proportional to the square root of the doping concentration: $\Delta n_{eff} \approx \sqrt{N_A}$. On the other hand, higher doping concentration will cause more optical loss due to absorption. To solve this dilemma, the doping profile in the active region can be tailored such that only part of the active region is doped or doped heavily, while other part of the active region remain intrinsic or lightly doped [68–70]. A schematic view of the device is shown in Fig. 2.7-a. As shown in the figure, the doping concentration is higher only in the region close to the junction. Other regions are lightly doped. As a results, the optical loss can be reduced significant, while the modulation efficiency is very high. Another benefit of this doping profile is that the depletion region can be confined inside the high concentration region. Since any excess of depletion region over the active region doesn't contribute to the effective index change, the device can achieve much better linearity [69]. Further, as shown in Eq. 2.1 and 2.2, the change of free hole concentration has larger effect in change of refractive index and less effect in change of absorption coefficient compared to the change of free electron concentration. Therefore, it is desirable to have more p-type dopant in the active region than n-type dopant, which means shift the junction position to the n-type doped side (Fig. 2.7-b). Some research groups have demonstrated improvement in performance by using this kind of “offset” junctions [23, 71].

Another important trade-off in MZM design is about the waveguide. The waveguide should be designed to support only the fundamental mode, so the size (height and width) of the rib waveguide should be minimized to exclude any

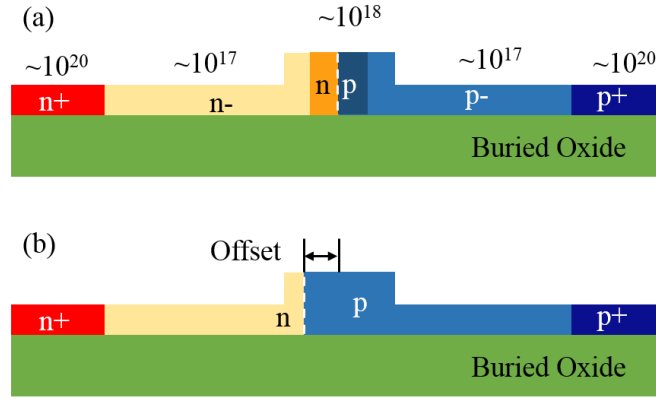


Figure 2.7: (a) Cross-sectional view of the PN junction with higher doping near the junction region. (b) Cross-sectional view of the PN junction with offset junctions. The junction locations are marked by white dashed lines.

higher order optical modes. Also, small center region leads to small parasitic capacitance. However, small center rib region results in poor overlap between the carrier depletion region and the optical mode thereby leads to poor modulation efficiency. It also results in large series parasitic resistance, thus large loss to the electrical driving signal. What's more, the device size is also limited by the capability of the fabrication facilities.

The design of a high speed TWE needs to consider the following three points [71]. Firstly, the velocities of the electrical and optical signals should be matched because the velocity mismatch will cause modulation efficiency deterioration. Secondly, the radio frequency (RF) loss of the TWE should be minimized. The loss of the TWE comes from two parts: the intrinsic loss of the transmission line, including the skin effect and the dielectric loss; the loading of the PN junction. The skin effect is the tendency that the current density is largest near the conductor surface due to the circulating eddy currents. The skin effect causes the effective cross section of the conductor to reduce at high frequency, thereby causes the effective resistance. The loss due to skin effect is proportional to $\sqrt{f}$. The dielectric loss is proportional to f . As we will discuss later, the series resistance of the loaded PN junction always leads to large loss to electrical driving signal, which

is dominant part of the total loss from the TWE. Minimization of the loss requires co-design of the PN junction and the transmission line electrode. Thirdly, the characteristic impedance of the phase shifter should be 50Ω . This is not a fundamental requirement because the MZM will be eventually integrated closely with the custom driver, in which case the short length of electrical connection and custom driver design would make it possible to choose any impedance as long as it advances the overall system performance. It might be even better to design a high impedance transmission line because that will reduce the current needed to generate required voltage swing. Nevertheless, for discrete device and testing purpose, the characteristic impedance of the TWE needs to be designed to match the 50Ω input impedance of most the RF test instruments.

Given no velocity mismatch, the accumulated phase shift is related to the driving voltage by:

$$\Delta\phi = \frac{V_0\pi}{V_\pi L_\pi} \int_0^l e^{-\alpha x} dx = \frac{V_0\pi}{V_\pi L_\pi} \frac{1 - e^{-\alpha l}}{\alpha} = \phi_0 \frac{1 - e^{-\alpha l}}{\alpha l} \quad (2.11)$$

where $\phi_0 = \frac{V_0\pi}{V_\pi L_\pi} l$, α is the frequency dependent attenuation coefficient, l is the length of the electrode. As frequency $f \rightarrow 0$, the attenuation coefficient $\alpha \rightarrow 0$. So we have $\phi \rightarrow \phi_0$. The EO conversion bandwidth corresponds to the frequency where the phase drops 3 dB, $\Delta\phi = \frac{1}{\sqrt{2}}\phi_0$. Solving for αl yields: $\alpha l = 0.74$. The electrical driving signal is $V = V_0 e^{-\alpha l} = 0.48$, which drops about “6.4 dB” from the DC value. This is the “6dB” rule of thumb frequency referred in publications about TW-MZM, *i.e.* the RF 6-dB bandwidth is close to the EO response 3-dB bandwidth. If we consider velocity mismatch between the electrical and optical signal, the phase shift expression is modified as [72]:

$$\Delta\phi = \phi_0 e^{-\alpha l/2} \left[\frac{\sinh^2(\alpha l/2) + \sin^2(\xi l/2)}{(\alpha l/2)^2 + (\xi l/2)^2} \right]^{1/2} \quad (2.12)$$

where $\xi = 2\pi f(n_{eff}^m - n_{eff})/c$ and n_{eff}^m is the effective mode index of the transmis-

sion line, n_{eff} is the effective mode index of the optical waveguide. As Eq. 2.12 show, in case of perfect impedance match, the frequency response of the loss and velocity mismatch determines the final EO conversion 3-dB bandwidth.

Last but not least, one of the challenge in designing a MZM lays in the fact that there is a lack of a unified platform includes fast and accurate simulation tools for electrical an optical co-simulation. All the design parameters need to be optimized in the context of integrated electrical and optical simulation to advance performance of the complete devices. Lumerical has started the effort and the demand for high speed optical link will drive it forward.

2.2 Equalization in Optical Interconnects

Equalization is a common technique in wireline communication to compensate for the large RF loss in channel arises from the skin effect and dielectric loss [73]. Conversely, in optical interconnects, the channel is almost flat in frequency domain. The waveguide on a PIC is usually in the order of mini-meter or centimeter at most, and single mode fiber connecting racks in a data center is up to 1 kilometer. The dispersion effect haunts the long-haul optical communication is very weak in interconnect systems because of short distance. Nevertheless, equalization can still find its application in optical transceiver circuit design. For instance, the relentless scaling of bandwidth often outpaces the development of photonic devices. Over-driving a “slow” photonic device to achieve high data rate has been investigated by many research groups and companies [35, 74, 75]. Take VCSEL based optical interconnect as an example, recent development of 400 Gbit Ethernet requires 50 Gb/s serial data rate in NRZ format [74]. However, the state of art VCSEL has a 3-dB bandwidth close to 30 GHz [76]. Commercial products are much slower than this. To overcome this limitation, different equalization techniques have been exploited. Recently, Kuchta and his co-workers have

demonstrated 71 Gb/s error free operation on a VCSEL based optical link by over-driving a VCSEL of 26 GHz bandwidth [75]. The key to this success was an equalized VCSEL driver. On the other hand, in the receiver side, equalization enables new receiver circuit architecture that can minimize noise and save power [36, 77]. Therefore, even though the channel in optical interconnects doesn't suffer from frequency dependent loss or dispersion, equalization is still a helpful technique that has been frequently applied in optical transceiver circuit design. In this chapter, we present a review of all equalization techniques that has been applied in optical transceiver circuit design. The equalization techniques discussed here limit to analog techniques in the analog front-end of an optical transmitter or receiver. Digital equalization through digital signal processing will not be covered.

2.2.1 Inter-Symbol Interference

When a perfect square pulse passes through a link with large dispersion or limited-bandwidth, the output pulse will have its energy spread out in time domain. Some of the energy will leak into the time slot (or unit interval (UI)) of adjacent pulses and causes problems. This is called “inter-symbol interference”. Consider the representative pulse response shown in Fig. 2.8, the pulse has leading ISI and trailing ISI and they are referred to *pre-cursors* and *post-cursors*, respectively. The goal of equalization is to cancel all the leading or trailing ISI. In wireline communication, equalization goal is stated in a slightly different way: to flatten the magnitude of a channel response till Nyquist frequency and remove time-domain ISI [73]. It should be noted that the requirement of flattening the frequency response till Nyquist frequency is not always followed in equalizer design for optical transceiver circuits, but removing time-domain ISI is a shared target.

Equalization can be realized in both electrical and optical domains. When realizing in optical domain, photonic devices are used to cancel the ISI in an optical pulse. In the next two sections, I will review the equalization techniques

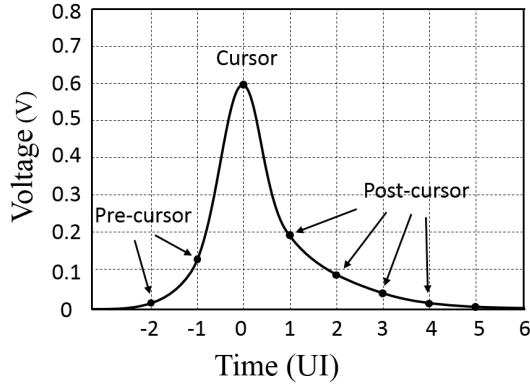


Figure 2.8: Conceptual pulse response showing pre-/post- cursor ISI.

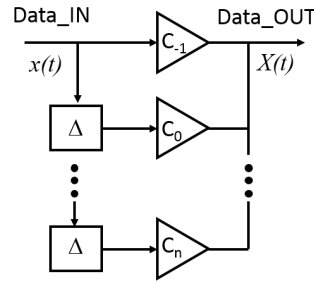


Figure 2.9: Block diagram of a FIR equalizer.

implemented in both domains.

2.2.2 Electrical Equalization

General speaking, there are two ways to cancel ISI. One way is to pre-distort the pulse before transmitting it, by adding a series of negative pulse before and after it to clean the leading and trailing ISI. The other way is to feedback a correction pulse based on the decision on previous bit to cancel the trailing ISI caused by previous bit.

A realization of the first method is shown in Fig. 2.9. This structure is the same as a *finite impulse response (FIR)* filter. Therefore, this method is referred to as FIR equalization. In a FIR equalizer, the input data will go through many taps. The signal on each tap will be delayed and weighed and then they will be

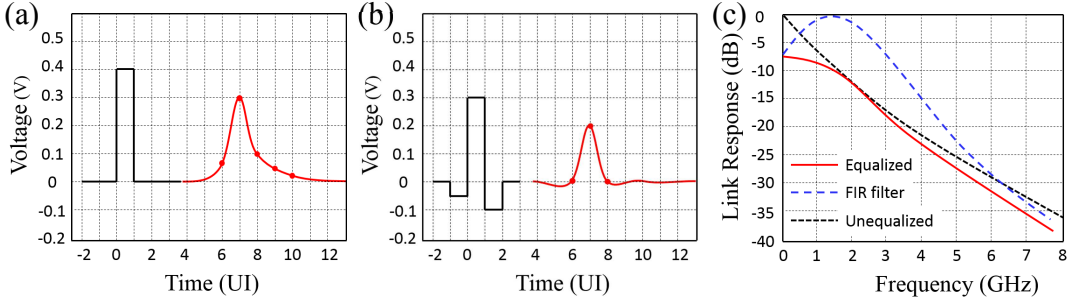


Figure 2.10: An example of FIR equalization.

summed up at the output. The output signal is given by:

$$X(t) = .. + C_{-1}x(t + \Delta) + C_0x(t) + C_1x(t - \Delta) + ... \quad (2.13)$$

where Δ is the delay, $C_{-1,0,1}$ is the weight of each tap. If Δ is a bit period T_b , the coefficient $C_{-1,1}$ will be equal to the ISI at -1 and 1 UI (Fig. 2.8). An example of the FIR equalization is displayed in Fig. 2.10. A pulse response exhibits one positive pre-cursor and a few positive post-cursors (Fig. 2.10-a red curve). To cancel the pre-cursor ISI, one tap with a $-T_b$ delay has to be added. The signal is inverted to cancel the positive ISI. Meanwhile, another tap with T_b delay should also be added and inverted to cancel the major positive post-cursor ISI. The resulting input pulse is shown as black curve in Fig. 2.10-b and the pulse response after passing through a link (red curve) exhibits no ISI. This method is sometimes referred to as *pre-emphasis* because of the waveform pre-distortion [35].

The frequency response of the link is shown in Fig. 2.10-c. The link has a large frequency dependent loss (black line) at the beginning. FIR filter suppresses the low frequency gain (blue line) while leave the high frequency part intact and thus generates a flattened response at low frequency. In time domain, the low frequency attenuation originates from the limited voltage swing in a driver circuit. For example, in a current mode driver, the output stage of the driver is a transconductance amplifier. It is important to maintain a high output impedance

by keeping the transistor in active region. Therefore, the minimum voltage is determined by the lowest voltage at the drain to make sure the transistor is still in active region. At the same time, the maximum voltage is capped by the supply voltage. As a consequence, as shown in Fig. 2.10-b, the maximum output voltage swing is fixed and the extra taps are added at the expense of reducing the main tap weight.

The delay element can be implemented as either digital shift registers [78] or analog delay cells [77, 79, 80]. Digital shift register delay has many drawbacks. Firstly, on the receiver side, to use digital shift registers, clock recovery has to come before equalization. That means it needs to operate on raw channel output, which results in large jitter in the recovered clock. To solve this problem a source synchronized interface needs to be used at the expense of more power consumption. Second, jitter in the sampling clock reduces the effectiveness of equalization. Conversely, an analog delay cell doesn't rely on sampling clock and it is easy to realize delay of a fraction of a bit period. It should be noted that a fractionally space equalizer overcomes the aliasing problem and exhibits better performance [81]. An example of FIR equalizer with analog delay cell was demonstrated in ref. [77]. Mammei *et al* presented a FIR equalizer in receiver using tunable active delay cells to compensate for dispersion in MM fiber. The 7-tap equalizer achieved successful operation up to 25 Gb/s.

One popular FIR equalizer structure in optical transmitter is *feed forward equalizer (FFE)*, whereas only two taps with fractional delay are utilized: one main tap and one pre-emphasis tap. In the time domain, the equalizer generates overshoot in the rising edge and undershoot in the falling edge of a transmitted pulse to increase the slope of both edges. This structure was firstly demonstrated in 2012 by Rylyakov *et al.* in a VCSEL based optical link [82]. In this work, the FFE was implemented in both transmitter and receiver. Error-free operation at 40 Gb/s was demonstrated for the first time by over-driving a 25 Gb/s VCSEL. Since

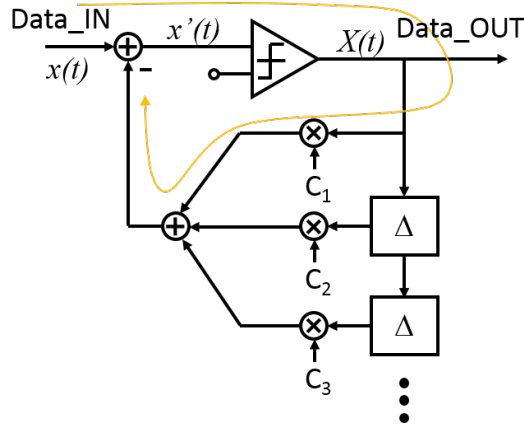


Figure 2.11: Block diagram of one realization of DFE. The yellow link shows the critical timing path.

then, FFE drivers have been frequently used to over-drive slow photonic devices. In [75], Kuchta *et al.* demonstrated the highest serial data rate (71 Gb/s) in a VCSEL based optical link using an FFE driver. In [83], Chen *et al.* presented a PAM-4 VCSEL transmitter using an FFE driver and achieved 56 Gb/s. Chen *et al.* [80] demonstrated an FFE microring modulator driver and Akiyama *et al.* [48] demonstrated an FFE driving scheme for a MZM.

In an FIR equalizer, the prior knowledge about the link is unknown. Therefore, the coefficient on each tap has to be adaptive and a training sequence is required generally to adjust the coefficients based on adaptive algorithms. In addition, if analog delay cells are used, the delay in each cell are often made tunable, too.

However, FIR filter has a disadvantage that it only attenuates the low frequency response while doing nothing to high frequency part. What's worse, an FIR equalizer amplifies the high frequency noise and therefore, it usually lead to deteriorated SNR.

In the second method, ISI is estimated based on the decision on previous bit. After that, a correction signal is fed back to the input to cancel the post-cursor ISI. This method is called *decision feedback equalization (DFE)* [73]. One realization of the DFE is shown in Fig. 2.11. Each tap in the feedback path cancels one

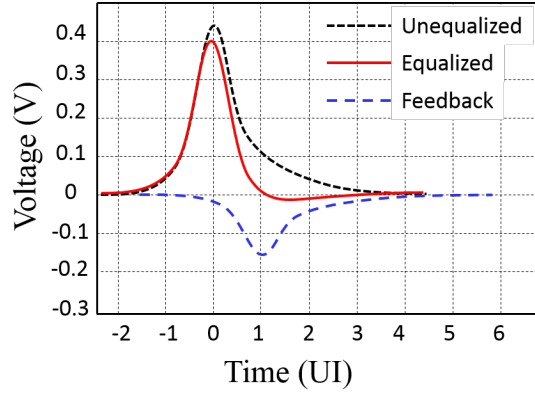


Figure 2.12: An example of DFE equalization.

post-cursor ISI. The input to the decision circuit is given by:

$$x'(t) = x(t) - C_1X(t - \Delta) - C_2X(t - 2\Delta) - C_3X(t - 3\Delta) + \dots \quad (2.14)$$

This method is sometimes referred to as *infinite impulse response (IIR)* because it employs a IIR filter. The IIR filter boosts the high frequency response and therefore, flattens the overall link response. Meanwhile, the decision stage removes the noise from data and so it doesn't amplify the high frequency noise like in a FIR filter [73]. As a consequence, a DFE generally has better noise performance than a FIR equalizer. A representative pulse response (black curve) and the feedback signal (blue curve) is shown in Fig. 2.12. The negative feedback cancels the major post-cursor ISI at 1 UI and results in a equalized pulse with little ISI (red curve).

A DFE is often found in a receiver design [36, 84]. In ref. [36], a new optical receiver based a DFE loop was demonstrated. Traditionally, a large bandwidth transimpedance amplifier (TIA) is required for high speed link. The large bandwidth trades often with transimpedance gain and consumes a lot of power. In addition, it leads to large integrated thermal noise and deteriorates SNR. In contrast to traditional design, ref. [36] replaced the TIA with a large resistor for

high transimpedance gain. The simple resistor had the advantages of low power consumption, large gain and low thermal noise. A DFE loop was then utilized to cancel the ISI caused by RC limited bandwidth. The optical receiver implemented using a 130 nm SiGe process achieved -7 dBm sensitivity ($BER = 10^{-12}$) at 15 Gb/s.

Despite its superb noise performance, DFE has a few design issues [73]. Firstly, as discussed above, it only cancel post-cursor ISI and therefore, usually a FFE is added at the same time to cancel the pre-cursor. Second, the effectiveness of ISI cancellation is based on the assumption that the decision on previous bit is correct. If error occurs, it will exacerbate the ISI instead of canceling it. Thirdly, in order to cancel the first post-cursor ISI, the feedback loop latency in the critical timing path (Fig. 2.12) must be within a bit period. The latency is the main speed limiting factor in a DFE. To overcome this issue, decision look-ahead schemes have been proposed [85,86] at the expense of much more power consumption and circuit complexity.

Other than these two major categories, there is another group of equalization technique that focuses on frequency response directly. It is called *continuous time linear equalization*. By using a stage with opposite frequency response as the channel, it can cancel the high frequency roll-off in a channel response. There are many techniques can be used to generate an opposite frequency response, such as passive or active RC network [87], RC source degeneration [88], zero-forcing by negative feedback [89], frequency peaking by inductor [90], and frequency peaking by negative resistance [91]. Moreover, the co-design of electrical circuits and photonic devices inspires new equalization techniques. One interesting technique applied in a TWMZM is to design a TWE that has a characteristic impedance increase with frequency [92]. The gain of the TWMZM driver is proportional to the characteristic impedance. As a result, the high frequency loss of a TWE can be compensated by higher gain. However, the drawback of this design is evident:

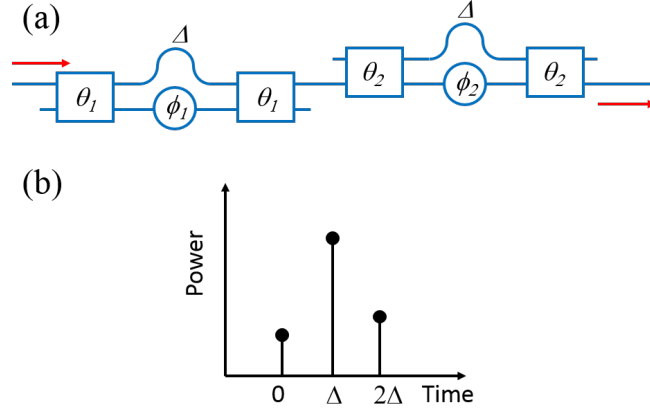


Figure 2.13: (a) Schematic of type I optical equalizer; (b) Time domain pulse response of the equalizer.

impedance matching is only satisfied in certain frequency.

2.2.3 Optical Equalization

Equalization can also be realized in optical domain, whereas optical pulse is shaped through direct time-domain addition or subtraction. The goal of the optical equalization is the same as electrical equalization, *i.e.* to cancel the time domain ISI of a pulse signal. Nevertheless, one difference between electrical and optical domain equalization is that optical pulse signal doesn't have negative ISI and a negative pulse can't be generated to cancel the positive ISI. One way to cancel the ISI in optical signal is to generate pulses of 180° phase difference to cancel the ISI in original pulses.

An optical equalizer based on the above idea was demonstrated in [37]. The schematic of the equalizer is shown in Fig. 2.13-a. It basically consists of two MZIs, each with a built-in optical path delay (Δ), a tunable phase ϕ and two 2×2 tunable couplers. The transfer function of this equalizer is given by:

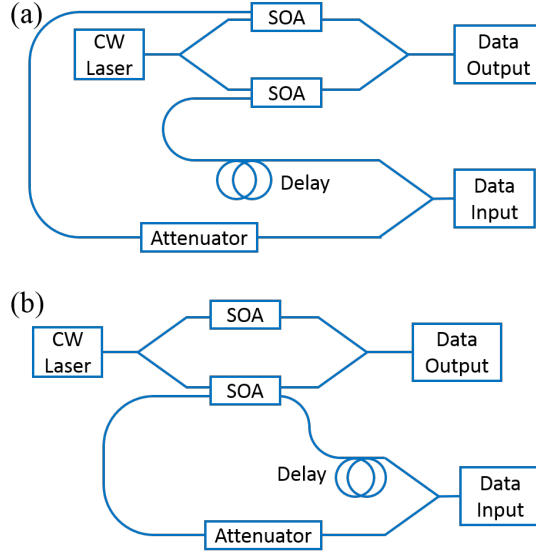


Figure 2.14: Schematic of type II optical equalizer: (a) two tap subtraction; (b) two tap addition.

$$\begin{aligned}
 h(t) = & -\sin(\theta_1)^2 \cos(\theta_2)^2 e^{j\phi_1} x(t) \\
 & + (\cos(\theta_1)^2 \cos(\theta_2)^2 + \sin(\theta_1)^2 \sin(\theta_2)^2 e^{j\phi_1 + j\phi_2}) x(t - \Delta) \\
 & - \cos(\theta_1)^2 \sin(\theta_2)^2 e^{j\phi_2} x(t - 2\Delta)
 \end{aligned} \tag{2.15}$$

where $\theta_{1,2}$ is the phase difference in the 2×2 couplers in first and second MZI, respectively. The coupling ratio is proportional to $\theta_{1,2}$ if $\theta_{1,2} \ll 1$. As shown in this equation, the pulse response consists of three taps: a main tap delayed by Δ , a pre-cursor tap and post-cursor tap delayed by 0 and 2Δ , respectively (Fig. 2.13-b). The amplitude of the optical signal in each tap is controlled by $\theta_{1,2}$ and its relative phase is controlled by $\phi_{1,2}$. A π phase shift between the main tap and pre-/post-cursor tap can be achieved by tuning both ϕ_1 and ϕ_2 .

In ref. [37], two MZIs were fabricated in separate silica substrates and connected by gluing to fiber ribbon. In [93], a integrated version of this optical equalizer was demonstrated on a silicon photonic chip.

Another way of canceling ISI is to convert pulse power into phase which can be easily subtracted by using a MZI. After that, the combined phase can be converted to optical power again by using a MZI biased in linear region. This method was demonstrated in ref. [94]. The schematic of a two tap feed forward optical equalizer is shown in Fig. 2.14. The combiner is a MZI with a semiconductor optical amplifier (SOA) on each arm. The SOAs are working in current injection mode, whereas a lot of free carriers are accumulated in the PN junction. When data signals come in, they consume some of the free carriers through a mechanism called *stimulated emission*. The amount of carrier consumed is proportional to the power of optical pulses. Further, the depletion of free carriers modifies the refractive index of waveguides and therefore modifies the accumulated phase shift of the light in these arms. Thereby, optical signals in these two taps are converted into phases, which is subsequently subtracted (Fig. 2.14-a) or added (Fig. 2.14-b), and then converted to optical pulse again by using the multi-mode interferometer (MMI) at the output of the MZI.

Optical pulse equalization through frequency domain manipulation is also possible [95]. But the setup is usually very bulky and it is not easy to be integrated.

Optical domain equalization is believed to have superior noise performance over electrical domain counterpart [37, 96]. Further, with electrical circuits move closer to photonic devices, more equalization techniques can be developed utilizing both electronic and photonic properties.

Chapter 3

A Multi-Physics Design Flow for Electronic-Photonic Integrated Circuits

With the maturing of fundamental building blocks, more research efforts have been focusing on integrating a large number of components into an integrated circuit, *i.e.*, to make EPIC systems [97–99]. To this end, standardized design flow similar to what is available in the microelectronic industry are required. Earlier work by Lumerical has developed a design flow that combines optical device simulation with some steps in traditional electronic circuit design flow [40]. A joint effort between Lumerical and IME has demonstrated a library of compact models of photonic devices which combines experimental data and photonic device simulation results in a foundry provided PDK [41]. These tools, however, are still not comprehensive. A lot of important design issues have not been addressed. Of particular importance is how to analyze and design the interface between photonic and electronic circuits, which becomes the bottleneck in high speed optical interconnect systems.

In this work, we present a comprehensive EPIC design flow that includes not only the optical devices simulation, but the electromagnetic (EM) simulation and electrical circuit simulation as well. The design flow integrates various commercial simulators through multi-lateral file sharing. Moreover, we propose a new device model for accurate representation of real devices. The loading effect from photonic

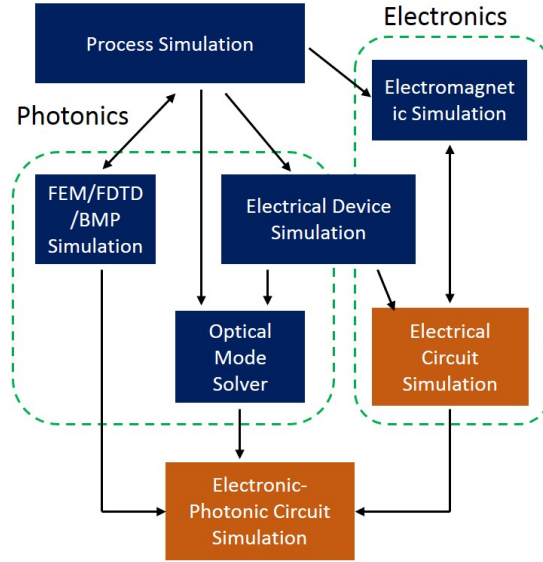


Figure 3.1: Proposed design flow of EPICs. Navy blocks are individual component modeling steps and orange blocks are circuit simulation steps. The green dashed box on the left is the photonic device modeling steps and the one on the right is the electronic part modeling steps. Arrows indicate the design flow. Iterations (represented by double headed arrows) may be required to optimize device/circuit performance.

devices on electrical driving signal, and the effect of attenuated electrical driving signal to optical signal have been taken into consideration. An example design of a TWMZM has been presented to illustrate the design flow.

3.1 Design Flow

A block diagram of the design flow is shown in Fig. 3.1. It starts from process simulation which could include lithography, etching, ion implantation and other fabrication steps. The process simulation defines the geometry of active and passive devices and it can also include the imperfections in fabrication process, such as lithographic smoothing of a nominally rectangular structure. In addition, it gives doping profile and defect distribution after ion implantation and rapid thermal annealing. The information obtained from process simulation will

be subsequently used for next step simulation, the results from which will help refine design parameters. For example, for passive devices, such as waveguide, the geometry defined in the lithography step is exported to a finite element method (FEM) or finite difference time domain (FDTD) simulator to obtain an optical S-parameter, based on which design parameters can be refined to achieve design targets. For active devices, like PN junction based phase modulator, both geometry and doping information are used for electrical device simulation to obtain electrical properties of the device. These properties will be sequentially passed to an optical mode solver to calculate its performance parameters, like modulation efficiency. Meanwhile, electronic part modeling starts from a high frequency EM simulation of electrodes. These electrodes serve as the interface between electrical circuits and photonic devices. Due to large frequency difference between electrical and optical signals, simulating the electrode and photonic device in a single simulation engine will be difficult. Therefore, to model an electro-optical interface, an electrical circuit model of the active photonic devices is firstly extracted from the small signal electrical device simulation and then this electrical circuit model is integrated with the electrical mode of the electrode (either a circuit model or an S-parameter) extracted from high frequency EM simulation. After that, both circuit models are then simulated together in an electrical circuit simulator, like ADS and Cadence Spectre. The electrical circuit simulation results give us the effect of photonic device loading on driving signals. Finally, the output of this simulation is exported to a photonic circuit simulator to evaluate the effect of deteriorated electrical signal on optical output in an optical modulator. Compared to the design flow reported by other companies, this provides a better representation of the electronic and photonic interface in a EO/OE device. On the other hand, for an opto-electronic interface, the modeling process will be almost the same. Only the data flow direction is reversed.

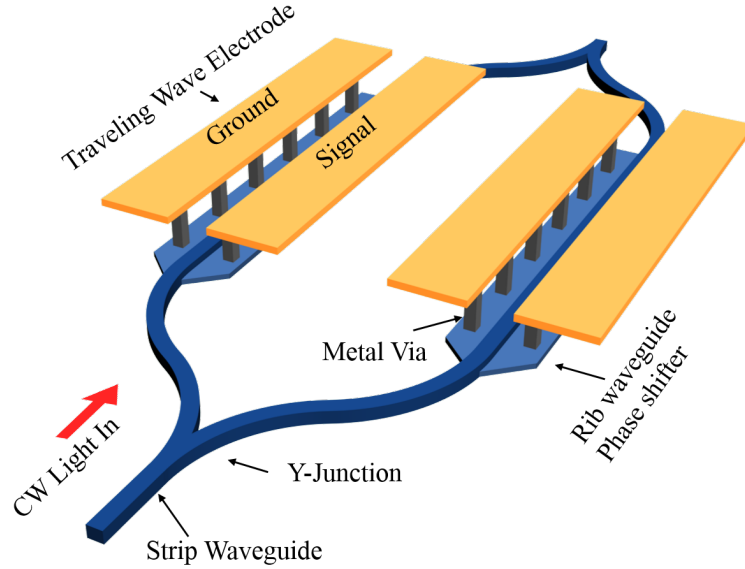


Figure 3.2: Schematic of a TWMZM based on PN junction. The input/output are made of strip waveguides and the phase shifters are built on rib waveguides. Notice that different colors used for rib and slab of rib waveguides are for display purpose only. They are both made of silicon. Long transmission lines are used to drive the phase shifters.

3.2 Design Example: Traveling-Wave MZM

To exemplify the multi-physics EPIC design flow, a TWMZM design using the design flow is discussed in this section. As shown in the schematic Fig. 3.2, a TWMZM can be considered as a mini-EPIC which includes both photonic components, such as rib-waveguide, MMI, and electrical components, such as transmission line, termination resistor. A successful design requires the detailed analysis and optimization of each individual building block, as well as the complete circuit. In this section, we will discuss the design of a TWMZM including all fundamental building blocks.

3.2.1 Silicon Waveguide

SOI is an ideal platform for silicon photonic devices. The most common SOI structure is Si-on-SiO₂ as shown in Fig. 3.3-a. A uniform layer of SiO₂ (a few micrometer), which is called BOX layer, is sandwiched between a top silicon layer (sub-micrometers) and the bulky silicon substrate (hundreds of micrometers). For an optical waveguide on SOI, the light is guided inside the top silicon layer through a mechanism called *total internal reflection*. The BOX layer has to be thick enough to prevent light from leaking into the substrate. The most widely used SOI wafer has 220-nm top Si layer and 2 μ m BOX [30, 52, 53]. In addition, that's the wafer used in IME and IMEC's silicon photonic process [44, 100]. Therefore, in this demonstration, we used a 220 nm SOI wafer for our TWMZM design. All the simulation or calculation done hereafter is for SOI with top silicon thickness of 220 nm.

Silicon waveguide is the most fundamental device in a silicon PIC. Two widely used silicon waveguides on a SOI platform are strip waveguide and rib waveguide. A strip waveguide is used for most of the routing and connections because of its compact size and low loss. Thanks to the large refractive index contrast between Si (approximately 3.5) and SiO₂ (approximately 1.46) a waveguide structure of sub-micrometer size is achievable. A typical strip waveguide is 450 nm wide and 220 nm high. The loss has been made smaller than 2 dB/cm [64, 100]. Waveguide bending can be made very small with negligible loss. Rib waveguides are used in many active devices. A rib waveguide is formed by partially etching down the silicon and leaving a ridge structure on top of a thin slab. This type of waveguide is preferred in many active silicon photonic devices because the slab provides contact for the PN junction embedded. The intrinsic loss of rib waveguide can be made as small as a strip waveguide. However, in a rib waveguide, the majority of the loss originates from p/n type doping, which is typically 10 times larger than its intrinsic loss. The top cladding of the waveguide could be either air or oxide.

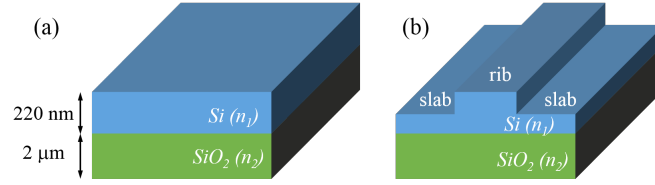


Figure 3.3: Schematic of (a) SOI wafer stack. (b) Rib waveguide on SOI wafer.

Most of the devices reported so far using a SiO_2 passivation layer on top. The function of top SiO_2 is twofold: to separate the metal electrodes from the silicon device below, and to smooth the waveguide surface to reduce loss. Therefore, in our design a SiO_2 top cladding is applied.

Single Mode Condition

Single mode condition must be guaranteed for waveguide structures because multi-mode waveguides have the problem of modal dispersion, which causes more loss and pulse broadening. The single mode condition for a strip waveguide can be estimated using the effective index method discussed in Ref. [101], if the width of a waveguide is much larger than its height. However, as the width shrinks, the assumption for effective index method no longer holds. Numerical calculation will be required to calculate the effective index and find the single mode condition. Finite difference method (FDM) method based on Ref. [102] is a widely-used method for numerical calculation of mode profile and mode index in a 3D waveguide structure. We apply this method here to study waveguide mode and find the single mode condition.

In a given technology, the height of the waveguide is usually fixed (220 nm in this example). The only parameter we can change is the width. Fig. 3.4 shows the effective index of different modes in a strip waveguide as a function of the waveguide width at wavelength $\lambda = 1.55\ \mu\text{m}$. A mode will survive in this waveguide if its effective index is larger than the cladding index. As we can see, even when $w \leq 200\ \text{nm}$, there are still fundamental transverse electric (TE) and

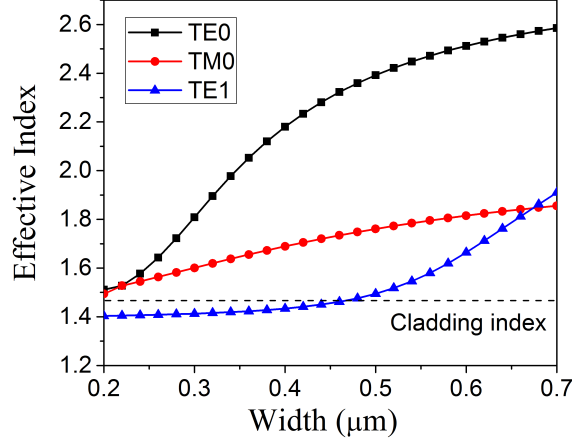


Figure 3.4: Effective index of the fundamental TE, TM and second order TE mode as a function of waveguide width in a strip waveguide with 220 nm height. The dashed line indicates the index of the cladding (SiO_2)

transverse magnetic (TM) mode in a waveguide. However, when w is relatively small, the TM mode index is close to the index of the cladding (the dashed line in Fig. 3.4) and the guiding is weak. So, more loss is expected for a TM mode. As w increases to 450 nm, the 2nd order TE mode (TE1) appears and its effective index crosses the effective index curve of the fundamental TM mode at around $w = 650$ nm, which echoes other modeling work [103]. The most common waveguide width designed on a 220 nm SOI platform is from 450 to 500 nm [64,100], which is a balance between mode confinement and requirement for single-mode condition. Fig. 3.5 shows the mode profile for fundamental TE and TM mode at $w = 500$ nm. The effective index for TE and TM modes are 2.39 and 1.76, respectively. The figure shows that the TE mode has large overlap with the sidewalls, while the TM mode interacts more with the top and bottom interfaces. Moreover, the TE mode is better confined than TM mode as expected and therefore it experiences much smaller loss. As we can see, 500×220 nm makes a good single mode TE waveguide at $\lambda = 1.55 \mu\text{m}$ and therefore, this dimension has been adapted in this design.

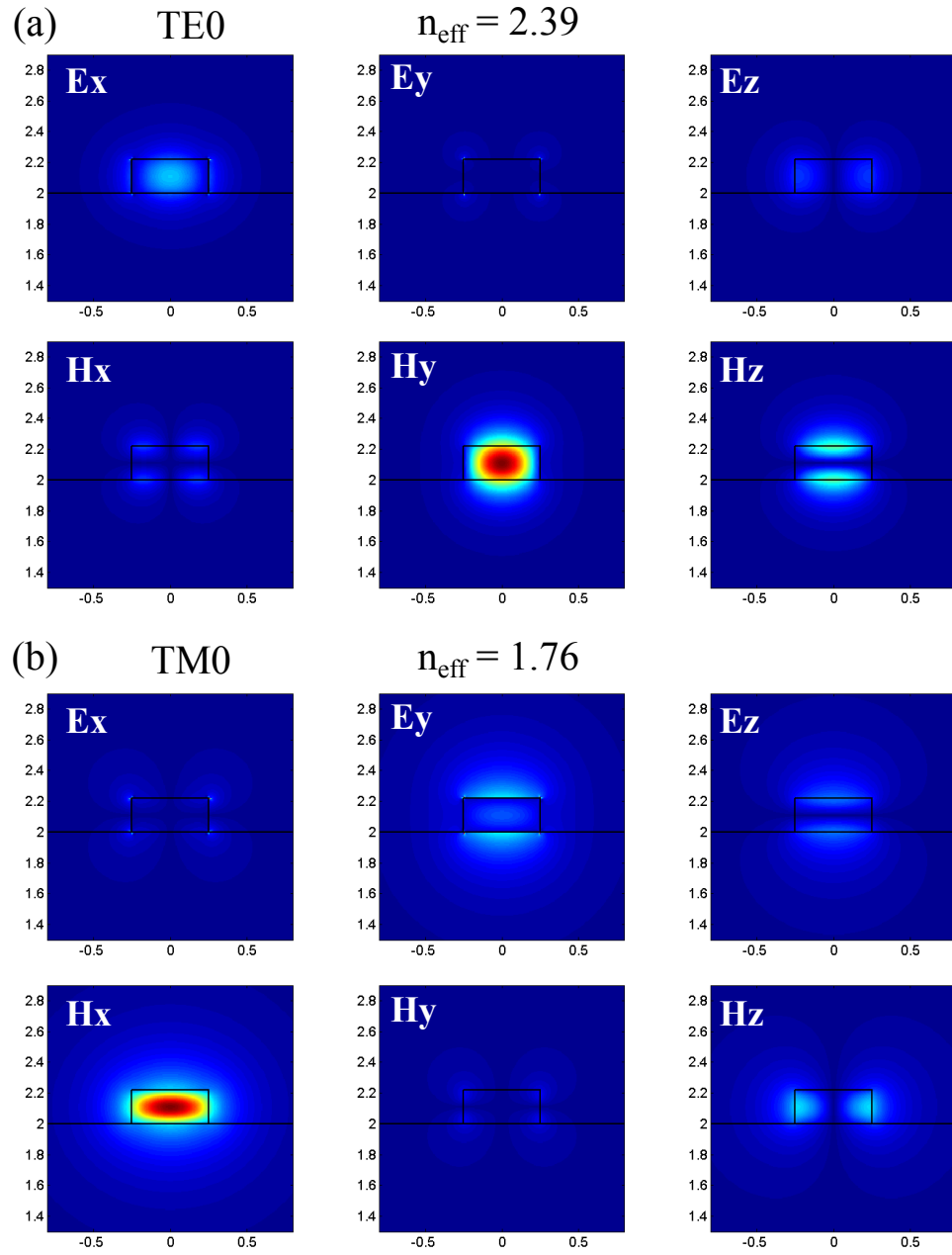


Figure 3.5: Simulated electric and magnetic field distributions of the fundamental TE(a) and TM(b) mode of a strip waveguide on SOI substrate. The waveguide dimension is $500 \times 220 \text{ nm}^2$. The black lines indicate the boundary of the rib waveguide.

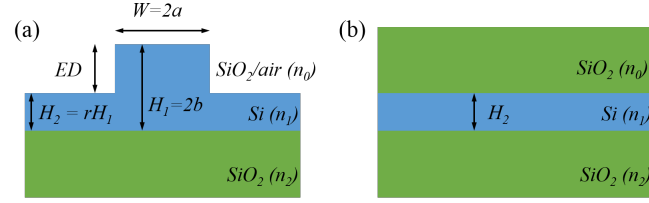


Figure 3.6: Cross-sectional view of (a) a SOI rib waveguide. (b) a planar waveguide.

The single mode condition for a rib waveguide was firstly discussed by Soref [104] in 1991. It argued that a rib waveguide doesn't have the same restrictive single mode condition as a strip waveguide because the guidance on the two sides of a rib waveguide is not as strong as in a strip waveguide. Higher order optical mode with lower mode index can leak into the slab, leaving only the fundamental mode propagating in the waveguide. Intuitively speaking, the single mode condition for a rib waveguide is that *the higher modes inside the rib region in the vertical direction should have an effective index smaller than the effective index of fundamental mode of the slab in vertical direction* [104].

Referring to the cross-sectional parameters of the rib waveguide defined in Fig. 3.6-a, a mathematical express of this condition is derived for a *large* rib waveguide that satisfies the following two conditions:

$$2b\sqrt{n_1^2 - n_2^2} \geq 1 \quad (3.1)$$

$$0.5 \leq r \leq 1 \quad (3.2)$$

The single mode condition is given as:

$$\frac{a}{b} \leq 0.3 + \frac{r}{\sqrt{1 - r^2}} \quad (3.3)$$

Nevertheless, a single mode rib waveguide may also be achievable for $r < 0.5$. But the analysis becomes complicated and an analytical expression is currently

not available. Numeric simulation is required to find out the critical dimension for single mode rib waveguides in this case.

Considering a rib waveguide fabricated on SOI wafer with 220-nm thick silicon layer, we have $2b\sqrt{n_1^2 - n_2^2} = 0.44 < 1$. So the *large* rib waveguide condition is no longer valid. The single mode condition criteria in Eq. 3.3 is not applicable in our device. Therefore, we have to resort to a numeric solution to determine the rib waveguide single mode condition.

Further, the mode cut-off condition for a planar waveguide is given as [101]:

$$V_m(TE) = \frac{m\pi}{2} + \frac{1}{2}\arctan\sqrt{\delta} \quad (3.4)$$

$$V_m(TM) = \frac{m\pi}{2} + \frac{1}{2}\arctan\left(\frac{n_1^2}{n_0^2}\sqrt{\delta}\right) \quad (3.5)$$

where $V = k_0 d \sqrt{n_1^2 - n_2^2}$, $k_0 = 2\pi/\lambda$, and λ is the wavelength of light in vacuum, $2d$ is the core layer thickness, and $\delta = \frac{n_2^2 - n_0^2}{n_1^2 - n_2^2}$, n_0, n_1, n_2 are the refractive index of top cladding, waveguide core, and bottom cladding.

In our design (Fig. 3.6-b), since both top and bottom cladding is SiO_2 , we have $\delta = 0$, thus $V_m(TE) = V_m(TM) = m\pi/2$. Then the single mode condition for the planar waveguide can be obtained by setting $m=1$, which is expressed as $V < \pi/2$. For 220-nm Si layer sandwiched between the top SiO_2 and BOX layers and $\lambda = 1.55 \mu\text{m}$, we have $V = 1.38 < \pi/2$, which means in vertical direction, there is only one optical mode. Therefore, the original criteria proposed by Soref doesn't hold. In this discussion, the Soref's criteria is modified a little. By considering the lateral confinement of the rib waveguide we argued that for a fundamental mode rib waveguide *the higher optical mode inside the rib region should have an effective index smaller than the effective index of fundamental mode of the slab*. That is $\bar{n}_{RibWG, m>0} < \bar{n}_{slab, 0}$, where $\bar{n}$ means effective index.

Based on the criteria above, we can draw a line delimiting the single mode and multi-mode domains by finding the maximum rib width at any given etching

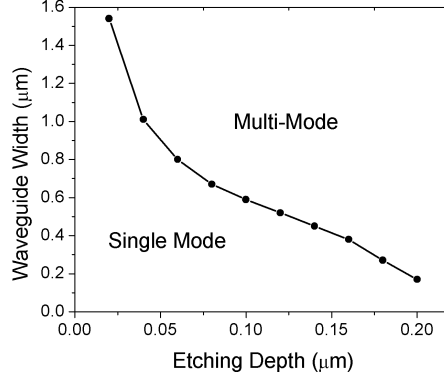


Figure 3.7: Maximum width of rib waveguides for each value of etching depth.

depth, and then sweeping the etching depth. The optical mode effective index of the rib waveguide at particular width and etching depth combination is calculated by using a vector FDM solver [102]. All the calculated mode indexes are then compared to the mode index of the planar waveguide. The criteria above is used to determine whether the waveguide is single mode or multi-mode. Fig. 3.7 shows the maximum rib width as a function of the etching depth for a 220 nm thick silicon layer. As shown in the Fig. 3.7, the maximum width of the single mode rib waveguide can be as wide as 1.5 μm , which is much larger than that of a single mode strip waveguide. As an example, both electric field and magnetic field profile for the fundamental mode of the waveguide with 0.5 μm width and 120 nm etching depth are plotted in Fig. 3.8. As shown in the figure, most of the energy is confined within the rib region, implying a high confinement factor for the fundamental mode. Strictly speaking, this mode is neither transverse electric (TE) nor transverse magnetic (TM) mode as neither E_z nor H_z is zero. The effective mode index is $n_{eff} = 2.533$ which is higher than the effective index of the planar waveguide of 100 nm thick core layer ($n_{eff,planar} = 2.155$). According to the modified Soref's criteria, this is a confined mode.

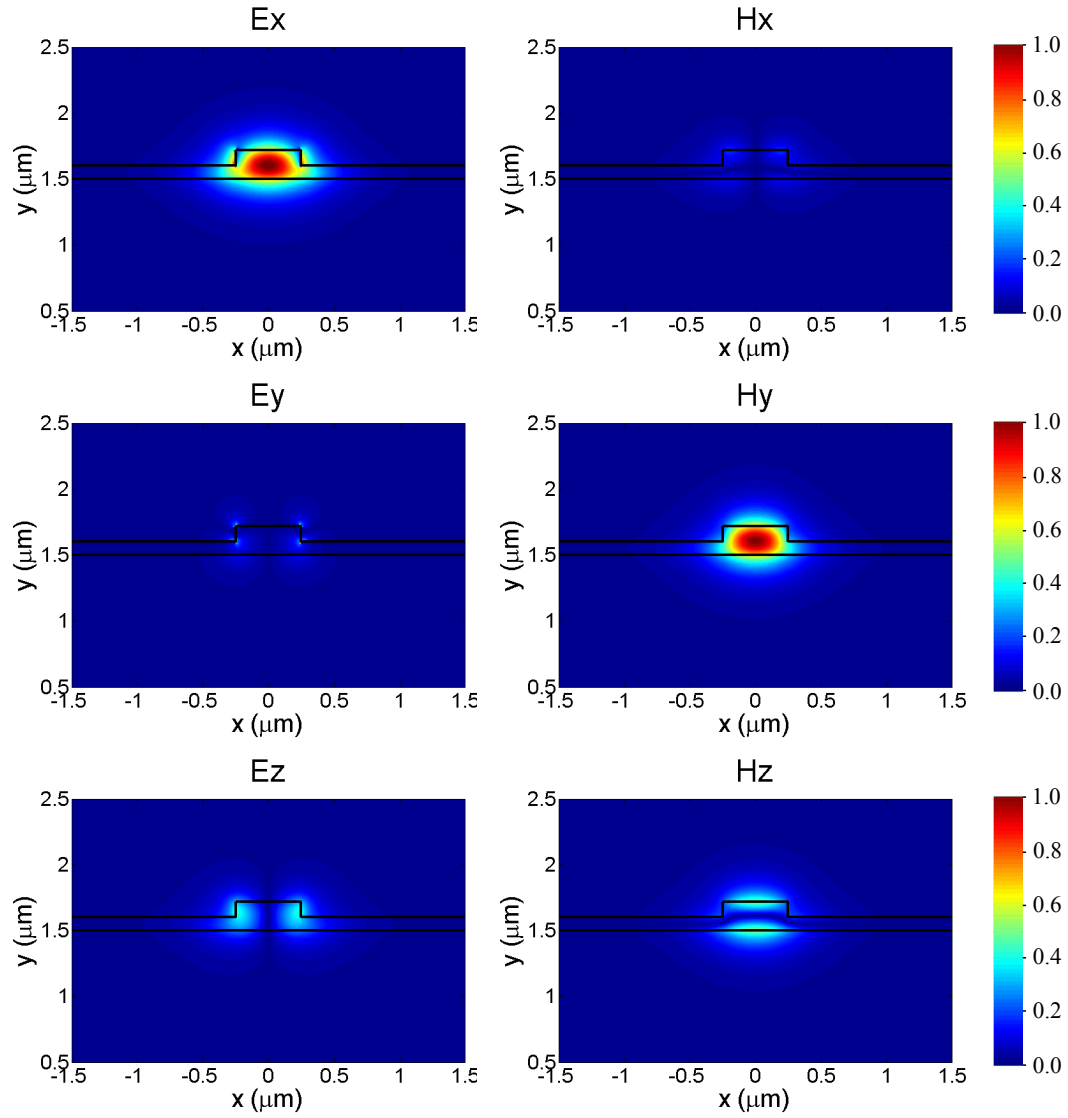


Figure 3.8: Simulated electric and magnetic field distributions of the fundamental optical mode of the rib waveguide on SOI substrate. The rib thickness is 220 nm, the width is 500 nm, and etching depth is 120 nm. The black lines indicate the boundary of the rib waveguide.

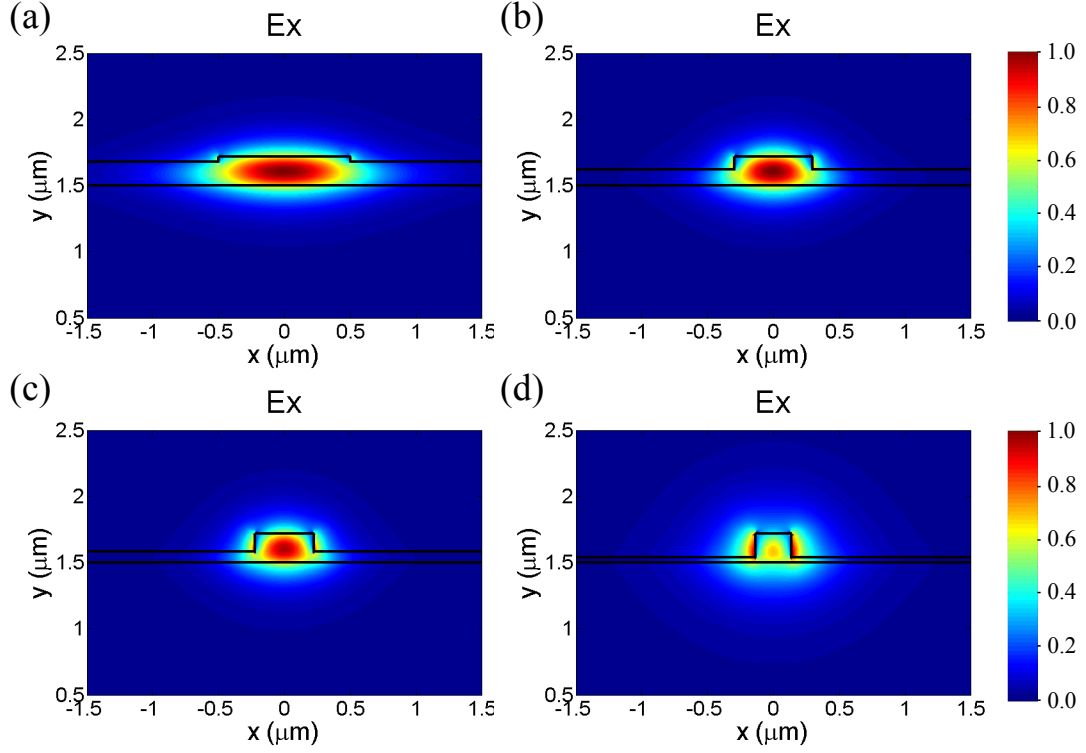


Figure 3.9: Simulated E_x distributions of the fundamental optical mode of rib waveguides on SOI substrate. (a) 40 nm etching depth, and 1 μm width. (b) 100 nm etching depth, and 590 nm width. (c) 140 nm etching depth, and 450 nm width. (d) 180 nm etching depth, and 270 nm width. The black lines indicate the boundary of the rib waveguide.

Field Confinement in the Rib Waveguide

For a PN junction based phase modulator, since most of the carrier concentration change occurs around the junction region located inside the rib, field confinement inside the rib region then determines how much interaction between the free charge carriers and the optical mode. Fig. 3.9 shows the E_x distribution of the fundamental mode of four rib waveguides with geometry defined by the boundary line in Fig. 3.7. As can be seen, moving along the line from small (Fig. 3.9-a) to large (Fig. 3.9-d) etching depth, the field confinement first increases then decreases as electric field spreads more outside the rib region. This behavior suggests that there may exist an optimal device size for best field confinement.

The confinement factor of the rib waveguide is a parameter defined as the ratio of the optical power in the desired region over the total optical power. To achieve a large modulation efficiency, the optical mode should be confined as much as possible to enhance the interaction between the free carriers and the optical mode. The region we used to define the confinement factor locates at the middle of the rib and it is 200 nm wide as shown in Fig. 3.10-a because we will see later that over the whole voltage swing, the depletion region will not change more than 200 nm. Fig. 3.10-b shows the confinement factor as a function of the waveguide width at different etching depth. All the curves end at the maximum width defined in Fig. 3.7. It clearly shows that a maximum confinement factor exists for all different etching depth and an optimal etching depth and waveguide width combination exists for best power confinement. In addition, the choice of waveguide geometry has to be considered together with other design parameters. One of the directly consequence of large etching depth is thin slab which in turn results in large series resistance for the PN junction. As a balance, the geometry of the rib waveguide is chosen to be 120 nm etching depth and 500 nm width. Another benefit of the 500 nm width is that it will also make the strip-to-rib transition much easier. The optical mode index is $n_{eff} = 2.533$, and the group index is $n_{g,eff} = 3.66$.

Thereby, we have defined the width of the most fundamental structure on a Si photonic chip. The process simulation at the beginning of our design flow can be used to optimize the fabrication process to achieve the design parameters and it will also help to understand the effect of imperfect waveguide wall to the optical mode profile and optical loss.

3.2.2 Y-Junction

Y-junction is a widely used passive component in a silicon PIC for optical power splitting and combining. In a TWZM, Y-junctions split and combine the optical

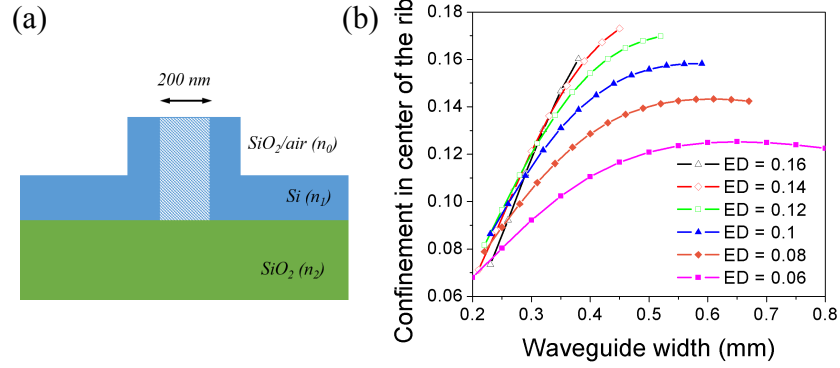


Figure 3.10: (a) Region for confinement factor calculation (b) Confinement factor as a function of waveguide width for different etching depth.

power at the input and output. For most of the passive device, we care about its transfer function at different wavelength, *e.g.*, reflection and transmission spectra. The data can be summarized in an optical S-parameter. FEM/FDTD simulation is usually carried out to obtain the optical S-parameter. In this TW-MZM design example, we design a basic Y-junction formed by two flipped sinusoidal bends (S-bends) and a feed waveguide [105]. A schematic of this Y-junction is shown in Fig. 3.11-a. There are several design parameters associated with this structure. As a demonstration here, a single design parameter is considered, which is the length of the bending section of this Y-junction, L_s . The bending radius (r) and angle (θ), and the length of the straight section (d) at middle of the S-bend are all calculated based on L_s and other parameters. To further simplify the analysis, the spacing between upper and lower arms are fixed. A 3D FDTD simulator, Lumerical FDTD solution is utilized for the study of optical S-parameter at different L_s . The strip waveguide design is the same as we discussed previously, $500 \times 220 \text{ nm}^2$. The spacing between the two arms is set to $36 \text{ }\mu\text{m}$ to save computation time and memory. In real devices, this value is usually around 100 to $200 \text{ }\mu\text{m}$, depending on the size of driving electrodes. Fig. 3.11-b plots the optical $|S_{21}|^2$ and $|S_{31}|^2$ over a wavelength range from 1.3 to $1.6 \text{ }\mu\text{m}$ at $L_s = 30 \text{ }\mu\text{m}$. No difference was observed for $|S_{21}|^2$ and $|S_{31}|^2$ indicating a perfect symmetry be-

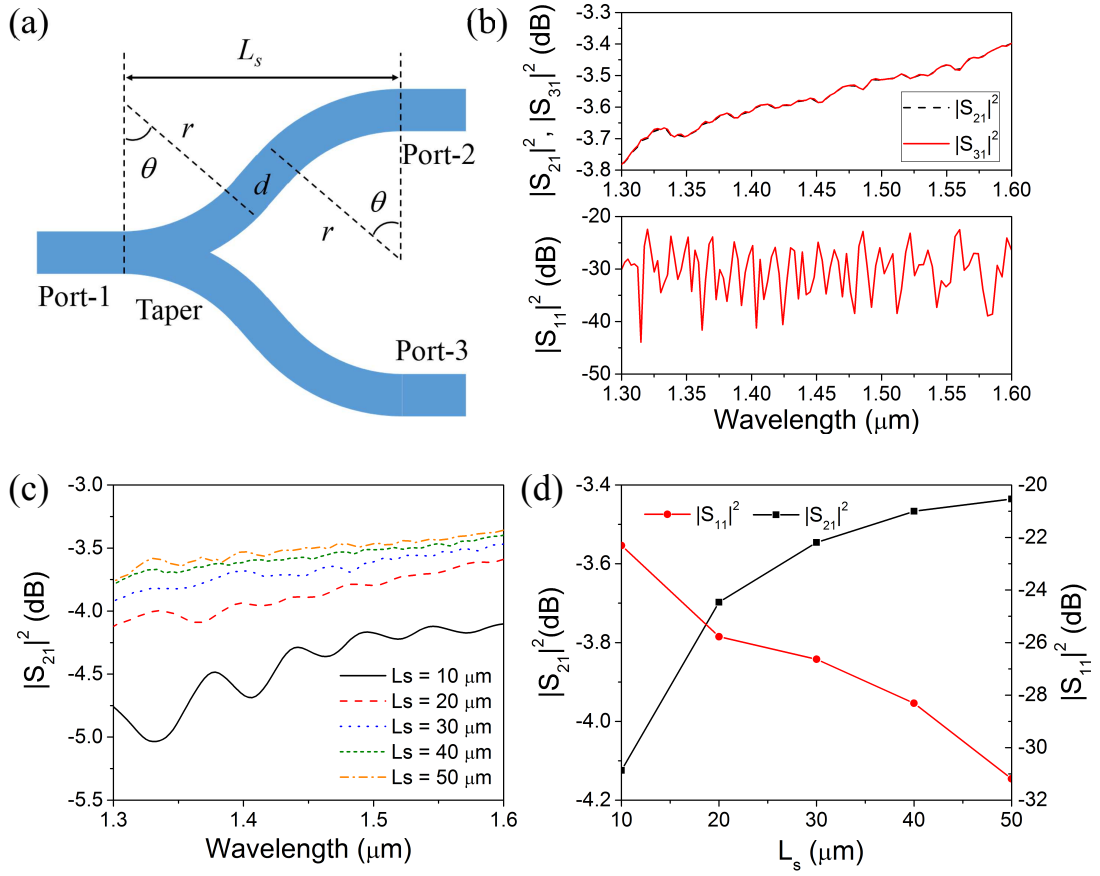


Figure 3.11: (a) Schematic of the Y-junction based on two S-Bends. (b) Simulated S-parameter of the Y-junction for $L_s = 30 \mu\text{m}$. Upper: $|S_{21}|^2$ and $|S_{31}|^2$; Lower: $|S_{11}|^2$. (c) Simulated $|S_{21}|^2$ of the Y-junction for different L_s values. (d) $|S_{21}|^2$ and $|S_{11}|^2$ at $\lambda = 1.55 \mu\text{m}$.

tween the two branches. However, in real devices, process variation will introduce power imbalance and process simulation calibrated with experimental data will be needed to fully characterize the device. Moreover, as shown in the figure, the loss at shorter wavelength is higher which is due to higher order TE modes being excited at that wavelength. When light enters the bending region, part of the energy is transferred to higher order mode and get lost. Shorter the wavelength, more the number of modes being excited and hence higher the loss. Fig. 3.11-b also plots the reflection spectra ($|S_{11}|^2$) of this device. The reflection is below 20 dB over the entire wavelength range.

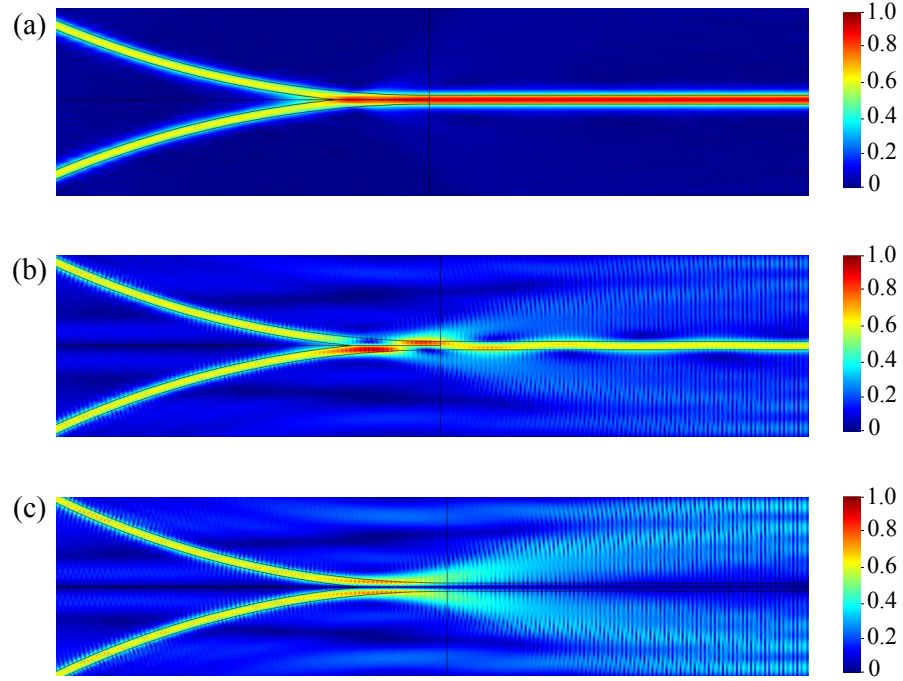


Figure 3.12: Simulated optical power combination using Y-junction at three phase differences between two input ports. (a) The phase difference is 0. (b) The phase difference is $\pi/2$. (c) The phase difference is π .

The $|S_{21}|^2$ at different L_s is shown in Fig. 3.11-c. The optical loss increases with the decreasing L_s as we expected. As shown in Fig. 3.11-d, $|S_{21}|^2$ is increased by 0.7 dB as L_s increases from 10 to 50 μm at $\lambda = 1.55 \mu\text{m}$, and the average reflection decreases almost by 10 dB. Further increase in L_s doesn't result in more loss reduction, but will take more chip area.

When Y-junction is applied as an optical combiner, the transfer function is a little bit complicated. To study its behavior at the output, a 2-D model of this type of Y-junction is simulated using FEM simulator COMSOL. (Fig. 3.11-a). Effective index method is adapted to build this 2-D model, whereas the core index is set to the effective index of a 2-D slab waveguide. The width of the waveguide is 500 nm, which is the same as that of the strip waveguide discussed previously. The radius of curvature of S-bend is 50 μm . The feeding waveguide is 20 μm long.

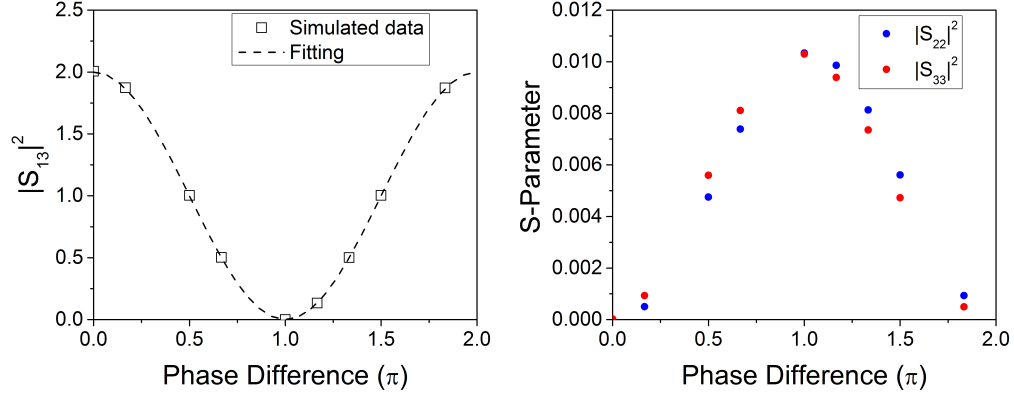


Figure 3.13: (a) Simulated power transmission versus the phase difference between two input ports. The dashed line is the fitting function $P = P_0(1 + \cos(\Delta\phi))$. (b) Simulated reflection at Port-2 (Black dots) and Port-3 (Red dots) as a function of the phase difference between two input ports.

In power combining, Port-2 and Port-3 are excited with the same power but different phase. It is found that the output power is a function of the phase difference between the two inputs. This is the fundamental working principle of a MZM. Fig. 3.12 shows the electric field distribution at the Y-junction as a function of phase differences between the two input ports. When the two input lights are in phase, their power simply add up, *i.e.* $|S_{13}|^2 = |S_{12}|^2 = 2$. With $\pi/2$ phase shift difference, we observe a lot of power radiating into the cladding and being absorbed by the boundary wall (scattering boundary condition). Higher order mode is excited when the two inputs just meet, which has a meandering shape and gradually diminish because of leakage. In addition, an interference pattern is observed which suggests the existence of weak back reflection. The calculated $|S_{13}|^2 = |S_{12}|^2 = 1$, which means only half of the input power remains at the output. With π phase difference, we can see all power radiates into the cladding and nothing is propagating inside the waveguide. The calculated $|S_{13}|^2 = 0$. The output power ($|S_{13}|^2$) and reflectance ($|S_{22}|^2$ and $|S_{33}|^2$) are plotted as a function phase difference between the two input ports. The $|S_{13}|^2$ can be fitted perfectly by: $P = P_0(1 + \cos(\Delta\phi))$ where P_0 is the power at input port. The reflectance

is inversely related to the transmission and it is very weak in general which is in accordance with the observed electrical field profile.

The passive device design follows the flow on the left hand side of the complete design flow chart. Theoretical calculation plus some numerical simulation provides the baseline for the design. Then process simulation helps to optimize the device fabrication and suggests any process variation. FED/FDTD simulation takes the geometry obtained from process simulation, including any process variation, and uses it for optical S-parameter extraction. Device optimization mainly focus on optical S-parameters and other related performance metrics, such as optical loss, Q-factor and so on. Several iteration may be required to fully optimize the device.

3.2.3 PN Junction Design

An active device's property can be modulated by electrical signal applied. Therefore, the design should include optimization of both optical part and the electrical part. In this TWMZM design example, phase modulator is an active devices which changes its refractive index (both real and imaginary parts) with applied voltage signal. A reverse-biased PN junction based phase shifter is widely used in many MZM design, demonstrating good performance and compatibility with the CMOS process. Here we present the detailed design process of a PN junction based phase shifter to demonstrate the design for active devices.

Many different types of junctions have been explored so far. Reed *et al.* [31] provides a comprehensive survey on the performance of all kinds of junctions. In this design, we are going to use the basic PN junction because of its simplicity and good performance [31]. We are going to optimize its doping profile in terms of the modulation efficiency, optical loss and parasitics.

Doping concentration is one of the most important design parameters. It is directly related to almost all the performance merits that we concern, such as modulation efficiency, optical loss, parasitics. To determine the doping concen-

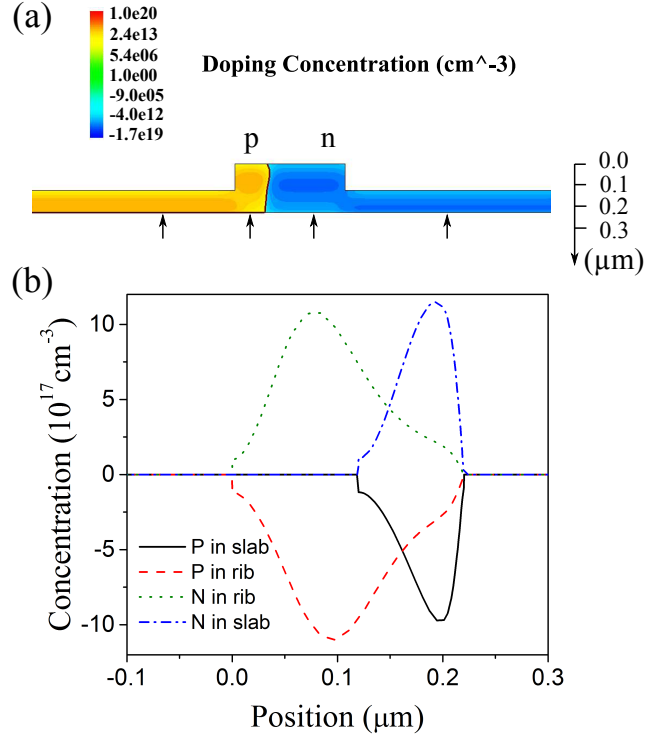


Figure 3.14: (a) Doping profile of the PN junction after process simulation. The arrows marks the location of the vertical cut lines. (b) Doping profiles along 4 vertical cut lines. Negative values mean p-type doping.

tration, let's first look at the modulation efficiency. The phase difference between the two arms of a MZM is given by:

$$\Delta\phi = \Delta n_{eff} k_0 L = \Delta n_{eff} L \frac{2\pi}{\lambda_0} \quad (3.6)$$

In order to achieve a phase shift of π , we must have $\Delta n_{eff} \frac{L}{\lambda_0} = 0.5$. For L in the order of 1 mm, $\Delta n_{eff} = 7.75 \times 10^{-4}$. According to Eq. 2.1 and Eq. 2.2, the doping concentration should be in the order of 10^{17} cm^{-3} . This will result in a change of absorption coefficient of $\Delta\alpha = 1.4 \text{ cm}^{-1}$, *i.e.* an additional 1.2 dB/mm loss.

To achieve the required Δn_{eff} , based on the estimation above and the fact that depletion region will be only part of the waveguide region, we estimated an average doping of $5 \times 10^{17} \text{ cm}^{-3}$. The same doping concentration for both n and

p side doping is adopted for simplicity. Additionally, Eq. 2.1 and Eq. 2.2 suggest that the p-type carrier has larger effect in the refractive index change and less effect in the absorption effect change. Therefore, it will be desirable to have an offset junction with more p-type doping in the rib region. In this design example, two different junction locations are exploited to investigate the effect of junction location to the performance of the devices. One has its junction located at the very center of the rib waveguide and the other one has its junction offset by 100 nm from the center.

The design of the PN junction involves detailed simulation using a suit of device simulator and optical mode solver. Before we discuss the simulation results, it is necessary to introduce the simulation flow. Again, the first step is the device fabrication process modeling using a 2-D process simulator Sentaurus Process. Previously, we have obtained the geometry of the rib waveguide. Here, the ion implantation and diffusion are simulated by employing an advanced analytical implantation model and a depth dependent lateral straggling. An analytic damage and point defect calculation is also utilized, which are calibrated to a large number of experiments. The generated files contain exact impurity distribution after ion implantation and annealing. These files are subsequently imported into a 2-D device simulator Sentaurus Device for electrical properties calculation. The device simulator uses a stationary solver which solves the coupled Poisson equations and the charge continuity equations for both electrons and holes numerically. It gives out the electrical field, charge and current distribution at different bias voltages. It can also output small-signal current and voltage results for parasitics extraction. The extracted free carrier distribution is then translated into the distribution of refractive index using Eq. 2.1. Finally, the effective mode index and optical loss are calculated using the FDM mode solver. Effective index calculated at different bias voltages gives the modulation efficiency and the extracted parasitics will affect the electrical signal modulating the devices.

The doping profile of an offset PN junction after the process simulation is shown in Fig. 3.14-a. The junction is about 100 nm offset from the center, resulting 350 nm p-doped region and 150 nm n-doped region. Fig. 3.14-b shows both the n-type and p-type(negative value) doping profile along vertical lines cutting the slab and the rib regions. Noticed that the zero position starts from the top of the waveguide. Both the rib and the slab are designed to have the same amount of peak doping concentration. Owing to the thickness difference, the locations of peaks are different in the rib and the slab regions. The peak doping concentrations for both n-type and p-type doping are about $1.1 \times 10^{18} \text{cm}^{-3}$. The contact regions for the PN junction are located $1 \mu\text{m}$ away from the edge of the rib as a compromise between series resistance and the optical loss. Doping concentrations at the contact regions are about $1 \times 10^{20} \text{cm}^{-3}$.

In electrical device simulation, we study both dc and small signal behavior of a PN junction. In the dc simulation, the bias voltage ramps from -1 to 8 volt, where positive means reverse bias and negative means forwards bias. The IV curve shows that the junction doesn't break over the entire ramping range. The carrier concentration at different bias voltages are shown in Fig. 3.15. Electrons (left) and holes (right) are summed up and shown together in this figure. As we can see, with increasing bias voltage, the carriers are pulled away from the junction region, increasing the width of depletion region. It is noted that the effective light-carrier interaction region is about 200 nm, for a voltage swing from 0 to 6 V. This is where an optical mode should be confined in. The number of charges being removed approximately equal to the product of doping concentration ($N_{A,D}$) and depletion region width W .

The extracted carrier distribution is then converted to distribution of refractive index (Δn) and optical loss ($\Delta \alpha$) based on equation Eq. 2.1 and Eq. 2.2, respectively. The new effective refractive index of the waveguide can be calculated again using the FDM optical mode solver [106] and an effective optical loss

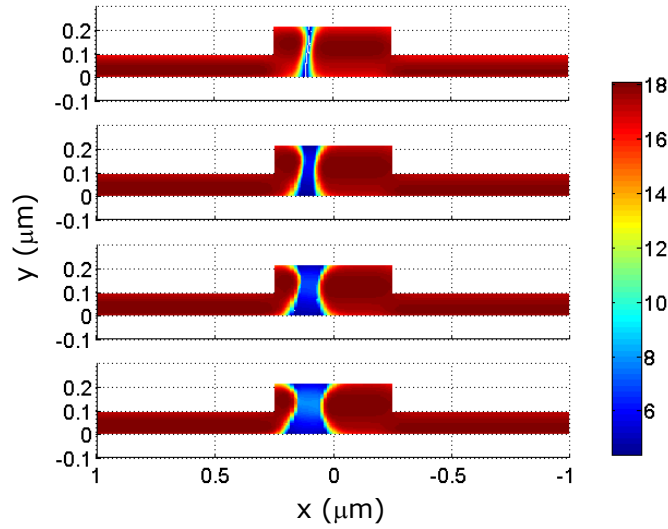


Figure 3.15: Carrier distribution at different bias condition. From top down: $V_R = 0, 2, 4$ and 6 V. The carrier concentration is plotted in log scale. So, the scale bar represents the order of magnitude. Electrons (left) and holes (right) are summed up and shown together in this figure.

can be calculated by:

$$\Delta \alpha = \frac{\iint_s \Delta \alpha(x, y) |E(x, y)|^2 dx dy}{\iint_s |E(x, y)|^2 dx dy} \quad (3.7)$$

These two values are calculated at different bias voltages and they are plotted in Fig. 3.16 against the bias voltage. As we can see, the change of effective refractive index ($n_{eff}(V) - n_{eff}(0)$) is on the order of 10^{-4} . Therefore, it can be considered as small perturbation to its original mode profile. The Δn_{eff} increases with reverse bias voltage and at 8 V, Δn_{eff} is about 2×10^{-4} . The PN junction breaks down at about 9 V. Moreover, the Δn_{eff} versus V curve is not linear, implying a modulation efficiency changes with bias voltage. The modulation efficient $V_\pi L_\pi$ can be calculated as $V_\pi L_\pi = \frac{\pi \Delta V L}{\Delta \phi}$ [44], where ΔV is change in bias voltage and $\Delta \phi$ is change in phase. L is the length of the phase shifter. The phase change $\Delta \phi$ is given by $\Delta \phi = \Delta n_{eff} k_0 L$. Given $k_0 = \frac{2\pi}{\lambda_0}$, the modulation efficiency $V_\pi L_\pi = \frac{\lambda_0}{2} \cdot \frac{\Delta V}{\Delta n}$. The modulation efficiency is inversely proportional to the slope of Δn versus ΔV curve. It is calculated and plotted also in Fig. 3.16. At 0 V bias,

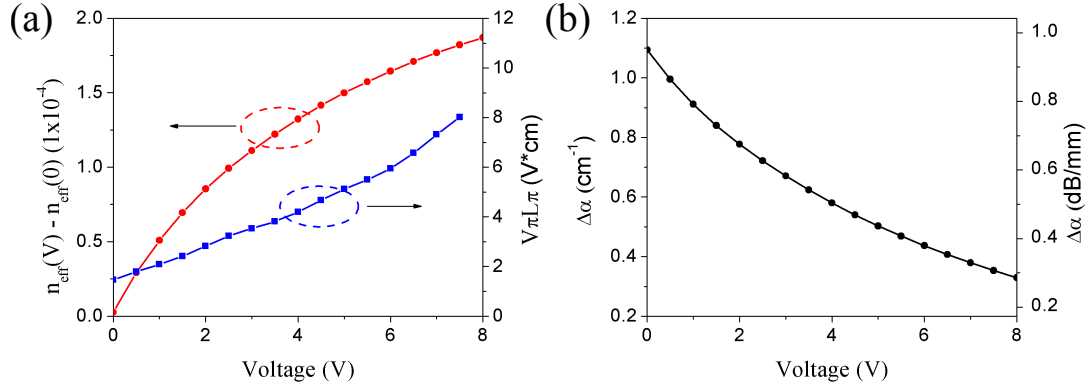


Figure 3.16: (a) Simulated effective mode index change ($n_{eff}(V) - n_{eff}(0)$) and calculated $V_{\pi}L_{\pi}$ of the phase shifter. (b) Simulated absorption coefficient change of the phase shifter at different reverse bias.

$V_{\pi}L_{\pi}$ is as low as $1.3 \text{ V} \cdot \text{cm}$, comparable or lower than the state of arts. As the bias voltage increases, the modulation efficiency decreases because the depletion region moves away from the center where most of optical power is confined. This nonlinearity causes optical pulse distortion in large signal operation, which will be discussed later in next chapter. Moreover, the change of absorption coefficient is plotted against the bias voltage as shown in Fig. 3.16-b. The loss in dB due to the change of absorption coefficient is shown on the right y-axis. The maximum additional loss is 2.1 dB/mm , which is much larger than the intrinsic loss of the waveguide (usually around 1 dB/cm). The optical loss decreases with the bias voltage as more free carriers being removed from the junction.

In the small signal simulation, a small periodic voltage excitation is added on top of the dc bias, and the current is monitored to calculate its admittance [107]. At each frequency ($f = \omega/2\pi$), the current is given by:

$$\delta I = Y\delta V = A\delta V + i\omega D\delta V \quad (3.8)$$

The coefficient matrix A and D are extracted and used to calculate its parasitic parameters. A reverse biased PN junction can be well-described by a series RC

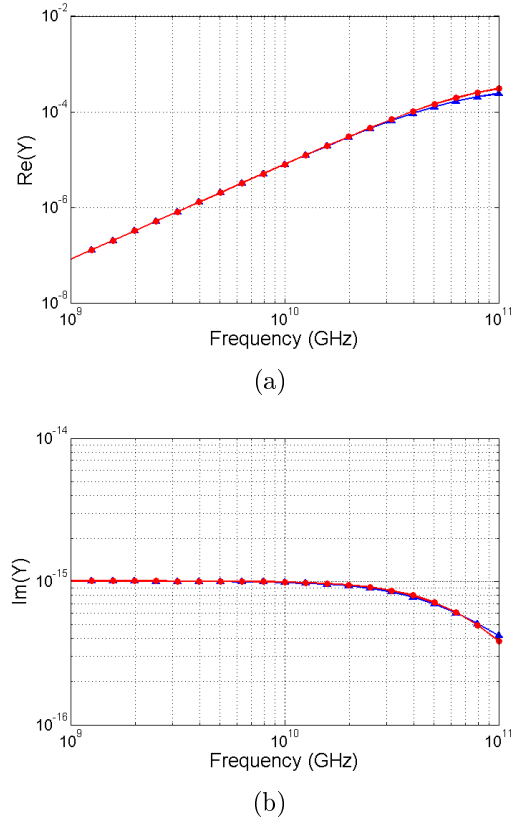


Figure 3.17: Calculated based on simple RC model (blue) and simulated (red) (a) real part, (b) imaginary part of admittance of the PN junction.

model. A fitting to the simulated admittance data gives the value for R and C . In Fig. 3.17, the calculated admittance based on extracted RC value and the simulated admittance are plotted. As we can see, the series RC model works very well, especially at low frequency. For frequency above 30 GHz, a small discrepancy is observed. However, this is already beyond the frequency range of interest for most of devices today. The parasitic RC is the major bandwidth limiter of a PN-junction based MZM. In a TWMMZM, capacitance alone doesn't limit the bandwidth since the capacitance will be absorbed into the transmission line, but it affect the mode index of the transmission line and therefore the velocity mismatch between optical and electrical signals. On the other hand, the resistance will directly affect the frequency response of driving electrode, which in turn affect

the EO response of modulators. To reduce the resistance, one can either increase the thickness of the slab, or increase the doping at the slab. Not surprisingly, both come with prices. The first method changes the rib waveguide design and affects its confinement factor, while the second method increases the optical loss and the complexity of the devices, leading to low yield and high cost. Given the complexity, the optimization of RC has to be done together with electrode design and has to be verified in a circuit level simulation.

The extracted parasitic series resistance and capacitance are plotted against the bias voltage as shown in Fig. 3.18. The parasitic resistance changes from $8.7 \, \Omega \cdot mm$ at $0 \, V$ bias voltage to $7.4 \, \Omega \cdot mm$ at $8 \, V$ bias voltage and the parasitic capacitance changes from $365 \, fF/mm$ to $179 \, fF/mm$ for the same voltage range. This device exhibits an RC bandwidth of over 40 GHz over the entire bias range. Overall performance of the designed device is comparable to the state of art (Tab. 2.2).

Optimization of the doping has to consider many different parameters, such as modulation efficient, optical loss, parasitic resistance and capacitance. They all trade with each in actual design. For instance, to increase the modulation efficient, one can introduce more dopant into the junction since it is proportional to the total charges being removed for a given ΔV . However, introducing more charges will cause more optical loss and capacitance.

3.2.4 Traveling-Wave Electrode Design

The driving electrodes of a TWMZM are transmission lines. There are three criteria a TWE design should satisfy. Firstly, the electrical driving signal should propagate in the transmission line at the same speed as the optical signal. Secondly, the loss arising from the silicon substrate and the loading PN junction should be minimized to maximum EO bandwidth. Finally, the characteristic impedance of the TWE should match that of most of the RF test instruments

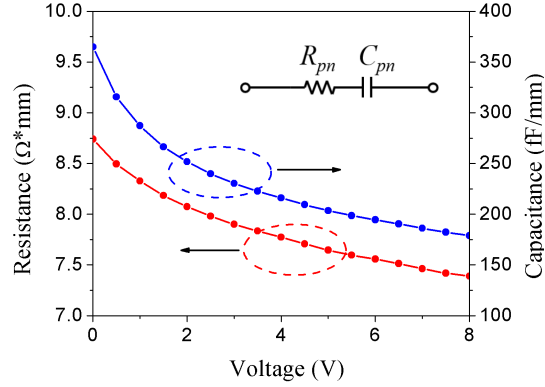


Figure 3.18: Extracted small-signal model parameters of the PN junction. The inset shows the RC circuit model of the PN junction.

which is 50Ω .

Transmission Line

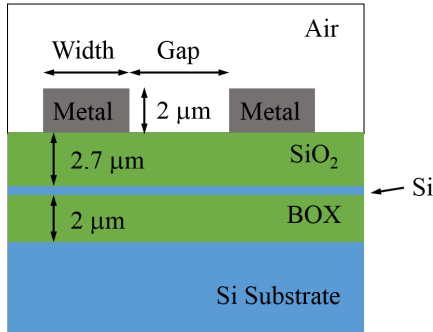


Figure 3.19: Dielectric layer stackup for the TWE.

An on-chip transmission line has many different varieties, among which coplanar strip is the most popular configuration because only one metal layer is required and the footprint is only half of that of a coplanar waveguide configuration. The cross-section of the coplanar strip configuration is shown in Fig. 3.19. The dielectric layer stackup is in accordance to the IME's silicon photonic process [108]. About $2.7 \mu\text{m}$ thick SiO_2 is deposited on top of the SOI wafer on which the MZM device is fabricated. Aluminum metal layer of $2 \mu\text{m}$ thick is deposited on top of

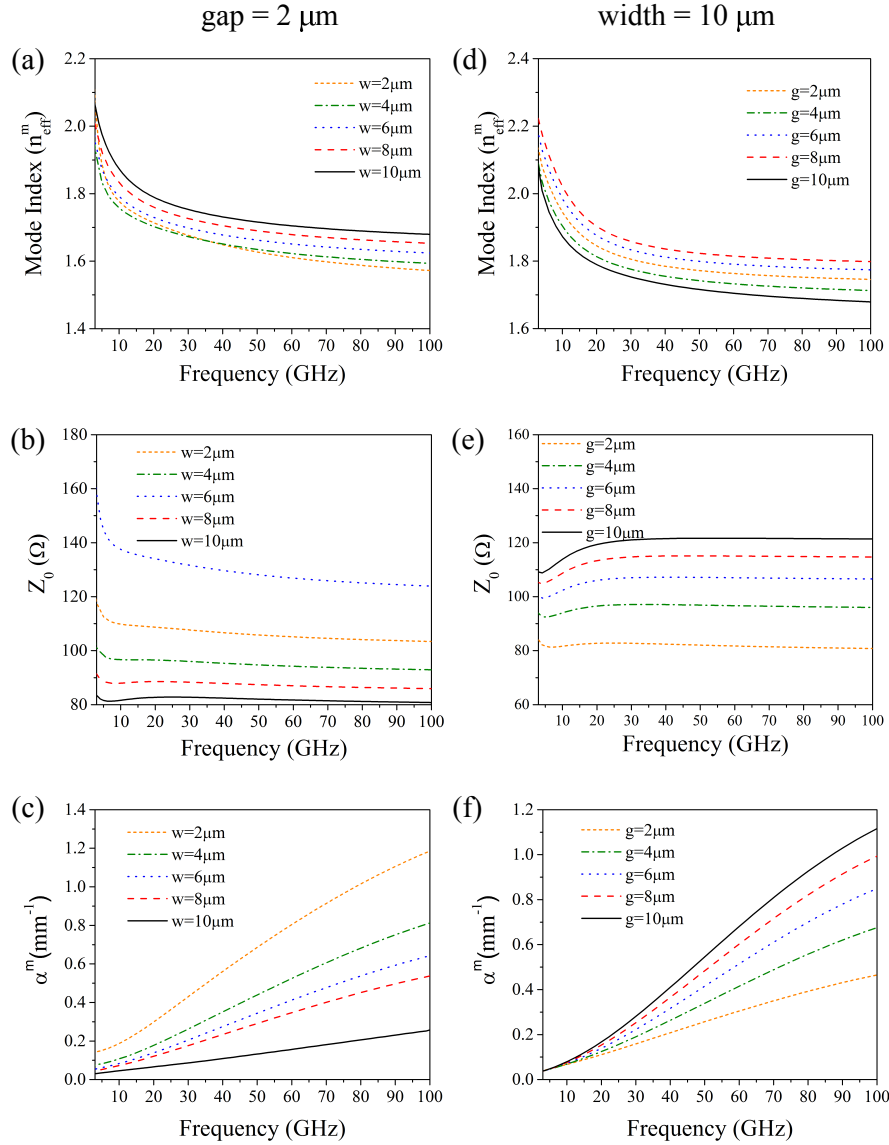


Figure 3.20: (a)-(c): Simulated effective index, characteristic impedance (Z_0), attenuation coefficient (α^m) of the transmission line with gap = $2 \mu\text{m}$ and width changes from 2 to $10 \mu\text{m}$. (d)-(f): Simulated effective index, characteristic impedance (Z_0), attenuation coefficient (α^m) of the transmission line with width = $10 \mu\text{m}$ and the gap changes from 2 to $10 \mu\text{m}$. The length of the transmission line simulated is $300 \mu\text{m}$.

the SiO_2 layer and then etched to form the metal traces. The gap and metal trace width are varied to optimize the performance of the TWE. Before we moved on to numerical calculation, it is instructive to look at the characteristics based on the

first principle analysis. Firstly, the characteristic impedance of the transmission line is given as: $Z_0 = \sqrt{L_0/C_0}$ [109], where L_0 and C_0 denotes the inductance and capacitance per unit length, respectively. Therefore, reducing the gap and increase the width of the metal traces help reduce the impedance. Further, for a lossless line, the velocity of the electrical wave in the transmission line is expressed as $v = 1/\sqrt{L_0C_0}$ [109]. Again, the velocity is inversely proportional to the square-root of the total capacitance loading. As a consequence, the loading of the PN junction parasitic capacitance reduces the characteristic impedance and slows down the electrical wave in the transmission line. Finally, the loss of the transmission line is characterized by the attenuation coefficient, which is given by the real part of the propagation constant: $\alpha^m \approx \frac{1}{2}(R_0/Z_0 + G_0Z_0)$, where R_0 and G_0 are the resistance and admittance per unit length of the transmission line, respectively. The PN junction loading contributes to the admittance, hence leads to additional loss. We will show later that this is the dominant part of the loss of a TWE.

A high frequency EM simulator SONNET is used to calculate the properties of the transmission line. The dielectric stackup in Fig. 3.19 is used for the simulation. The simulated transmission line is 300 μm long. Fig. 3.20 shows the simulated sensitivity of the impedance, characteristic impedance, and effective index on the metal traces gap and width. The results are in agreement with the first principle analysis. In addition, it is noted that the effective index increases with the width of the metal traces which can be explained by more spreading of the electrical field into the silicon substrate. Moreover, the effective index of the unloaded transmission line is smaller than the group index of the optical signal ($n_g = 3.66$). However, after loading a PN junction, the effective index will increase.

PN Junction Loading

The design of the TWE has to take both the metal traces and the junction parasitics into account. In [53], an analytic method was employed to evaluate the performance of the TWE with loading. However, given the complexity of both electrical and optical system, the analytic method may not be accurate enough to account for all effects in the design. Therefore, in this design, we use the equivalent circuit method, wherein the transmission line and the PN junction are simulated separately to extract a circuit model for each device. The extracted circuit models are then combined [44, 71, 110], and based on which the characteristic impedance, effective index and the attenuation coefficient are then calculated. As we discussed in Sec. 3.2.3, the PN junction can be modeled as R_{pn} and C_{pn} connected in series. The R_{pn} and C_{pn} values change with bias voltages as shown in Fig. 3.18. In this design, the dc reverse bias is set to $V_R = 2$ V for optimal performance and adequate voltage swing and the junction parameters are $R_{pn} = 8.1 \, \Omega \cdot \text{mm}$ and $C_{pn} = 252 \, \text{fF/mm}$. For the transmission line, the well-known RLGC model can be used. The RLGC values at different frequencies are extracted from the high frequency EM simulation. Together, the PN junction loaded transmission line can be modeled by the equivalent circuit as shown in Fig. 3.21-a, where the PN junction loading is represented by the series RC in the dashed box. The complex propagation constant is expressed as [109]:

$$\gamma = \sqrt{(R_{tl} + j\omega L_{tl}) \left[\frac{1}{R_{eff}} + G_{tl} + j\omega(C_{eff} + C_{tl}) \right]} \quad (3.9)$$

$$Z_0 = \sqrt{\frac{R_{tl} + j\omega L_{tl}}{\frac{1}{R_{eff}} + G_{tl} + j\omega(C_{eff} + C_{tl})}} \quad (3.10)$$

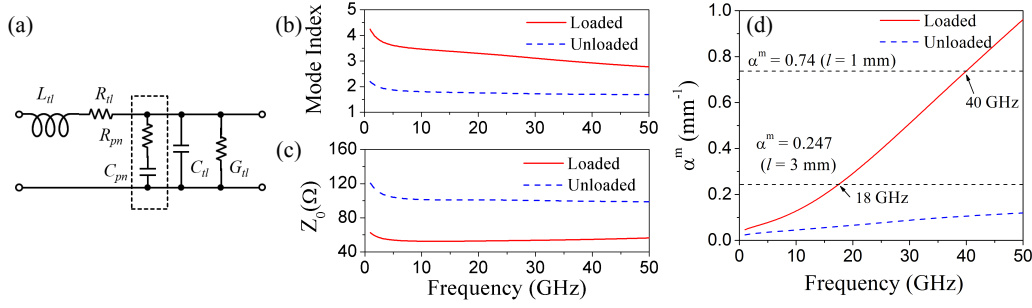


Figure 3.21: (a) Circuit model of the loaded transmission line. The series RC model of the PN junction is marked by the dash box. (b) Mode index (n_{eff}^m) and characteristic impedance of the unloaded and loaded transmission lines; (c) Loss of the unloaded and loaded transmission lines. For a 1-mm long phase shifter, the 3-dB EO bandwidth corresponds to $\alpha^m = 0.74$, which is about 40 GHz.

where R_{tl} , G_{tl} , L_{tl} , and C_{tl} are the extracted resistance, admittance, inductance and capacitance per unit length of the transmission line itself. And

$$R_{eff} = R_{pn} \left(1 + \frac{1}{\omega^2 C_{pn}^2 R_{pn}^2} \right) \quad (3.11)$$

$$C_{eff} = \frac{C_{pn}}{1 + \omega^2 C_{pn}^2 R_{pn}^2} \quad (3.12)$$

The mode index and the attenuation coefficient are calculated as: $n_{eff}^m = \text{Im}(\gamma)c/\omega$, and $\alpha^m = \text{Re}(\gamma)$, respectively, where ω is the angular frequency.

The design target is that *the loaded transmission line has an effective mode index about 3.66 about characteristic impedance around 50 Ω*. The metal traces gap and width are tuned to achieve the target velocity and impedance matching, as well as to minimize the electrical loss. In addition to the criteria mentioned above, the metal trace design has to satisfy the minimum width and gap limited by the practical lithography technology. The final transmission line has metal traces gap of 2 μm and width of 5 μm . The calculated effective index, characteristic impedance, and attenuation coefficient of the transmission line before and after the PN junction loading are shown in Fig. 3.21-b,c. At 25 GHz, the mode index

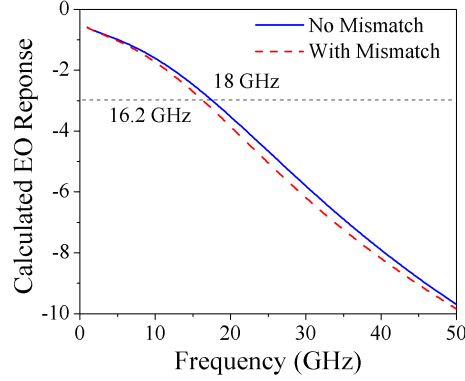


Figure 3.22: Calculated EO response with and without velocity mismatch for a 3-mm long TWE.

is $n_{eff}^m = 3.2$ and the characteristic impedance is 53Ω . The velocity mismatch results from the effective mode index difference between electrical and optical signal affects the EO response. The calculated EO response for a 3 mm long TWE according to Eq. 2.11 and Eq. 2.12 with and without velocity mismatch are plotted in Fig. 3.22. As can be seen, the velocity mismatch results in about 10% sacrifices in 3-dB bandwidth. Further increase of this effective index is difficult, unless the loading capacitance is increased which will affect the modulation efficiency and other performance merits. One way to improve the bandwidth is to bias the device at higher voltage, both R_{pn} and C_{pn} will decrease which leads to smaller loss and higher bandwidth. However, large bias voltage leads to smaller modulation efficiency (Fig. 3.16). For the same extinction ratio (ER), it requires longer devices which again deteriorates the electrical loss.

In general, the junction loading has a large effect on the performance of the TWE. Due to the large parasitic capacitance, the loaded transmission line has its effective mode index almost doubled and the characteristic impedance almost halved compared to the unloaded ones. From Fig. 3.21-c, it is clear that the loss due to R_{pn} is the dominant source of electrical loss at high frequency. As we discussed in previous section, the EO 3-dB bandwidth is determined by $\alpha^m l = 0.74$.

Given the length of the device, EO 3-dB bandwidth can be directly found from the loss vs. frequency curve. For example, for a MZM with $l = 1 \text{ mm}$, the 3-dB EO bandwidth is over 40 GHz. This is comparable to that of the depletion mode MZM by Thomson, which achieved 50 Gb/s data rate [52]. Thereby, the designed device achieved a good tradeoff between modulation efficiency, bandwidth and optical loss.

Thereby, we finish the device modeling part of this design flow. The electrical and photonic circuit simulation will be demonstrated in the following EPIC designs. We will also verify the design flow by comparing the simulated results with the measurement results from silicon photonic foundries.

Chapter 4

A Multi-Function Mach-Zehnder Modulator for Pulse Shaping and Generation

4.1 Introduction

MZM has always been one of the most critical building blocks in optical systems and photonic integrated circuits. Recently, silicon MZMs have demonstrated good performance with increasing data rate or decreasing driving voltage [22, 29, 31]. Lately, research in silicon MZMs has been expanded from optimizing the photonic device structures into new areas such as integrating circuit techniques for performance improvement, and constructing new MZM circuits for advanced signaling, modulation, and multiplexing schemes. In the latter area, Dong *et al.* demonstrated 50 Gb/s quadrature phase-shift keying modulation using nested MZM [111]. Plant *et al.* explored pulse amplitude modulation in conjunction with polarization domain multiplexing using nested MZM and polarization control devices to achieve 4 bits per symbol in a long-haul optical fiber communication system [112]. This work illustrates our efforts in both areas, with the focus on circuit technique innovation.

MZM is of great interest in manipulating temporal characteristics of optical pulses [94, 113]. Pulse shaping, particularly equalization, is a common circuit technique employed in wireline communication systems to mitigate the ISI due to the

limited bandwidth of the link. Similar equalization techniques can be applied in either electrical or optical domain to improve the performance of MZM. For example, [48] demonstrated $0.29 \text{ V} \cdot \text{cm}$ $V_\pi L$ and 12.5 Gb/s modulation rate by using driving signals with pre-emphasis. An optical equalizer was demonstrated in [37] using cascaded MZMs and complicated phase control to achieve a 40 Gb/s modulation speed. When equalization is carried out in the electrical domain, there is increased bandwidth requirement on the MZM driver circuit. Optical equalization is therefore preferred, if done without introducing new electrical circuit constraints.

Optical pulse generation using MZM has also been explored, which offers advantages such as ease of fabrication and flexible pulse repetition rate compared to the conventional optical pulse generation by using mode-locked laser. For example, in [114], 2.4 ps optical pulse generation was demonstrated using a single stage MZM, a bandpass filter and a dispersive fiber. Xia *et al.* [115] demonstrated pulse repetition rate multiplication using cascading MZMs and 3-dB couplers.

On chip implementation of pulse shaping and generation using silicon MZMs will be extremely interesting for future large-scale photonic integrated circuits and systems. Co-design of photonic devices and electrical components in these MZM circuits will lead to many new circuit techniques within the paradigm of EPIC.

In this work, we propose a new EPIC design based on a single MZM and supporting electronics. By driving the MZM using time-shifted signals of tunable amplitude, the circuit can be used to incorporate pre-emphasis functionality to its conventional EO modulation capability without a pre-emphasis MZM driver. Further, it can generate optical pulse with tunable pulse width at double clock rate. Compared with previous designs [37, 94, 113–115], the proposed EPIC has the advantages of multi-functionality, simplicity and compatibility with silicon photonics technologies. In addition, we present the complete design flow of the silicon photonic circuits from devices simulation, compact models building to the

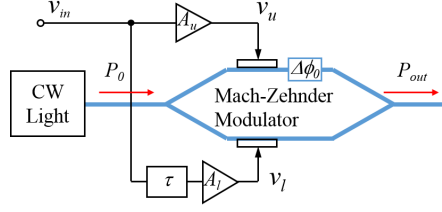


Figure 4.1: Schematic of the proposed modulator circuit.

circuits modeling.

4.2 MZM Based EPIC Design

The proposed circuit is illustrated in Fig. 4.1-a. It mainly consists of a dual-drive MZM and supporting driver circuits. Instead of applying synchronized differential signals with equal amplitude to the two arms as in a conventional MZM, the two arms are driven *independently* by two signals (v_u and v_l) with tunable amplitudes and relative delay. The two driving signals can be generated from a single input signal v_{in} using two drivers with variable gains (A_u and A_l) and a delay element (τ). In this case, the two driving signals are $v_u(t) = A_u v_{in}(t)$ and $v_l(t) = A_l v_{in}(t - \tau)$.

First, we choose to operate the circuit in region II of a MZM centered around $\Delta\phi_0 = \frac{\pi}{2}$, which can be implemented by adding a small section of waveguide biased at specific V_0 on the upper arm. Eq. 2.10 can be simplified:

$$P_{out}(t) = \frac{1}{2}P_0[1 - \sin(\Delta\phi_u - \Delta\phi_l)] \approx \frac{1}{2}P_0(1 + k_lv_l - k_uv_u) \quad (4.1)$$

where we assume $\Delta\phi_u = k_uv_u$ and $\Delta\phi_l = k_lv_l$, and they are small. $k_{(u,l)} = \frac{\pi}{V_{\pi(u,l)}}$ are modulation efficiency of the upper and lower arms.

Eq. 4.1 shows that the output signal is directly determined by the weighted difference of the two driving signals, $k_lv_l - k_uv_u$. By generating the driving signals

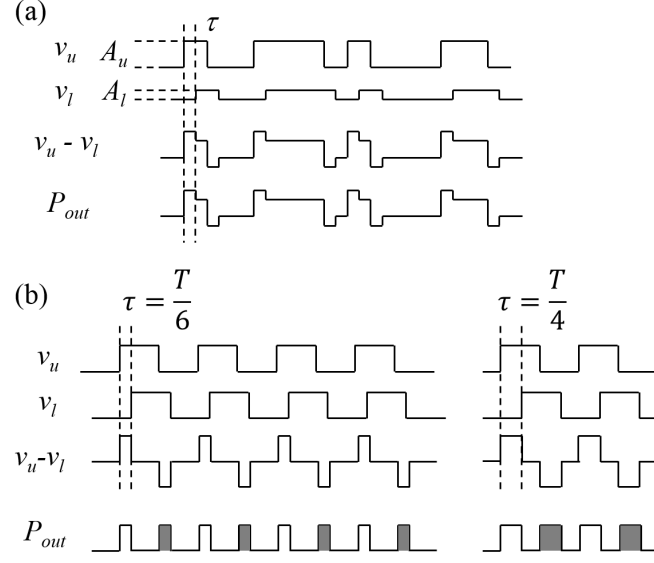


Figure 4.2: (a) Pulse shaping based on two-tap pre-emphasis, assuming $k_l = k_u$. (b) Pulse generation at both rising and falling edges with $\tau = \frac{T}{6}$ and $\tau = \frac{T}{4}$. T is the period of the clock. Shallow pulses are 180° out of phase. $A_u = A_l$ and $k_l = k_u$ are assumed.

v_u and v_l independently from v_{in} , the output signal's pulse shape can be changed. As shown in Fig. 4.2-a, assume $k_l = k_u$, when relative delay and gain values are applied to two electrical signals, we can achieve pre-emphasis on $v_l - v_u$, which will in turn shape the optical pulse through Eq. 4.1. As a result, the pre-emphasized optical signal is generated directly from unpre-emphasized driving signal, eliminating the need for a pre-emphasis driver and saving power.

Secondly, when the MZM operates in the quadratic region ($\Delta\phi_0 = \pi$), and given $\Delta\phi_u - \Delta\phi_l$ is small, Eq. 2.10 becomes:

$$P_{out} = \frac{1}{2}P_0[1 - \cos(\Delta\phi_u - \Delta\phi_l)] = P_0\sin^2\left(\frac{\Delta\phi_u - \Delta\phi_l}{2}\right) \approx P_0\left(\frac{k_u v_u - k_l v_l}{2}\right)^2 \quad (4.2)$$

Interestingly, the MZM performs the XOR operation on the two driving signals, and can be used as an optical pulse generator. To achieve that, the amplitude of the driving signals are set equal, $A_u = A_l$, and the modulation efficiency of both

phase shifters are also equal, $k_u = k_l$. The XOR operation of the MZM results in an optical pulse at both the rising and the falling edges as shown in Fig. 4.2-b. Therefore, the pulse repetition rate is twice of the input clock frequency. The pulse width is determined by the relative delay between the two driving signals. Due to the different signs of $v_l - v_u$ at the rising and the falling edges, the pulse at rising-edge and falling-edge are 180 degree out of phase [116]. Therefore, they can be easily separated by a coherent detector if needed, adding more flexibility to this circuit.

4.3 Photonic Circuit Design and Analysis

Based on the device parameters of the MZM from device and EM simulation in Section 3.2, the proposed photonic circuit is designed and analyzed using an electrical circuit simulator ADS, and a photonic circuit simulator Lumerical INTERCONNECT [40].

The schematic of the prototype circuit is shown in Fig. 4.3, in which green and blue lines represent the optical and electrical signal path, respectively. A distributed model of the TWMZM is built using multiple sections of waveguides and electrodes as shown in Fig. 4.3, to overcome the limitation of INTERCONNECT. Each 300 μm long electrode is modeled as a loaded transmission line in ADS and its S-parameter is then exported into INTERCONNECT. For the phase modulator, the extracted device parameters are used to define the compact model in INTERCONNECT. There is one assumption made here that the effective index versus bias voltage curve in Fig 3.18 which is obtained under DC operation is used here for RF simulation. The total length of the phase modulator is 3 mm and the optical loss is less than 1 dB. The optical losses of the Y-junctions at the input and output of the MZM are both set to 0.5 dB [64, 117] and the optical loss of the undoped waveguide is assumed to be 3 dB/cm (conservative estimation

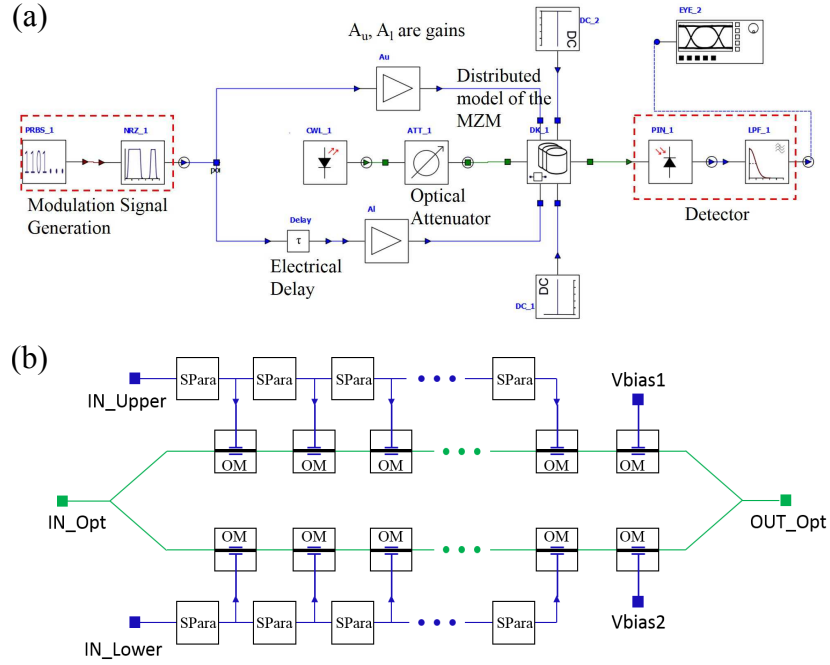


Figure 4.3: (a) Schematic of the simulated multi-function MZM circuit. (b) Schematic of the distributed model for the MZM. S-parameter boxes are used to represent the $300\ \mu\text{m}$ long loaded transmission lines. The S-parameter is obtained from ADS simulation. The initial phase difference is implemented as an extra phase shifter on both arms and controlled by two DC sources.

based on the devices fabricated in IME silicon photonic process [64]). An optical attenuator with 3 dB attenuation in front of the laser source accounts for total waveguide optical loss. A fixed phase difference is implemented as extra phase shifters on both arms which are controlled by DC sources. Given the $V_\pi L_\pi$ of the PN junction and assuming a single-drive voltage $V_{pp} = 4\ \text{V}$, the MZM can achieve 9.6 dB ER. The electrical part of the circuit consists of two drivers with gain A_u and A_l . Both are triggered by the same pseudo-random bit sequence generator. A delay element is added in front of the lower driver. An ideal PIN photodetector ($0.5\ \text{A/W}$) with RC filter is used to emulate a bandwidth limited photodetector. The optical output is measured using a virtual oscilloscope.

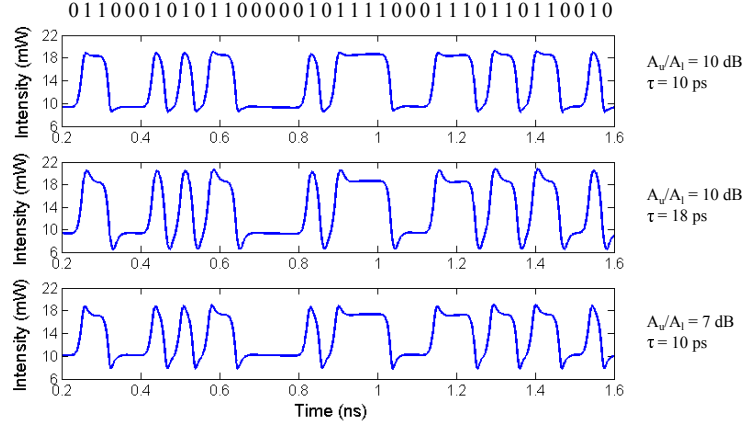


Figure 4.4: Simulated optical pulse waveforms. The data rate is 28 Gb/s. The data is shown on top. From top down: $A_u/A_l = 10$ dB, $\tau = 10$ ps; $A_u/A_l = 10$ dB, $\tau = 18$ ps; $A_u/A_l = 7$ dB, $\tau = 10$ ps.

4.3.1 MZM with Built-In Pre-Emphasis

For the pre-emphasis application, an initial $\pi/2$ phase difference is inserted. The simulated optical waveforms are shown in Fig. 4.4. It shows the successful generation of the pre-emphasized optical signals. The strength of the pre-emphasis can be controlled by varying the delay τ and gain ratio between the two signals, i.e., A_u/A_l , as demonstrated in the figure.

Pre-emphasis is an equalization technique employed to mitigate the ISI due to bandwidth-limited transmitters, detectors, or chromatic dispersion, in an high speed optical link. In order to study the effectiveness of the optical pre-emphasis, we simulated two scenarios in which limited bandwidth results in ISI: modulator bandwidth limited and photodetector bandwidth limited. For comparison, the same MZM is also used in the conventional way of driving, in which it is driven by a single TWE while the other is biased with a DC voltage. From Fig. 3.21-c, the 3-mm long TWE has 3-dB EO bandwidth about 18 GHz. In the first scenario, the driving signal is selected to be 50 Gb/s. Apparently, the conventional way of driving can't operate at such high data rate. The eye diagram shows a small eye opening and large jitter (Fig. 4.5-a). The BER is estimated from the

simulated eye diagram and it is only 10^{-3} . In contrast, as shown in Fig. 4.5-a when pre-emphasis is applied by employing the proposed design, the output optical eye diagram shows enlarged eye opening and reduced jitter. The estimated BER now reaches 10^{-12} , which suggests successful operation at 50 Gb/s. In the second scenario, the data rate is set to 28 Gbps. To emulate the photodetector bandwidth limiting effect, the RC filter cut off frequency is set to 10 GHz. For the conventional design, the output eye-diagram exhibits large eye-closure and pronounced deterministic jitter. The estimated BER is about 10^{-5} . On the other hand, using the proposed design, significant improvement is observed. The eye is enlarged substantially and both the RMS jitter and deterministic jitter are reduced. The BER is estimated to be $< 10^{-15}$, showing significant improvement in the signal quality by incorporating the pre-emphasis function (Fig 4.5-b). Therefore, the proposed MZM design with built-in pre-emphasis can significantly improve the data rate or relax the demanding bandwidth requirement for the detector design as we demonstrated.

4.3.2 MZM for Double-Rate Pulse Generation and RZ Modulation

As discussed in Section 4.2, the MZM exhibits a nonlinear EO transfer function when the initial phase shift is biased at π , and this operation region can be utilized for a variety of pulse generation and modulation applications.

In the first case we are investigating here, the input electrical signal is a clock signal, and the MZM functions as an optical pulse generator with pulse repetition rate twice of the input clock frequency. To achieve this, the gain values of the two driver and the modulation efficiency of the phase modulators are set to equal and a delay is added to the two driving signals. Fig 4.6-a shows the simulation results with a 14 GHz sinusoidal clock input. The delay to the lower arm driving signal v_l is $\tau = 18$ ps. Optical pulses with a high ER are generated at both rising and

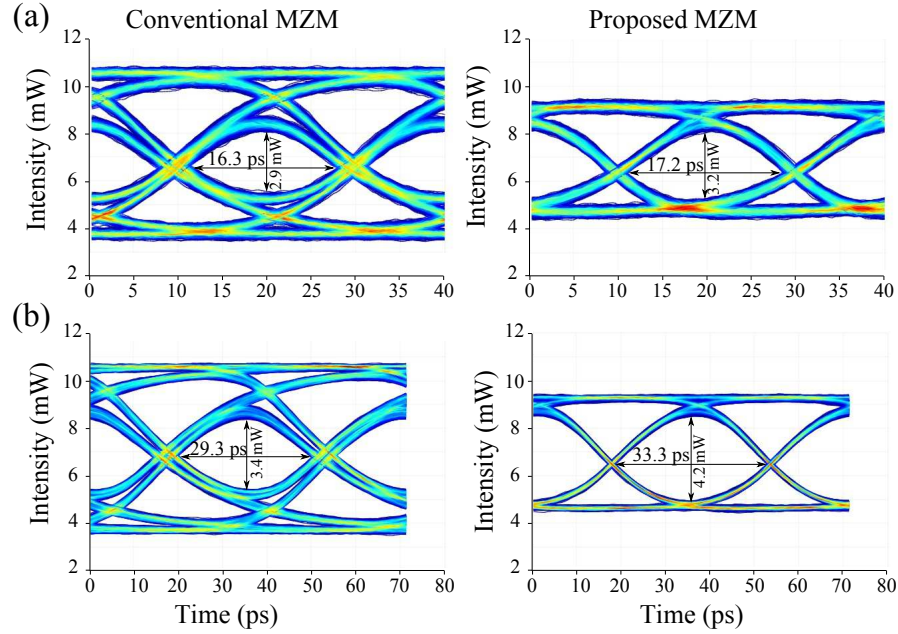


Figure 4.5: ISI compensation by pre-emphasis in optical domain: (a) compensation of the ISI due to modulator bandwidth limitation; data rate is 50 Gb/s; the delay and weight ratio in the simulation are $\tau = 7$ ps and $A_u/A_l = 10$ dB. (b) compensation of the ISI due to photodetector bandwidth limitation; data rate is 28 Gb/s, the bandwidth of the detector is set to 10 GHz; The delay and weight ratio in the simulation are $\tau = 17$ ps and $A_u/A_l = 10$ dB.

falling edges of the input clock. This shows a good match between the simulation result and theoretical analysis in Section 4.2. The power spectrum density is shown in Fig. 4.6-b, which confirms the successful generation of 28 GHz optical clock signal. Frequency spurs at 14 GHz and 42 GHz are due to the nonlinear n_{eff} vs. V curve, which are 25 dB lower and 29 dB lower than the desired signal power. They can be further suppressed if the amplitude of the driving signal is reduced.

The optical pulse train generated by the MZM can be further modulated to generate optical return-to-zero (RZ) signals. RZ signaling is sometimes preferred over NRZ in optical communications, especially in optical time division multiplexing systems due to its tolerance to fiber nonlinearity and chromatic dispersion [118, 119]. In this demonstration, an ideal MZM component in the Lumerical

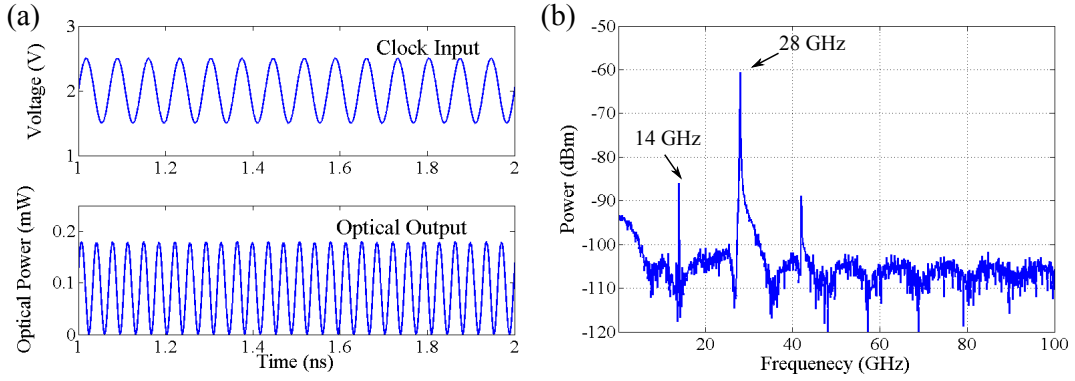


Figure 4.6: Optical pulse generation and frequency multiplication simulation results with sinusoidal input. (a) Waveform of the input 14 GHz electrical clock signal (upper) and output optical signal (lower) at $\tau = 18$ ps. (b) Spectrum of the output optical signal showing the main peak at 28 GHz and two frequency spurs.

INTERCONNECT is cascaded to the new MZM, and modulated by the input digital bits (Fig. 4.7-a). A square input clock signal is used and the delay is set to $\tau = 16$ ps to generate short optical pulses. As shown in Fig. 4.7-b, 28 Gbps optical RZ data is successfully generated at the output of the 2nd MZM.

4.4 Electrical Circuit Design

The delay (τ) and gain (A_u and A_l) parameters determine the waveforms of v_u , v_l and hence P_{out} . Any delay and gain variations lead to the output waveform distortion and possibly data errors. It is critical to integrate the driving electrical circuit together with the photonic devices as a complete EPIC. It can be implemented either using a monolithic silicon photonic technology or as a 3-D heterogeneously integrated chip stack. To illustrate the feasibility, we show a passive-only design as follows. The delay and gain can be implemented using transmission lines, as shown in Fig. 4.8-a. At the Y-junction, 44% of the power will be delivered to each arm which corresponds to 2/3 of the input voltage swing [109]. The delay is generated by the transmission line length difference

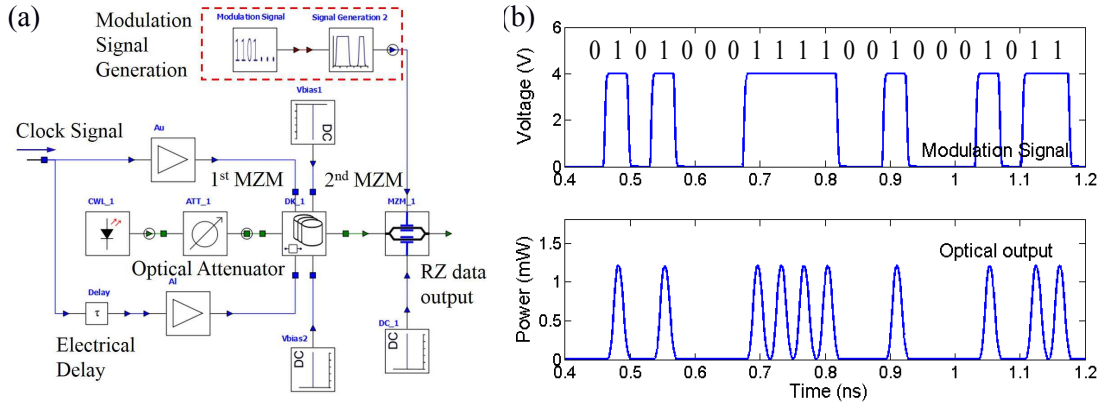


Figure 4.7: Optical RZ data generation simulation results. (a) Circuit schematic for the RZ data generation. The 2nd MZM is an ideal component in Lumerical INTERCONNECT. (b) Generation of 28 Gb/s RZ data. Upper: electrical modulation signal; lower: optical data taking RZ format.

between the two arms, and implemented in the meandering sections of *both* arms. Meandering both transmission lines balances the power loss at the corners. The gain factors can be implemented as one or a series of T pad attenuators. As shown in Fig. 4.8-b, the passive driving circuit exhibits low loss and wide bandwidth.

4.5 Prototype Fabrication

A prototype circuit is fabricated using IMEC 50G Si photonic process. The layout is shown in Fig. 4.9. The circuit uses a standard phase shifter in the device library, which is based on a PN junction in reverse bias. The length of the phase shifter is 2.5 mm. Based on the modulation efficiency given in the design manual, the $V_\pi = 5.6$ V at 0 V bias. The loss of this phase shifter section will be 3.1 dB at the same bias. Unbalanced arms are used for testing purpose. Dc phase bias is achieved by two thermal phase adjusters on both arms. The phase shifters are driven by co-planar waveguide transmission line electrode, which will be terminated externally. Grating couplers are used for the optical input and output.

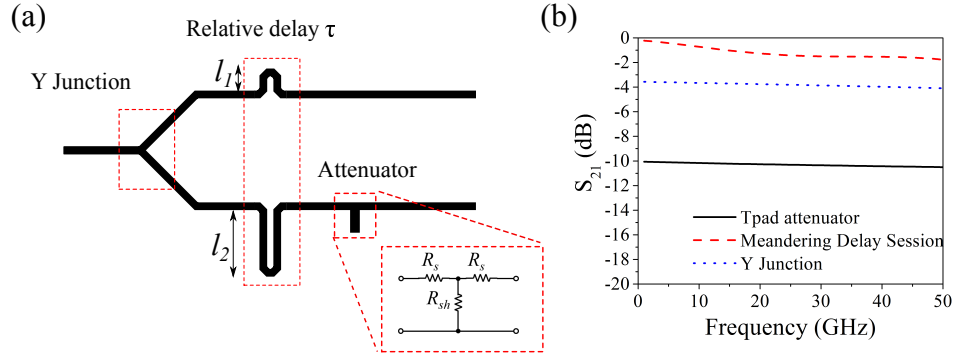


Figure 4.8: (a) Schematic of the passive-only driving circuit. The inset shows the Tpad attenuator circuit schematic. (b) Simulated S_{21} of the Y-junction, 1 mm long meandering delay line and the 10 dB T-pad attenuator using an EM simulator SONNET.

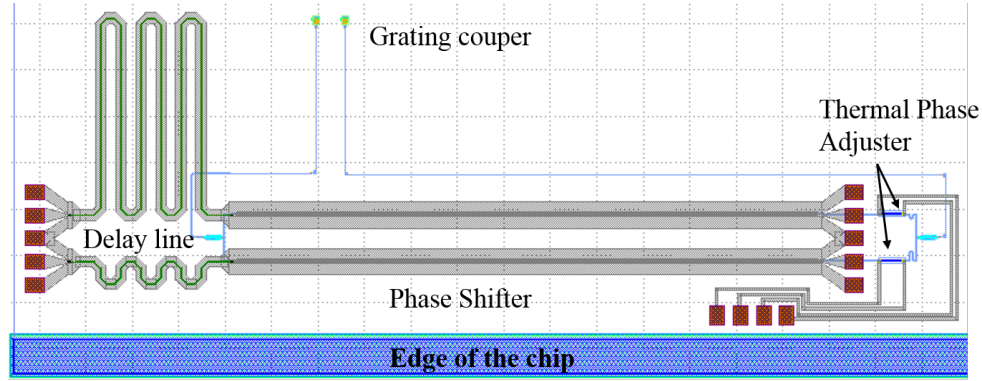


Figure 4.9: Layout of the multi function EPIC.

The electrical delay line is implemented on-chip by meandering transmission lines. To achieve good impedance matching and minimize the device footprint, microstrip lines are used. Meandering lines are implemented on both arms to balance the loss at turning corners in two arms. Simulation results shows the delay line gives about 30-ps delay. Resistive attenuator is not implemented on-chip because it drains a dc current which causes voltage drop on the signal line. That will change the bias voltage on the phase shifter. Therefore, the amplitude of the input power will be tuned externally.

Chapter 5

A Feedback Equalized Traveling-Wave Electrode Design for Mach-Zehnder Modulator

5.1 Introduction

As we discussed previously, the electrical signal propagating in the TWE of a TWMZM suffers from large RF loss due to PN junction loading (Fig. 5.1 inset). The high frequency roll-off of electrical signal is usually the major bandwidth limiting factor of a TWMZM. As an example, the transfer functions of a 1-mm and a 3-mm long TWE are calculated using the TWE model in Chapter 3 and plotted in Fig. 5.1. As shown, the TWE has a 6-dB bandwidth of 40 GHz for the 1-mm long device and 18 GHz for the 3-mm long device. Due to the weak electro-optical effect in silicon, long devices are usually adopted to achieve large ER, which will directly trade with 6-dB bandwidth. As data rates continue to increase, the limited bandwidth of the TWE becomes the bottleneck of the whole optical link. To break this trade-off and compensate for high frequency roll-off, various equalization techniques have been investigated in driver and receiver circuit design.

Analog two-tap FFE, also known as pre-emphasis, is the most popular equalization technique applied in an optical driver circuit. An FFE driver is often used to over-drive a bandwidth-limited VCSEL to achieve a high data rate. In [82],

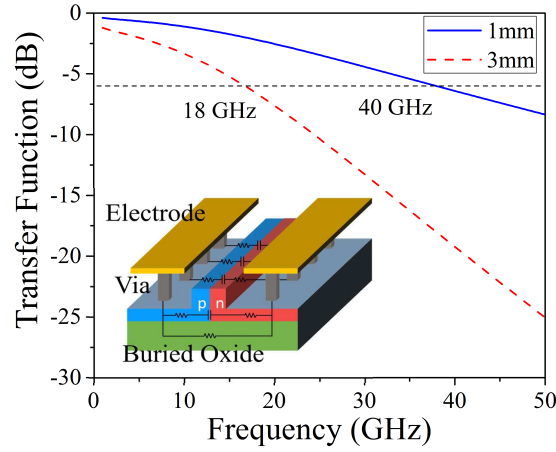


Figure 5.1: Transfer functions of TWEs with 1-mm and 3-mm length. Inset: a schematic of a transmission line loaded with a PN junction.

Rylyakov *et al.* demonstrated a FFE driver for VCSEL which achieved 40 Gb/s ($BER < 10^{-12}$) operation with a 25 Gb/s grade device. In [48], a pre-emphasis driving scheme was used to drive a forward-biased PN junction based MZM. The 250 μm long device itself had a bandwidth smaller than 1 GHz. However, with a pre-emphasized driving signal, the author demonstrated 12.5 Gb/s operation with $ER = 8$ dB. On the other hand, the pre-emphasis driver suffers from voltage headroom limitations and implementation of a pre-emphasis driver requires a pre-emphasis tap with accurate and tunable analog delay lines, which increases circuit complexity and total power consumption. In addition, the FFE driver only attenuates the dc gain without providing any high frequency boost. As a consequence, the ER of the optical signal is sacrificed.

DFE is another equalization technique that has been widely adopted in receiver design. Compared to FFE, DFE has the advantage of high frequency gain boost and noise mitigation, thus improving SNR substantially. However, a DFE has never been implemented in a driver partially because a decision function is not necessary in a driver and signal degradation has not occurred at this stage of a link.

In this chapter, we exploit the concept of feedback equalization in a DFE to implement a new equalized TWE design for TWMZM. The new equalization technique takes advantage of the natural time delay in a TWE to form an IIR filter in a conventional TWMZM. Several equalization stages are cascaded for optimal speed and ER performance. We present a detailed design of the equalized TWE based on the aforementioned EPIC design flow, including the electrical and photonic blocks. The prototype circuit is modeled from the device to system level. The simulation results show that the equalization technique can significantly improve the operation speed of a TWMZM, while maintaining a high ER.

5.2 Theoretical Analysis

The proposed scheme is shown in Fig. 5.2-a. In the time domain, a delayed RF signal is sampled and then fed back to the input signal. The resulting signal is pre-distorted with over-shoot in the rising edge and under-shoot in the falling edge. The signal pre-distortion process looks similar to that of an FFE driver. The frequency domain behavior of this circuit is analyzed using the model in Fig. 5.2-b. v_i , v_f , v' , and v_o are the input signal, feedback signal, sum signal and output signal, respectively. k and α^m are the propagation constant and loss coefficient of the transmission line, respectively. The feedback circuit is characterized by a dc voltage gain A_f and a phase ϕ . The transfer function of this circuit is given by:

$$A_v = \frac{v_o}{v_i} = \frac{e^{-jkl}e^{-\alpha^m l}}{1 + A_f e^{-j(kl+\phi)}e^{-\alpha^m l}} \quad (5.1)$$

The magnitude is then given by:

$$|A_v| = \frac{1}{\sqrt{e^{2\alpha^m l} + A_f^2 + 2A_f e^{\alpha^m l} \cos(kl + \phi)}} \quad (5.2)$$

From this first order analysis, the voltage gain depends on the loss factor, feedback

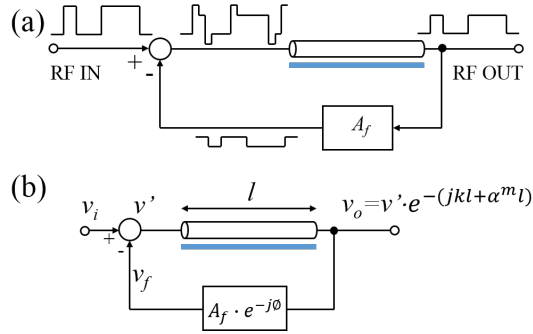


Figure 5.2: (a) Schematic of the proposed feedback equalization for TWEs. (b) Mathematical model for frequency analysis

gain, as well as the feedback path phase shift ϕ . It is noted that the transfer function suggests a resonance peak at the frequency where $kl + \phi = \pi$. At the resonance frequency, feedback signal is added constructively to the input signal and boosts the overall frequency response near that particular frequency. At the same time, the negative feedback suppresses dc gain. Both effects lead to a flattening of the transfer function. As an example, Fig. 5.3 shows the transfer function of a loaded transmission line before and after equalization. In this plot, we use the frequency dependent propagation constant k and loss coefficient α^m modeled in Chapter 3. The dotted blue curve shows the transfer function when transmission line loss is absent ($\alpha^m = 0$). As shown, the feedback provides a loss at low frequency and a gain at resonance frequency, thus equalizing the transfer function of the TWE. The peak frequency corresponds to the resonance frequency, which is one of the most important design parameters in this feedback equalization. It is determined by the transmission line propagation constant, length, and the phase of the feedback circuit. In this example, the dc voltage gain is set to 0.15 and the phase is given by: $\phi = \omega T_{dy}$, where T_{dy} is the signal delay in the circuit, which is assumed to be 2.5 ps.

The EO transfer function can be calculated from Eq. 5.2. Based on the deriva-

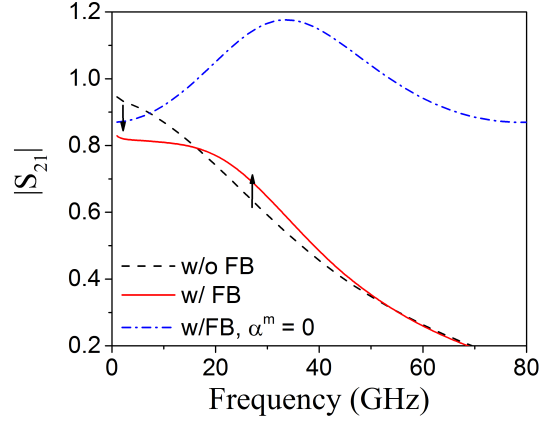


Figure 5.3: Transfer function of the TWE with and without an equalization. The dc voltage gain is set to $A_f = 0.15$ and the delay in the feedback circuit is assumed to be $T_{dly} = 2.5$ ps.

tion in Eq. 2.11, by replacing v_0 with v' , we have:

$$\begin{aligned}\Delta\phi &= \left|\frac{v'}{v_i}\right|\phi_0 \frac{1 - e^{-\alpha^m l}}{\alpha^m l} \\ &= \phi_0 |A_v| \frac{e^{\alpha^m l} - 1}{\alpha^m l}\end{aligned}\tag{5.3}$$

Assuming linear relationship between optical power and phase shift, the EO frequency response can be expressed as:

$$|S_{21,EO}| = |A_v| \frac{e^{\alpha^m l} - 1}{\alpha^m l}\tag{5.4}$$

The reflection of the equalized TWE is another important performance parameter. Assuming a good impedance match at the source and load, the S_{11} of an equalized transmission line is given by [120]:

$$S_{11} = S_{22} = \frac{-A_f e^{-j(\phi+kl)}}{e^{\alpha^m l} + A_f e^{-j(\phi+kl)}}\tag{5.5}$$

The magnitude is given by:

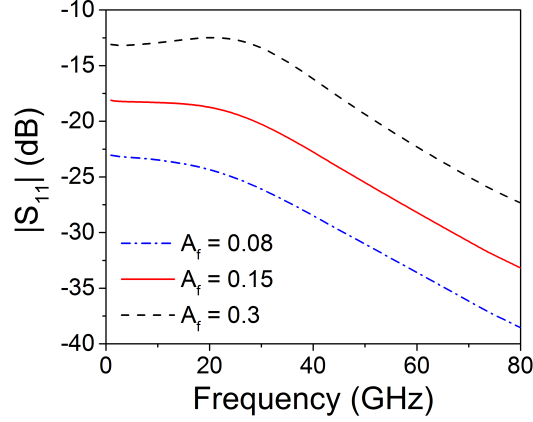


Figure 5.4: Calculated $|S_{11}|$ for equalized TWE at feedback gain $A_f = 0.08$, 0.15 , and 0.3 .

$$|S_{11}| = \frac{A_f}{\sqrt{e^{2\alpha^m l} + A_f^2 + 2A_f e^{\alpha^m l} \cos(kl + \phi)}} \quad (5.6)$$

At low frequency ($f \rightarrow 0$), $|S_{11}| = A_f/(1 + A_f)$ and at resonance frequency ($f = f_{res}$), $|S_{11}| = A_f/(e^{\alpha^m l} - A_f)$. Thus, at low frequency, small A_f will be desired for lower S_{11} . As the frequency increases, the attenuation term, $e^{\alpha^m l}$, will increase to reduce the reflection. Example S_{11} plots using the above analytic model is shown in Fig. 5.4. The same TWE device model from Chapter 3 is used here again. To study the sensitivity of S_{11} to feedback gain A_f , three S_{11} plots are shown in this figure, which correspond to $A_f = 0.08$, $A_f = 0.15$, and $A_f = 0.3$. For small A_f , the S_{11} stays below -20 dB for the entire frequency range of interest. As A_f increases to 0.3 , peaking is observed in the S_{11} curve and maximum S_{11} increases to -12 dB at about 23 GHz. Therefore, to minimize the reflection, A_f should be kept as small as possible. In addition, the stability of the feedback system will also benefit from a small A_f .

Furthermore, for high speed operation, group delay variation (GDV) is another important parameter needs to be considered in optical transceiver circuit design. Ref. [121] suggests that GDV should be less than 10% of the bit period (*i.e.* ± 0.1

UI) within the frequency range of interest. The group delay (GD) is defined as the slope of the phase versus frequency curve. The phase of a PN junction loaded transmission line is given as $\theta = -n_{eff}^m l \omega / c$, where n_{eff}^m is the effective index of the transmission line and c is the speed of light in vacuum. As we can see, GD is directly proportional to transmission line mode index n_{eff}^m . As shown in Chapter 3, n_{eff}^m decreases monolithically with frequency and the slope increases after PN junction loading, *i.e.* more GDV is expected after PN junction loading. To study the effect of the feedback equalization on GDV, we first calculate its phase. From Eq. 5.1, we can get:

$$\begin{aligned}
 A_v &= \frac{-e^{-jkl}(e^{\alpha^{ml}} + A_f e^{j(\phi+kl)})}{(e^{\alpha^{ml}} + A_f e^{-j(\phi+kl)})(e^{\alpha^{ml}} + A_f e^{j(\phi+kl)})} \\
 &= \frac{e^{\alpha^{ml}} \cos(kl) + A_f \cos(kl) \cos(\phi + kl) + j A_f \cos(kl) \sin(\phi + kl) - j e^{\alpha^{ml}} \sin(kl) - j A_f \cos(\phi + kl) \sin(kl) + A_f \sin(\phi + kl) \sin(kl)}{e^{2\alpha^{ml}} + A_f^2 + 2A_f e^{\alpha^{ml}} \cos(\phi + kl)} \quad (5.7) \\
 &= \frac{e^{\alpha^{ml}} \cos(kl) + A_f \cos(\phi) + j(A_f \sin(\phi) - e^{\alpha^{ml}} \sin(kl))}{e^{2\alpha^{ml}} + A_f^2 + 2A_f e^{\alpha^{ml}} \cos(\phi + kl)}
 \end{aligned}$$

The phase is given by:

$$\psi = \arctan\left(\frac{-A_f \sin(\phi) + \sin(kl)e^{\alpha^{ml}}}{A_f \cos(\phi) + \cos(kl)e^{\alpha^{ml}}}\right) \quad (5.8)$$

Let's take a look at two special cases: dc ($f \rightarrow 0$) and resonance frequency ($f \rightarrow f_{res}$). At dc, $kl + \phi \rightarrow 0$, so

$$\begin{aligned}
 \psi &= \arctan\left(\frac{-A_f \sin(-kl) + \sin(kl)e^{\alpha^{ml}}}{A_f \cos(-kl) + \cos(kl)e^{\alpha^{ml}}}\right) \\
 &= kl
 \end{aligned} \quad (5.9)$$

At resonance frequency, $kl + \phi = \pi$, so

$$\begin{aligned}\psi &= \arctan\left(\frac{-A_f \sin(\pi - kl) + \sin(kl)e^{\alpha m l}}{A_f \cos(\pi - kl) + \cos(kl)e^{\alpha m l}}\right) \\ &= kl\end{aligned}\quad (5.10)$$

As we can see, at both dc and resonance frequency, the phase before and after equalization are the same.

The GD is defined as: $\Delta\psi/\Delta\omega$. Given a small change in frequency $\Delta\omega$ at dc, we can get from Eq. 5.8,

$$\Delta\psi \approx \frac{-A_f \Delta\phi + \Delta(kl)e^{\alpha m l}}{A_f + e^{\alpha m l}} \quad (5.11)$$

So, assuming $e^{\alpha m l} \gg A_f$, we have GD of the equalized TWE given by:

$$\frac{\Delta\psi}{\Delta\omega} \approx \frac{-A_f}{A_f + e^{\alpha m l}} \frac{\Delta\phi}{\Delta\omega} + \frac{\Delta(kl)}{\Delta\omega} \quad (5.12)$$

The GD of feedback circuit $\Delta\phi/\Delta\omega$ is usually positive, therefore at low frequency, the GD of an equalized TWE is smaller than that of a TWE without equalization (*i.e.* $\frac{\Delta\psi}{\Delta\omega} < \frac{\Delta(kl)}{\Delta\omega}$). In other words, the GD of an equalized TWE at low frequency is suppressed. On the other hand, since the phase of a TWE with and without equalization is the same at both $f = 0$ and $f = f_{res}$, and if we assume that these are the only two points at which the phase of equalized and non-equalized TWE are equal, it can deduced that the GD at the resonance frequency will be larger in the equalized TWE than in the non-equalized TWE. Combining with dc GD suppression, the feedback equalization also equalizes the GD versus frequency curve. As a consequence, we can potentially reduce the GDV of a TWMZM.

An example of GD equalization is demonstrated in Fig. 5.5. In this example, the same TWMZM model in Chapter 3 is used. The TWE is 1-mm long. The dc gain is $A_f = 0.08$ and 0.15 and feedback circuit GD is $T_{dly} = 2.5$ ps. Fig. 5.5-a

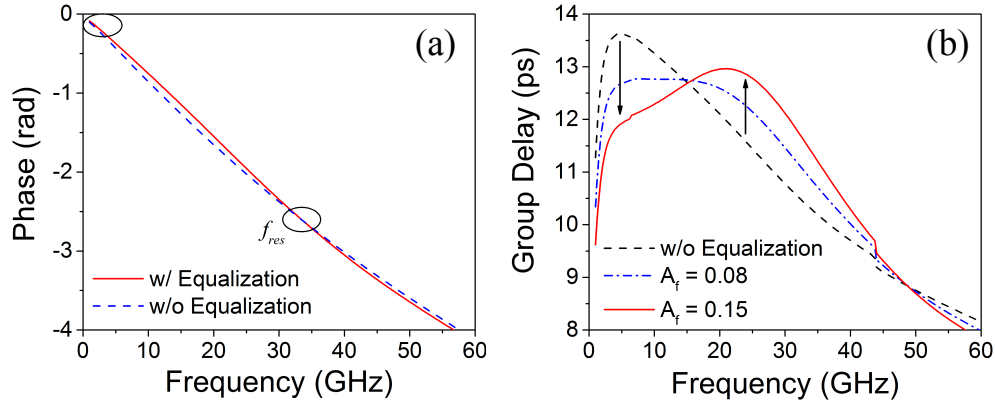


Figure 5.5: (a) Calculated phase with and without equalization. (b) (b) Calculated GD without equalization, with $A_f = 0.08$ and 0.15.

shows the phase as a function of frequency for the TWEs with (red) and without (blue) feedback equalization ($A_f = 0.15$). Just as derived above, the two phase curves have one cross point at $f \rightarrow 0$ and another cross point at f_{res} . There are other cross points at higher order resonance, but there is no cross point between dc and the first order resonance, f_{res} , which confirms our previous assumption. Fig. 5.5-b shows the GD for these two cases. As we can see, GD at low frequency is indeed suppressed while the GD at the resonance frequency is boosted. At $A_f = 0.08$, the overall GDV in this case is reduced from 3.2 to 2.6 ps for a frequency range from dc to 40 GHz, demonstrating the versatility of the feedback equalization. One has to note that GDV reduction depends on how much we distort the GD curve. Too much distortion (dc suppression or high frequency boost, e.g. $A_f = 0.15$) will lead to the opposite effect. Moreover, the parameters we tune to reduce the GDV will also affect the magnitude of the transfer function. A careful balance between GDV and 3-dB bandwidth is required.

5.3 Design of the Equalized TWMZM

To illustrate the concept of feedback equalization, a prototype feedback equalized TWMZM is designed. The TWMZM is designed using IMEC 50G Si photonic process and an active feedback circuit is designed using GlobalFoundries 130-nm SiGe process. Following our design flow in Chapter 3, the system is first simulated separately for photonics and electronics. Then the overall system performance is evaluated in a photonic circuit simulator.

5.3.1 MZM Design

The MZM phase shifter is an IMEC library device. It is based on a reverse-biased PN junction with junction located at the center of the rib waveguide. The rib height is 220 nm and width is 450 nm. The slab thickness is about 50 nm, and the BOX layer is about 2 μm . Co-planar waveguide transmission lines made from level 1 metal are used to drive this MZM. The dc and small signal characteristics of this PN junction at different bias voltages are required for circuit and system simulation. They are extracted based on process and device simulation (the same as Chapter 3). The simulated modulation efficiency, $V_\pi L_\pi$, and attenuation coefficient, α , are shown in Fig. 5.6. At 0-V reverse bias, the $V_\pi L_\pi$ is about 1.4 V·cm, showing very good match with the measured result from IMEC design manual. The modulation efficiency increases with bias voltage and reaches 2.2 V·cm, at $V_R = 2$ V. The attenuation coefficient at 0 V bias is about 1.6 cm^{-1} , which corresponds to about 13.7 dB/cm. The attenuation coefficient decreases with reverse bias as more carrier are pulled out of the waveguide center. These values agree very well with the measured value provided in the IMEC design manual, which validates our modeling process [122]. A compact circuit model for the phase shifter is created based on these parameters.

The parasitics of this PN junction will be presented as a load to driving elec-

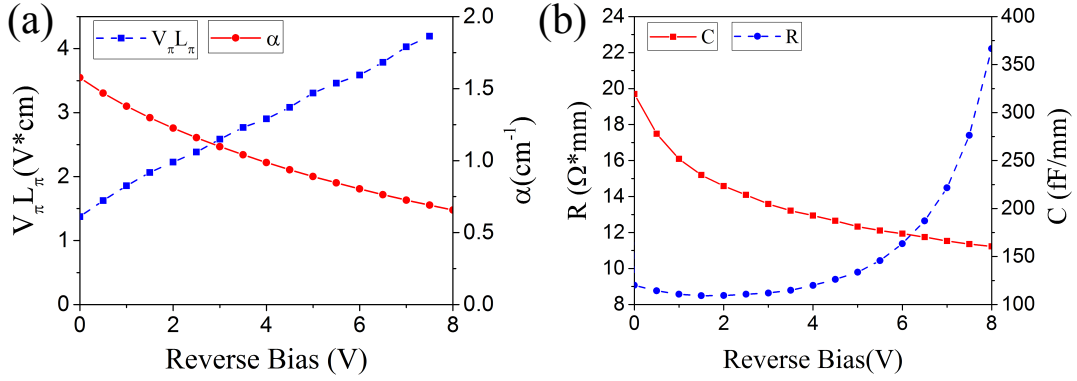


Figure 5.6: (a) Simulated modulation efficiency $V_{\pi} L_{\pi}$ and loss coefficient α at different reverse bias. (b) Extracted parasitic RC at different reverse bias.

trodes. An RC model is typically used to extract the parasitics. More complex model considering substrate parasitics can also be found in [53]. However, those parasitics tend to be small compared to PN junction parasitics. As we have shown in Chapter 3, an RC model can fit the small signal admittance very well. Therefore, only RC is considered in this design example. The extracted RC at different reverse bias are shown in Fig. 5.6-b. The resistance at $V_R = 0$ V is about $9 \Omega \cdot mm$ and capacitance is $320 fF/mm$.

The TWE is designed based on our specifications. Modeling follows the same process as in Chapter 3. The length of the phase shifter is chosen to be 3 mm for large modulation efficiency, so that a small driving voltage can be used. The electrode is first simulated in a high frequency EM simulator, SONNET. Then, S-parameter is extracted, exported to Cadence, and simulated together with the RC load. The electrode is designed to satisfy the three aforementioned performance requirements. In addition, the design also has to meet one special requirement: low dc resistance, for a reason that we will discuss later. The simulation results for the loaded transmission line is shown in Fig. 5.7. Fig. 5.7-a and -b are the mode index (n_{eff}^m) and characteristic impedance (Z_0) curves, respectively. As we can see, the mode index is about 4.6 to 4.4 from 10 to 30 GHz, which is larger

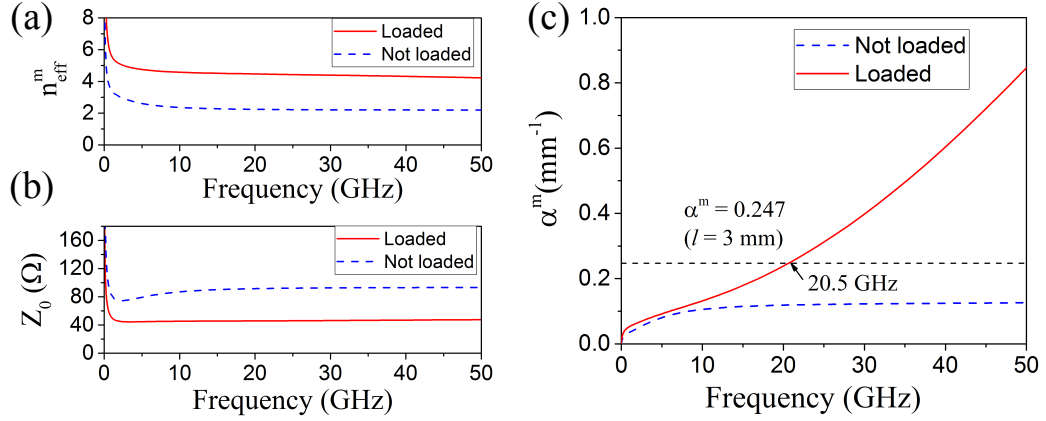


Figure 5.7: Simulated mode index n_{eff}^m (a), characteristic impedance Z_0 (b) before (blue) and after (red) PN junction loading. (c) Simulated loss coefficient α^m as a function of frequency before (blue) and after (red) PN junction loading. The frequency corresponding to $\alpha^m = 0.74/3$ is marked by the dashed line.

than the group index of the rib waveguide (3.8) estimated from the waveguide geometry. Based on the model in Eq. 2.12, the index mismatch will lead to 7.7% sacrifice in 3-dB EO bandwidth for a 3-mm long device, which is still tolerable. The characteristic impedance after loading is about 46 Ω , very close to the 50 Ω characteristic impedance requirement. Further, the attenuation coefficient (α^m) of the loaded transmission line is plotted in Fig. 5.7-c. As we can see, the loss is mainly from PN junction loading. The 3-dB EO bandwidth can be estimated from this curve for a perfectly matched TWE. Based on our previous discussion, the 3-dB EO bandwidth corresponds to a frequency where we have $\alpha^m = 0.74/3$, which is 20.5 GHz in this case. Such 3-dB EO bandwidth will not be enough for 50 Gb/s operation as we will see later. It should be noted that, the IMEC library device, which is claimed to be able to operate at 50 Gb/s, is only 1.5-mm long. Therefore, this result is reasonable.

The simulated S-parameters of the 3-mm long loaded TWE are shown in Fig. 5.8. Very good impedance matching is observed as S_{11} is below -10 dB most of the time. The S_{12} curve shows about 1.1 dB loss at low frequency, which is due

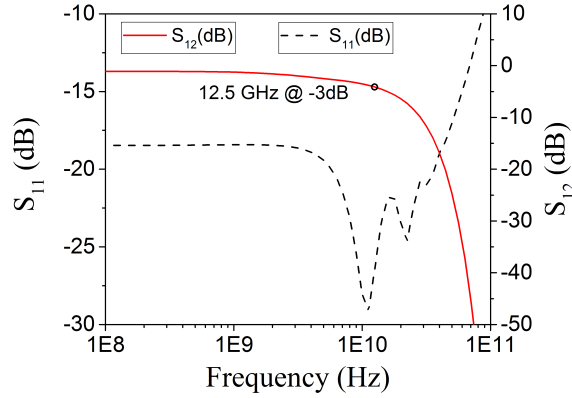


Figure 5.8: Simulated S-parameter of the designed TWE. The 3-dB bandwidth are marked on S_{21} curve.

to the dc resistance of the metal wire. This partially explained why we want to minimize the dc resistance of the signal line. The 3-dB bandwidth is about 12.5 GHz (Fig. 5.7-c).

5.3.2 System Optimization

The feedback equalization system is designed based on the above TWE. First, we have to determine the two most important design parameters: dc gain, A_f and resonance frequency, f_{res} . The first order mathematical model in Eq. 5.2 can be used. To simplify the design process, we consider only the magnitude of the transfer function. The effective mode index n_{eff}^m and attenuation coefficient α^m simulated above are used to model the TWE.

The transfer function of the feedback system without any loss on transmission line looks like a cosine wave with minimum at zero and maximum at the resonance frequency. This is the origin of the high frequency gain boost. To achieve high 3-dB bandwidth, the f_{res} should be placed as far as possible. On the other hand, good gain flatness has to be ensured for good settling behavior, which implies that f_{res} can't be arbitrarily far. The same trade-off applies to A_f , *i.e.* larger A_f results

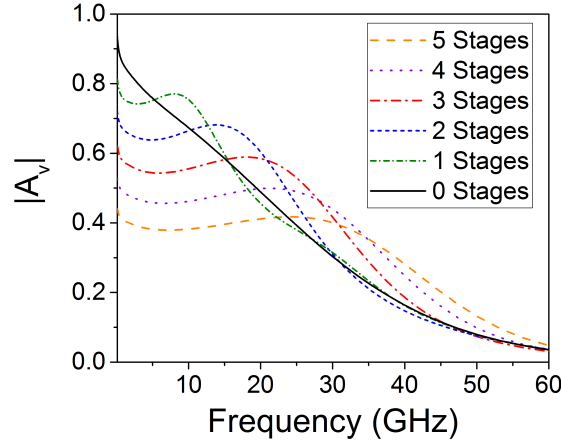


Figure 5.9: Calculated transfer function with optimized A_f for different number of stages.

in greater high frequency gain boost but leads to peaking in frequency response. Based on the criteria of maximum gain flatness and maximum 3-dB bandwidth, the values of A_f and f_{res} are optimized. From Eq. 5.2, f_{res} can be tuned by changing the effective mode index of TWE (n_{eff}^m), feedback circuit GD (T_{dly}) or TWE length (l). Among these three, n_{eff}^m is determined by many different factors as we discussed in TWE design section and T_{dly} is also a complicated parameter that cannot be tuned without affecting other design parameters. The length l is the only parameter that can be freely changed without any complications.

Therefore, in this design, the 3-mm long TWE is segmented into sections with equal length. The number of sections is varied to tune the TWE length. Each section is equalized separately and they are cascaded to form the 3-mm long equalized TWMZM. The cascaded structure not only enables the optimization of the equalization design, but also reduces the amount of signal pre-distortion required at each section since it only has to compensate for the loss of a short TWE. Therefore, unlike the FIR equalizer, the equalization technique will increase the usable output swing from the driver.

To obtain optimal A_f and f_{res} , the dc gain of the feedback circuit, A_f , is firstly

optimized for maximum gain flatness at different number of sections. Then, the optimal combination of A_f and f_{res} is chosen based on gain-bandwidth product (GBP) because both signal amplitude and bandwidth are important for a MZM. The optimization process for A_f is as follows. From Eq. 5.2, the system gain at dc and at resonance frequency f_{res} are firstly derived:

$$A_{v,dc} = \frac{1}{1 + A_f} \quad (5.13)$$

$$A_{v,f_{res}} = \frac{1}{e^{\alpha^m|_{f_{res}} l} - A_f} \quad (5.14)$$

where $\alpha^m|_{f_{res}}$ is the attenuation coefficient at resonance frequency. To minimize peaking in transfer function, it is desirable to have $A_{v,dc} = A_{v,res}$. From Eq. 5.13 and 5.14, we get:

$$A_f = \frac{e^{\alpha^m|_{f_{res}} l} - 1}{2} \quad (5.15)$$

Unfortunately, the transfer function doesn't peak at the resonance frequency. Therefore, using this value as a starting point, A_f is further optimized to have the dc gain of the transfer function equal to its peaking if that exists. Fig. 5.9 shows the transfer functions for different numbers of stages at optimal A_f . As can be seen, high frequency gain peaking is observed in all transfer functions but the peak amplitude is equal to the amplitude at dc, which is a condition forced in the calculation. Gain flatness is defined as the difference between the amplitude at dc and the lowest amplitude before the peaking. The gain flatness is 0.33 dB for one feedback stage. It becomes larger as the number of stages increases. Nevertheless, gain flatness of all transfer function is within 1 dB. It has been noticed later that the second order effect in a feedback circuit will flatten the transfer function in an actual circuit. As a result, the actual implementation could have no peaking in the transfer function. The dc gain of an equalized system decreases with the number of stages, while the 3-dB bandwidth increases with the number of stages. The

Number of Stage	3-dB BW GHz	Dc Gain	Gain*BW GHz	A_f
0	12.5	0.88	11.2	0
1	15.6	0.77	12.0	0.16
2	22.7	0.68	15.5	0.14
3	28.2	0.59	16.6	0.15
4	32.9	0.5	16.5	0.16
5	36.9	0.4	15.4	0.17

Table 5.1: System performance for different numbers of stages.

performance for different numbers of stages at optimal A_f is shown in Table 5.1 and plotted in Fig. 5.10. As shown in the table, the optimal value for A_f is generally about 0.15 and this value doesn't change much over different numbers of stages. The 3-dB bandwidth increases with the number of sections because f_{res} is moved to a high frequency. With 2 stages, the 3-dB bandwidth is almost doubled compared to an unequalized TWE. With 5 equalized sections, the 3-dB bandwidth can even be 3 times as high as that of the original TWE. On the other hand, the dc gain drops with increasing number of stages. With 3 equalization stages, the power is already 3-dB down. For a TWMZM, both signal amplitude and bandwidth are important to achieve high ER and SNR. The GBP at different numbers of stages is then plotted in Fig. 5.10-b. The value peaks at 3 stages and the peak value is 18.9 GHz. Further increasing the number of stages not only decreases the GBP, but also causes more power consumption. Therefore, in this design, we choose 3-stage cascaded design with a 1-mm long TWE in each stage. That gives us more than 100% increase in 3-dB bandwidth, but costs 3.5 dB dc amplitude drop. As we will show later, the 3.5 dB dc amplitude drop is not an issue because the actual optical power loss is much smaller. The resonance frequency can be estimated to be around $f_{res} = 30$ GHz and the corresponding A_f is 0.15.

From Eq. 5.6, the reflection coefficient for the optimized TWE can be calcu-

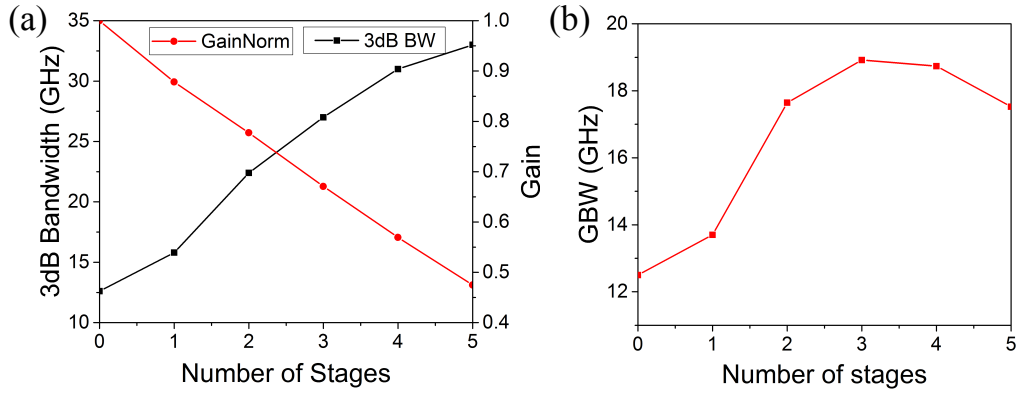


Figure 5.10: (a) Calculated gain and 3-dB bandwidth with optimized A_f for different numbers of stages. (b) Calculated GBP with optimized A_f for different number of stages. Zero stage means no equalization is applied.

lated. From previous discussions, we learned that at low frequency, the TWE has about 1.1 dB loss. Based on this value, S_{11} is calculated to be -18.6 dB at low frequency. This value will further decrease as the frequency increases. Note that this reflection coefficient is obtained based on NO reflection from original transmission line. However, for our TWE designed in the above section, we have about -18.5 dB reflection at low frequency. The total reflection will be about 3-dB above this, *i.e.*, -15.5 dB at low frequency, which is still an acceptable value.

The feedback has to be stable with enough phase margin to avoid potential oscillation or ringing in the time domain waveform. At resonance frequency, the phase shift of the loop gain is exactly π . By the Barkhausen stability criterion, the magnitude of the loop gain should be smaller than 1 when the phase shift is π . In the optimized TWE, the loop gain is given by: $A_f e^{-\alpha^m l}$. Due to the loss of the TWE and small feedback gain, this criterion is always satisfied. In the above optimal design, the loop gain is -26.5 dB, which guarantees stability.

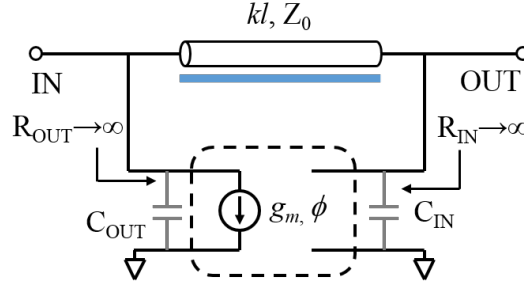


Figure 5.11: Schematic of the transconductance feedback network. The parasitic capacitance at the input and output are shown in grey.

5.3.3 Feedback Circuits

To work together with a TWE, the feedback circuit should satisfy several criteria. Firstly, the feedback circuit will add additional load to the original TWE and change its characteristics. To minimize this effect, the feedback circuit should have very large input and output impedance. Secondly, the feedback circuit should be unidirectional. In the feedback network, the signal should only travel from the output to the input of a TWE. These two criteria rule out the possibility for a passive network. Finally, this feedback circuit should operate at high enough speed and be able to tolerate a large voltage swing.

A transconductance amplifier satisfies all these criteria. It has a large impedance at both input and output and very good unidirectionality. SiGe process usually has a large cut-off frequency (f_T) and can tolerate a large voltage swing. Therefore, in this work, GlobalFoundries 0.13 μm SiGe BiCOMS process is used for the feedback circuit design. The high performance npn transistor has $f_T > 200$ GHz and the breakdown voltage for open base $V_{CEO} = 1.8$ V. This value will be higher if the base is not open. A schematic of the transconductance amplifier is shown in Fig. 5.11. It is characterized by transconductance g_m and frequency dependent phase ϕ . The low frequency gain is given by $A_f = g_m \cdot R_L$, where R_L in this case is $Z_0/2$. Parasitic capacitance associated with the transconductance amplifier C_{IN} and C_{OUT} will be presented as a load to the TWE (Fig. 5.11) and

affect the characteristics of the TWE. They can be absorbed into the transmission line in our mathematical model as additional phase. So the total phase on the TWE will be $kl + \omega (\tau_1 + \tau_2)$, where $\tau_1 = C_{OUT}Z_0$ and $\tau_2 = C_{IN}Z_0$. Assume a parasitic capacitance of 50 fF for both input and output capacitance, τ_1 and τ_2 can be calculated as: $\tau_1 = \tau_2 = 2.5 \times 10^{-12}$. At the resonance frequency of 30 GHz, the phase is about 7.5×10^{-2} rad. Assuming a linear relationship between additional phase and the resonance frequency, the parasitic capacitance at both input and output will result in a 5% change in resonance frequency. In addition, the parasitic capacitance will also affect the characteristic impedance of the TWE, causing more reflection. To reduce these parasitics, small size active devices must be used.

In this work, two transconductance feedback circuit designs are presented: linear feedback and limited feedback. In linear feedback, A_f is constant and doesn't change with input voltage swing. On the other hand, in limited feedback, the transconductance amplifier works as a switch and the feedback gain is controlled by the bias current.

Linear Feedback

The schematic of a linear feedback circuit is shown in Fig. 5.12. It has two stages: an emitter follower (EF) stage samples the voltage signal at the output of the TWE and drives an emitter degenerated common emitter (CE) stage. The CE stage is a widely used transconductance amplifier. The emitter degeneration linearizes its transconductance and improves its linear range. The EF stage provides large current drive capability. Its small output impedance makes it suitable to drive large capacitive load for very high speed operation. Therefore, it is often used to precede a CE stage. In this design, another reason for using an EF stage is that the input voltage and output voltage are at the same dc level as they both are tied to the same transmission line. During operation, large voltage swing at

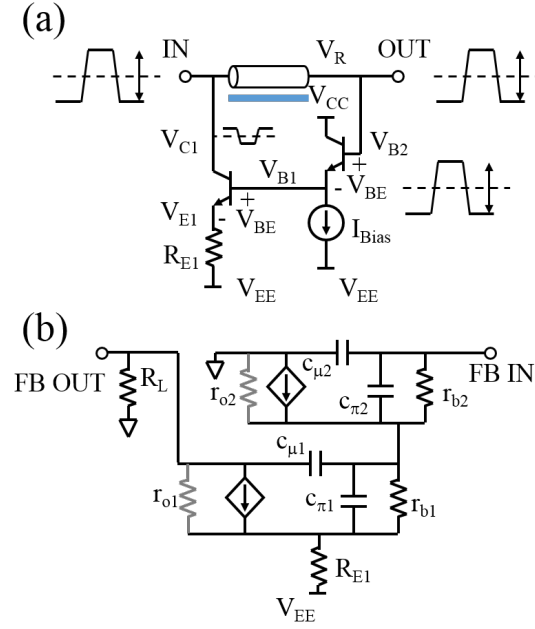


Figure 5.12: (a) Schematic of the linear feedback circuit; (b) Small signal model of the linear feedback circuit. The resistances due to channel length modulation effect are shown in grey.

the input and output of Q_1 may push it into the saturation region and reduce its speed. In order to maintain high speed operation, the input dc level is then down-shifted by V_{BE} using an EF. In this design, the maximum single-ended V_{pp} is assumed to be 1.4 V as a compromise between device stress and driving capability. The reverse bias on the TWE is 2 V as a balance between maximum voltage swing and bandwidth. At 2 V bias, the modulation efficiency of the TWE is $V_{\pi}L_{\pi} = 2.2 \text{ V} \cdot \text{cm}$ and the optical loss of the phase shifter is about 1.1 dB/mm. So, for a 3-mm long device, the total optical loss due to the phase shifter is 3.3 dB.

The value of V_{EE} is designed based on the total current drain by Q_1 and the degenerating resistor value. The low frequency feedback gain is given by: $A_f = G_m Z_0/2$, where G_m is the transconductance of this degenerated CE stage. Ignoring second order effect, $G_m = g_{m,1}/(1 + g_{m,1}R_E)$. Given a large $g_{m,1}$ for Q_1 ,

	Q_1	Q_2
$V_{be}(\text{mV})$	870	870
$I_{Bias}(\text{mA})$	4	7.2
$g_m(\text{mS})$	112	199
β	267	225
$C_\pi(\text{fF})$	73	130
$C_\mu(\text{fF})$	5.6	9
$r_O(\text{k}\Omega)$	75	52
$r_e(\Omega)$	4.5	2.6

Table 5.2: Small signal parameters for Q_1 and Q_2 in a linear feedback circuit.

G_m is approximately equal to $1/R_E$. Based on a 0.15 gain target obtained from system optimization, this value is calculated to be $R_E = 167 \Omega$. After simulation, the final R_E is chosen to be 160Ω to achieve the gain target. Further, to ensure high speed operation, Q_1 should not be switched off during voltage swing, *i.e.* V_{EE} should be equal to or smaller than the lowest possible voltage present at V_{E1} . For GlobalFoundries 0.13 μm SiGe BiCOMS process, $V_{BE} \approx 0.9V$. Therefore, V_{EE} is calculated as: $V_{EE} = 2 - 2 \times V_{BE} - V_{pp}/2 = -0.5V$. The V_{CC} is designed to be 2.5 V to make sure Q_2 will never be pushed to the saturation region. The design parameters of the linear feedback circuit is summarized in Table. 5.2.

The small signal model of this circuit is shown in Fig. 5.12-b. The two-stage amplifier forms a Darlington configuration. Its input resistance is improved through multiple stage degeneration. Neglecting all second order effect, the small signal input resistance is: $R_{in} = (\beta_2 + 1)[(\beta_1 + 1)(r_{e1} + R_{E1}) + r_{e2}]$, where β_1 and β_2 are the current gain of Q_1 and Q_2 ; r_{e1} and r_{e2} are the input resistance at the emitter of Q_1 and Q_2 , respectively. Given the parameter values in Table. 5.2, R_{in} is calculated to be $9.6 \text{ M}\Omega$. The output resistance is also enhanced by emitter degeneration. The output resistance is: $R_{out} = r_{O,1}(1 + g_{m,1}R'_{E,1})$, where $R'_{E,1} = R_{E,1}/r_\pi$, which is $1.5 \text{ k}\Omega$ in this design. The output resistance is $1.3 \text{ M}\Omega$. The high input/output resistance minimized the loading effect to the TWE.

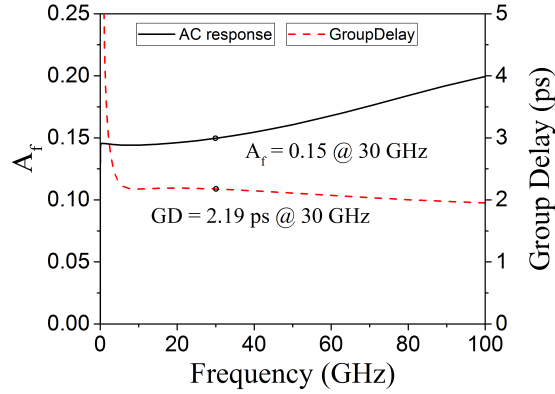


Figure 5.13: Simulated feedback gain and GD of the linear feedback circuit.

Parasitic capacitance loading can also be calculated. At the input, $C_{\pi,2}$ is bootstrapped at a relatively low frequency. The only capacitive load is $C_{\mu,2}$, which is a small value (9 fF in this design). At high frequency, $C_{\pi,2}$ is added to the picture, but it is connected in series with the load capacitor $C_{\pi,1}$, which immediately reduces its effect to about half. The degeneration resistance at the emitter further reduces the effective capacitance loaded at the input. At the output, the only capacitive loading is the miller capacitor $C_{\mu,1}$. The effective load present at the output node is $(1 + A_f)C_{\mu,1}$. Since the gain in our design is much smaller than 1, this effective capacitor is only slightly larger than $C_{\mu,1}$, which is 5.6 fF in this design. Given the small capacitance at the input and output, we expect negligible effect on the characteristics of the TWE and the resonance frequency.

The feedback circuit is required to provide constant gain A_f over a large frequency range. The ac simulation results in Fig. 5.13 shows the circuit has a gain slightly smaller than 0.15 at low frequency. It increases with frequency and becomes 0.15 at resonance frequency (30 GHz). This response is common for an EF stage with capacitive loading. If we extend the simulation to an even larger frequency range, inductive peaking will be observed. This inductive peaking behavior is expected when an EF is loaded with a capacitor [123]. To reduce the peaking, we can either reduce the size of Q_1 or increase the size of Q_2 . However,

by reducing the size of Q_1 , we may be at risk of biasing it over the peak f_T and sacrificing performance. On the other hand, increasing the size of Q_2 will add more power consumption and more capacitive load to the input. System level performance evaluation is required to determine which parameter is more critical than others. The GD is also plotted and shown in Fig. 5.13. As we can see, at the frequency range of interest, the GD stays relatively constant, justifying our previous assumption in the mathematical model analysis. The GD is 2.19 ps at 30 GHz.

Limiting Feedback

Although the feedback circuit in the previous section has a large bandwidth and good linearity, it is not differential, thus, it is susceptible to common mode noise. In this section, a differential feedback circuit design is presented. Given the large voltage swing at the input, a linear circuit is very difficult. The schematic of the limiting feedback circuit is shown in Fig. 5.14. It adopts the same EF-CE configuration as in the linear feedback. The difference is that the output CE stage is a differential pair. For an npn differential pair, it takes only $V_{pp} = 2 \ln 2 V_T = 36$ mV to completely switch it. Owing to the large voltage swing present at the inputs, Q_1 and Q_2 in the differential pair will work like switches. The output amplitude is controlled by the tail current, $V_{pp,out} = I_{EE} R_L$, where I_{EE} is the tail current and $R_L = Z_0/2$. This value will not change with the input voltage swing. The tail current I_{EE} depends on many different factors.

The feedback signal amplitude v_f is given by:

$$v_f = e^{-\alpha^m l} A_f v' \quad (5.16)$$

where $v' = v_i - v_f$ and v_i is the input signal. Substituting into Eq. 5.16, we get:

to minimize the source voltage. In this design, the source voltage is tied to the lowest voltage of the circuit, V_{EE} .

Since the source and body have to be tied to a lower potential ($V_{EE} = -0.5$ V) than the substrate ($V_{sub} = 0$), the NMOS device must be separated from the substrate. Therefore, triple well devices are utilized which have an additional buried n-type layer inserted between the local p-well and p-substrate. A ring of N-well is used to surround the P-well and completely isolate the local p-well from the substrate. The N-well needs to be tied to the highest voltage on the chip to ensure good isolation.

5.3.4 Equalized Traveling-Wave Electrode Design

As discussed in the system optimization section, 3 stages of equalized TWE will be cascaded to provide optimal performance. The schematic of the whole circuit is shown in Fig. 5.15. The waveguide and driving electrode have to be laid out in a meandering shape to minimize the effect of long metal contacts in the electrical circuit. In this section, we discuss the performance of the equalized TWE focusing on the electrical signal. Linear feedback is mainly used for the discussion. Limiting feedback will be similar and differences will be pointed out in particular.

The bias current to the feedback circuit must be supplied through the transmission line. Any resistance on the signal line will lead to an IR drop and change the actual bias on the circuits. To mitigate this effect, the resistance on the signal line has to be minimized. That's why in the TWE design, we chose to use a wide metal piece for the signal line. Based on the sheet resistivity provided by IMEC design manual, the resistivity of the signal line is calculated to be $4.1 \Omega/\text{mm}$. We know the current drain at point "A" is 4 mA. The voltage drop at this point will be 16.4 mV, which is still acceptable. To avoid further voltage drop down the line, another voltage supply is used at the other end. So, the maximum voltage drop

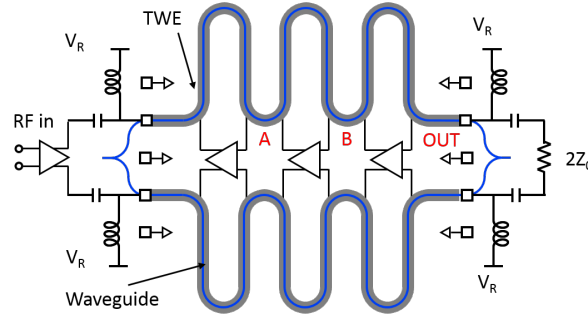


Figure 5.15: Schematic of the equalized TWE system. In order to minimize the voltage drop along the TWE, two voltage supplies are used at both ends. Point “A”, “B”, and “OUT” are the output for stage 1, 2, and 3, respectively.

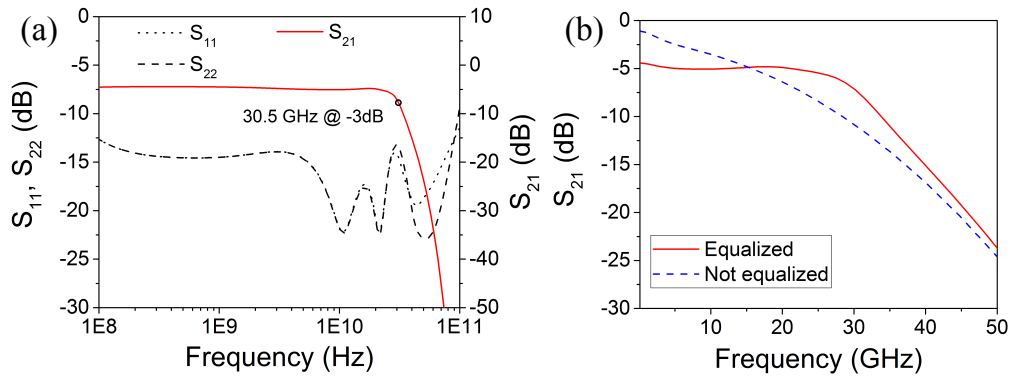


Figure 5.16: (a) Simulated S-parameter of the equalized TWE. The 3-dB bandwidth are marked out. (b) Comparison of S_{21} of the TWE with and without equalization.

will be 16.4 mV at point “A” and “B”.

The electronic circuit is simulated in Cadence Spectre with a TWE model consisting of sections of transmission lines (represented by a s2p file) and RCs loads extracted in the MZM design section. Dc operating point simulation shows the voltages at point “A” and “B” are both 1.982 V, which indicates an 18 mV IR drop. It is very close to the calculated 16.4 mV. The simulated S-parameter results for the TWE with linear feedback are shown in Fig. 5.16-a. As a reference, the unequalized S-parameters are also plotted in Fig. 5.16-b. From this figure we can see, the 3-dB bandwidth of the equalized TWE reaches 30.5 GHz, slightly larger

compared to calculated 28.2 GHz in Sec. 5.3.2. The low frequency response is 3.3 dB down compared to the unequalized TWE, which is smaller than the calculated 3.5 dB. Moreover, the frequency response curve looks very flat. The ripple is only 0.25 dB as compared to 0.71 dB in the calculation, thanks to the secondary effect in the feedback circuit. In summary, the actual circuit performance matches our mathematical calculation in Sec. 5.3.2.

Compared to the unequalized TWE (Fig. 5.16-b), the 3-dB bandwidth is improved by 2.4 times. Although a 3.3 dB dc drop is observed, this dc drop doesn't translate directly to optical modulation amplitude drop due to the traveling-wave nature of the modulator as we will show later. The reflection increases due to the feedback. S_{11} at low frequency is about -14.5 dB, which is close to the calculated -15.5 dB. The small discrepancy could be attributed to the parasitic capacitance at the input and output of the feedback circuit. The EO frequency response can be calculated based on Eq. 5.4. The EO response of the 3-stage cascaded design is given by:

$$|S_{21,EO}| = (|A_{v0}| + |A_{v0}|^2 + |A_{v0}|^3) \frac{e^{\alpha^m l} - 1}{3\alpha^m l} \quad (5.18)$$

where $|A_{v0}|$ is the transfer function of a single stage. The EO response before and after equalization is plotted in Fig. 5.17. The equalized transfer function exhibits boost at high frequency and suppression at low frequency. The dc suppression is about 2.2 dB which is smaller than the dc drop in EE response. The 3-dB EO bandwidth is improved from 23 to 38 GHz, *i.e.*, 65% increases.

To study the circuit sensitivity to process variation, a Monte Carlo simulation, including both process variation and mismatch, is carried out. Fig. 5.18-a and -b show the variation of low frequency gain $|A_v|$ and 3-dB bandwidth of the linear equalized TWE. The mean of the low frequency gain is $|A_v| = 0.3$ with a small standard deviation of 2.75×10^{-3} . Therefore, for 95% of trials, the low frequency gain only varies by 1.8%, indicating good robustness of this circuit to

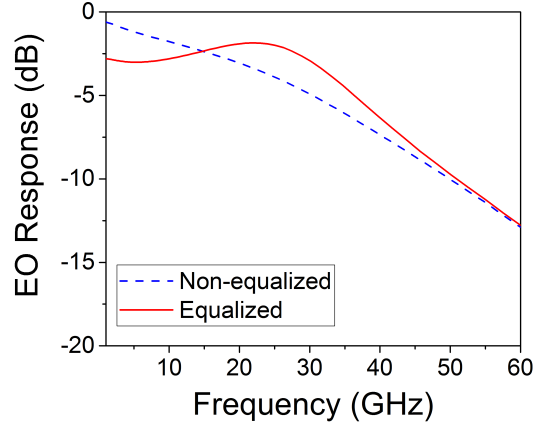


Figure 5.17: EO response of the TWMZM with and without equalized TWE.

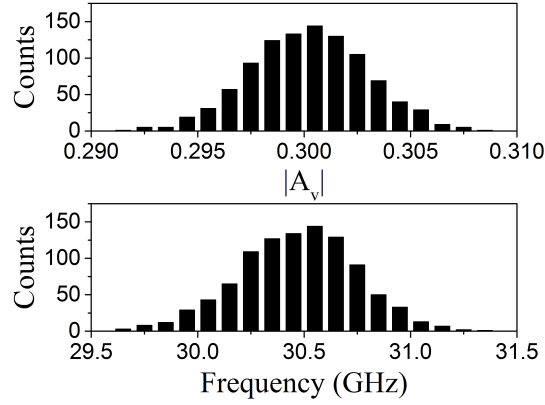


Figure 5.18: Monte Carlo simulation of the process variation and mismatch effects on the system gain (a) and 3-dB bandwidth (b).

process variation and mismatch. In addition, the bandwidth plot shows a mean 3-dB bandwidth of 30.5 GHz and 276 MHz standard deviation. Most of the trials (within $\pm 3\sigma$) exhibit only 2.7% variation in 3-dB bandwidth. At the same time, temperature is another important parameter that could cause performance variation. Simulation results show that A_v drops about (1%) and 3-dB bandwidth remains the same when temperature rises from 25 to 100°C. Therefore, we expect very consistent performance for the linear equalized TWE.

Since the limiting feedback has very different small signal and large signal be-

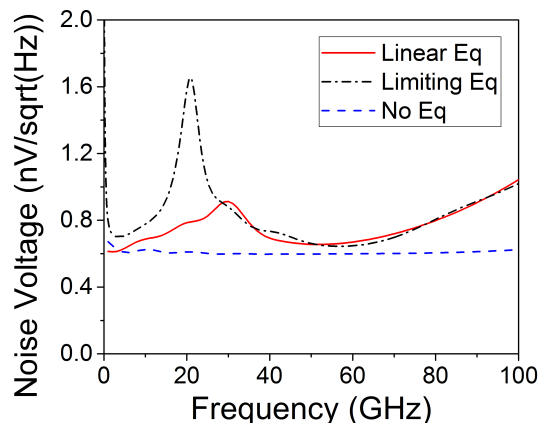


Figure 5.19: Noise spectra of the equalized TWE without equalization, with linear feedback equalization and with limiting feedback equalization.

havior, we can't draw much conclusion about its large signal performance from its small signal response. Therefore, S-parameter simulation results for the limiting equalized TWE are not discussed here.

As in many other equalization techniques, the equalization boosts the high frequency signal as well as high frequency noise. In a DFE, the noise is removed by the decision stage, which however, is not available in this feedback equalization. In addition, the feedback circuits add extra noise to the system. The total noise contribution from a non-equalized, linear equalized and limiting equalized TWE are simulated at 25°C and the noise spectra at “OUT” nodes are shown in Fig. 5.19. For the non-equalized TWE, the noise is mainly due to the thermal noise of parasitic resistance, which has constant power spectral density. The feedback circuit contributes additional noise and also amplifies the noise because of the feedback loop. Peaks are observed in the noise power spectrum at resonance frequencies. Owing to the large small-signal gain, the noise amplification is much larger in limiting equalization. The peak value for limiting equalization is $1.6 \text{ nV}/\sqrt{\text{Hz}}$, almost two times as high as that of the linear equalization ($0.9 \text{ nV}/\sqrt{\text{Hz}}$) and 3 times as high as that of non-equalized TWE ($0.6 \text{ nV}/\sqrt{\text{Hz}}$). Further, the noise

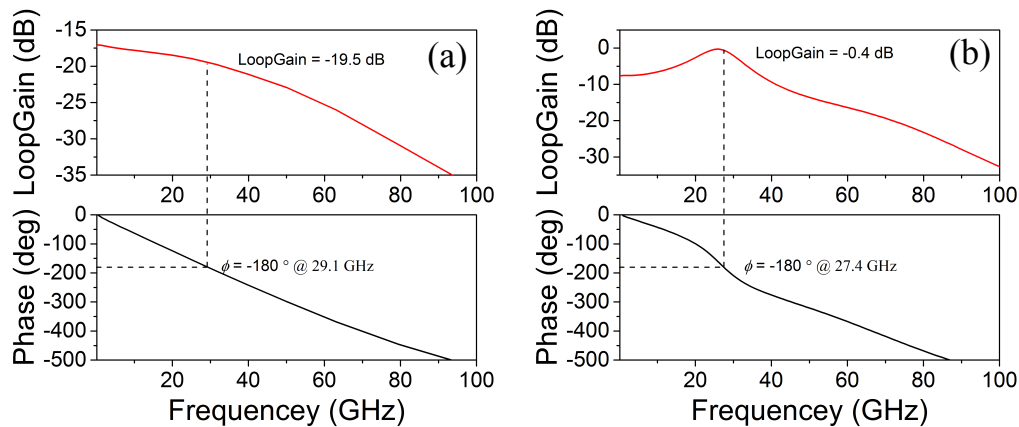


Figure 5.20: Loop gain and phase of the linear (a) and limiting (b) feedback equalization. The frequency where $\phi = -180^\circ$ is marked out.

spectrum curves increase rapidly at high frequency, because of the 2nd order resonance, which causes another peak at even higher frequency. Fortunately, this peak is irrelevant because it falls outside of the cut off frequency of the subsequent receiver stage. Based on the power spectral density curves, we can estimate the root-mean-square (RMS) noise voltage for these three cases by integrating the total noise power up to 50 GHz (assuming a receiver has a cut off frequency of 50 GHz). The RMS noise voltages for non-equalized, linear equalized, and limiting equalized TWE are: 136, 165, and 206 μV . So, the equalization increases the total thermal noise of the TWE by 21 % for linear equalization and 52% for limiting equalization.

As discussed previously, the feedback system should be stable because of small A_f and attenuation on the TWE. To study the stability quantitatively, the loop gain of the feedback equalization is simulated. Fig. 5.20-a and -b show the loop gain and phase of the linear and limiting feedback, respectively. Loop gain of linear feedback is below 0 dB for the entire frequency range of interest as expected. Its phase becomes -180° at 29.1 GHz, which is very close to the resonance frequency predicted by the mathematical model. Since the gain never goes above 0 dB, we use gain margin instead to characterize its stability. Gain margin is defined as

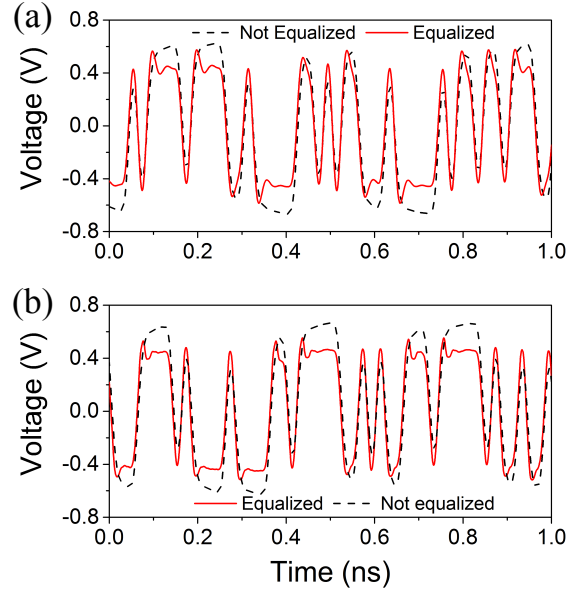


Figure 5.21: Simulated electrical waveforms at the output of the TWE without equalization and with linear (a), and limiting (b) feedback equalization. The data rate is 50 Gb/s and input voltage swing is $V_{pp} = 1.4$ V.

$GM = 0 - L_{\phi=-180^\circ}$, where $L_{\phi=-180^\circ}$ is the loop gain at the frequency where phase equals to -180° . This value must be positive for feedback to be stable and the stability improves as this number increases. For linear feedback, the gain margin is 19.5 dB, suggesting very good stability. Meanwhile, for limiting feedback, due to large small-signal gain, its loop gain is much larger than that of linear feedback. It reaches almost 0 dB at its peak. The gain margin for this circuit is only 0.4 dB, implying a possible stability issue.

Transient simulation with 50 Gb/s NRZ data is carried out to study its large signal performance. The input voltage swing is set to the maximum design voltage swing, $V_{pp} = 1.4$ V. The waveforms at the output of the TWE with and without equalization is shown in Fig. 5.21 for linear (a) and limiting (b) equalization. In both cases, equalization improves the signal rising and falling edges, so that it can settle at a much higher/lower voltage, and thus mitigating ISI. On the other hand, the maximum voltage the signal can reach is reduced after equalization

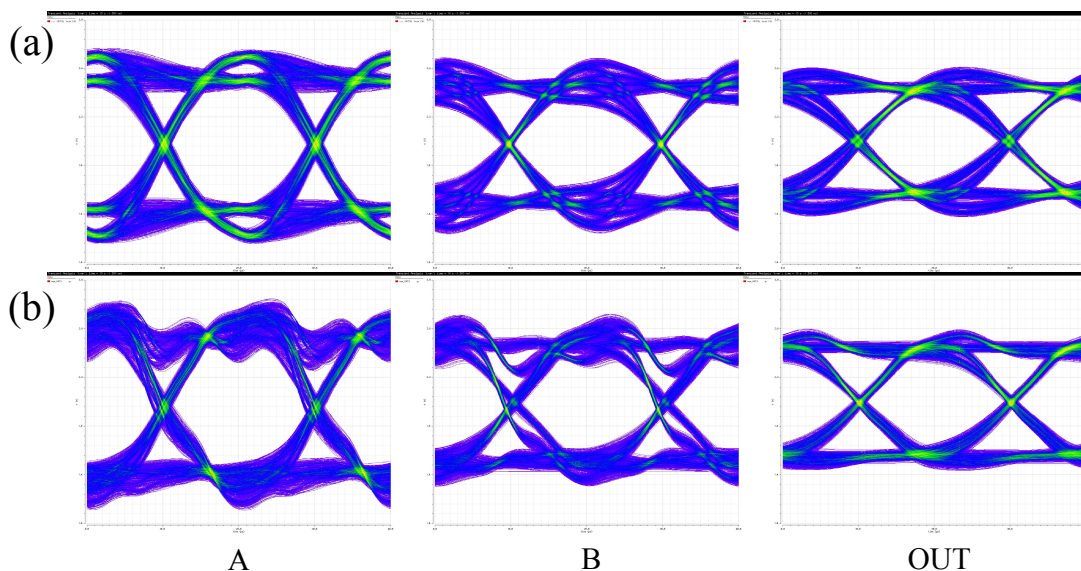


Figure 5.22: Simulated electrical eye diagrams at point “A”, “B”, “OUT” (Fig. 5.15) on the TWE for (a) linear feedback, (b) limiting feedback equalized TWE. The data rate is 50 Gb/s and input voltage swing is $V_{pp} = 1.4$ V.

as a manifestation of suppressed dc gain. As a result of dc gain suppression and attenuation, the voltage swing at the output is only $V_{pp} = 0.8$ V, while this number is > 1 V for an unequalized TWE. A little bit of ringing is observed on the equalized signal owing to the feedback. This ringing is more pronounced in limiting feedback equalization which could be attributed to its small gain margin as discussed previously.

The simulated eye diagrams with 50 Gp/s NRZ, at point “A”, “B” and “OUT” are shown in Fig. 5.22 for both linear (a) and limiting (b) equalization. In Fig. 5.22-a, eye diagrams show pre-emphasis at all three points. This pre-distortion of the waveform is a result of summation of output signal and input signal as illustrated in Fig. 5.2. When traveling in the TWE, the high frequency component experiences more loss, so the pre-emphasis gradually diminishes, but the amplitude of the eye diagram stays constant as it travels. For limiting equalization (Fig. 5.22-b), the eye diagrams exhibit a lot of ringing at point “A” and “B”, which could be accounted for by the small gain margin as well as impedance mismatch.

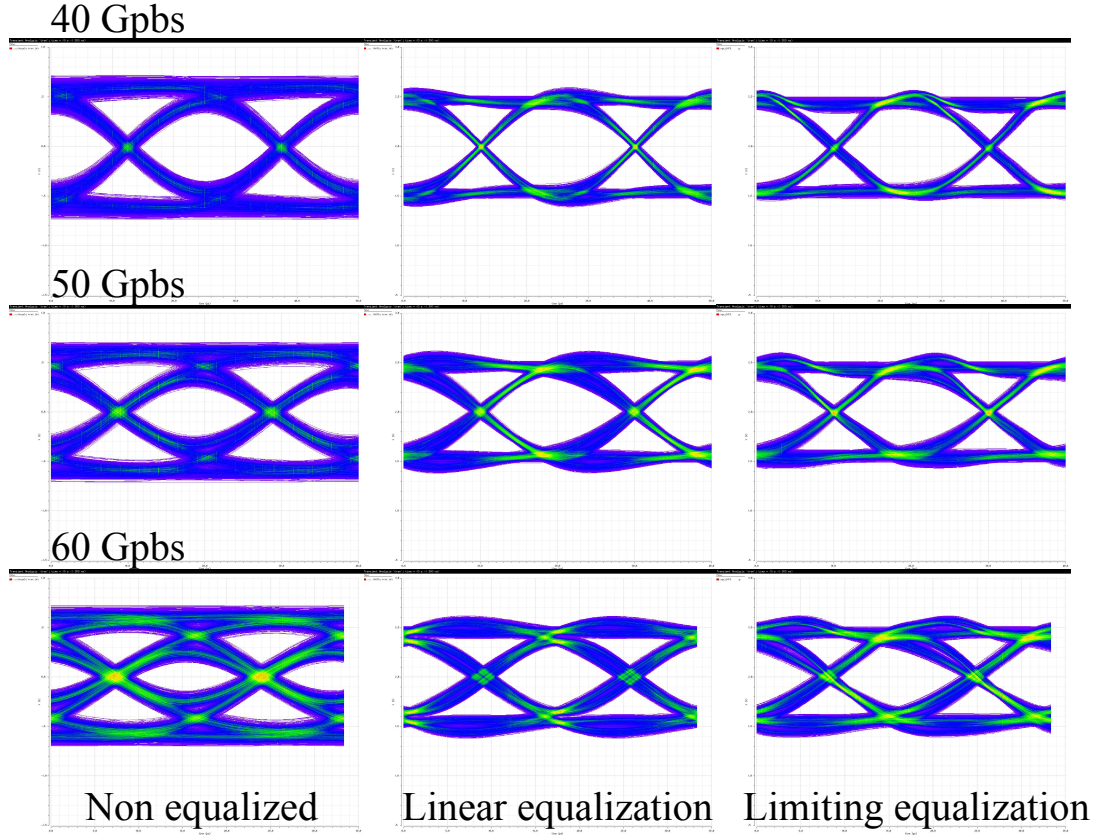


Figure 5.23: Simulated electrical eye diagrams at point “OUT” of the TWE at 3 data rates: 40, 50, 60 Gb/s for the non equalized (left), linear (middle) and limiting (right) feedback equalized TWE. The input voltage swing is $V_{pp} = 1.4$ V.

As discussed previously, the feedback will change the characteristics of the TWE. According to Eq. 5.6, the magnitude of S_{11} will increase with small-signal gain, A_f . Since the small-signal gain in the limiting feedback circuit is about 1.4, the magnitude of S_{11} could reach -5.1 dB at low frequency, much higher than that in the linear equalization.

Fig. 5.23 shows eye diagrams at the “OUT” node at 3 data rates: 40, 50 and 60 Gb/s for non-equalized, linear equalized, and limiting equalized TWEs. The figure clearly shows the improvement in signal quality after equalization. Even though the eye amplitude is reduced a little, in both linear and limiting equalization, eye height and width are improved significantly and jitter is also reduced substantially.

Table. 5.3 lists the detailed improvement in eye quality. The amplitude of eyes is reduced by about 10% for linear feedback and 12% for limiting feedback at 40 Gb/s. This is however much smaller than the 3.5 dB drop predicted in S-parameter simulation as shown in Fig. 5.16. The reason for this difference is that the ISI brings the mean value of level 0 and 1 towards the center in the unequalized TWE. As a consequence, the actual amplitude at a high data rate is smaller than its amplitude predicted by the frequency response. The eye height is calculated as: $\text{Level_1_mean} - 3\delta_{level1} - (\text{Level_0_mean} + 3\delta_{level0})$. At 40 Gb/s, the eye height is almost doubled and the eye width is increased by almost 3 ps with linear equalization. Higher SNR is observed compared to unequalized TWE at all 3 data rates with either linear or limiting equalization.

The power consumption for one stage of the linear feedback is 31.6 mW. The differential drive TWMZM will need 6 stages. So the total power consumption will be 190 mW. For the limiting feedback, one differential stage consumes 57 mW of power. So, it will need 171 mW of power for a cascaded 3-stage TWMZM.

5.3.5 MZM with Equalized Traveling-Wave Electrode

To evaluate the performance of the TWMZM with equalized TWEs, a TWMZM photonic circuit is built in Lumerical INTERCONNECT as shown in Fig. 5.24. As in our previous design example in Chapter 4, all the optical components, such as Y-junction, optical modulator, *etc.*, come from either modeling of IMEC library devices using our EPIC design flow or direct measurement results. Considering the traveling-wave nature of a TWMZM, a distributed model with 12 sections is built. Each section consists of an optical modulator block and a waveform block. In the waveform block, the electrical signal waveform at corresponding points on the TWE is directly imported from electrical circuit simulation. It should be noted that the optical modulator block in this tool doesn't include the optical time delay. Therefore, an artificial delay block is inserted after each optical mod-

Parameters	40 G			50 G			60 G		
	No Eq	Linear Eq	Limiting Eq	No Eq	Linear Eq	Limiting Eq	No Eq	Linear Eq	Limiting Eq
Eye Amp (mV)	967.6	872.6	851.5	875.5	815.9	806.4	787.2	770.1	760.1
Eye Height (mV)	262.8	559.7	602.3	32.85	347.4	406.3	-178.8	182.3	217.1
Eye Width (ps)	20.22	23.07	21.62	14.79	16.77	17.65	NA	11.38	12.76
Eye S/N	4.119	8.367	10.25	3.117	5.224	6.046	2.445	3.93	4.2

Table 5.3: Electrical eye quality comparison between non equalized, linear and limiting equalized TWEs at 40, 50 and 60 Gb/s data rates.

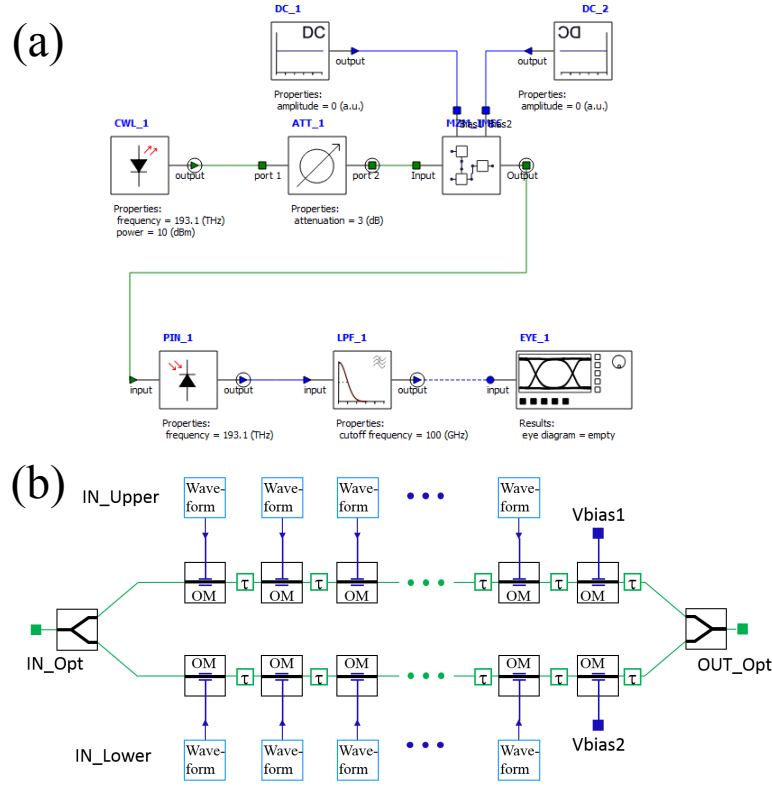


Figure 5.24: (a) Top level schematic for the photonic integrated circuit. The 3-dB bandwidth of the PD is intentionally set to 100 GHz to remove any effect of the limited receiver bandwidth on transmitter performance. (b) Schematic of the TWMZM. Delay cells (τ) are intentionally inserted to account for the optical signal delay, where $\tau = 3.17$ ps

ulator block with delay given by $\tau = l_{sec} \cdot n_g / c$, where l_{sec} is the length of each section, set to $250 \mu\text{m}$ and n_g is the waveguide group index. Based on n_g obtained previously, τ can be calculated as $\tau = 3.17$ ps. Two additional optical modulator sections are added to provide 90° bias phase. The detector is a PIN PD with 0.45 A/W responsivity. In this model, the PIN PD itself has a flat frequency response. Therefore, a low pass filter is added to mimic the parasitic RC time constant in a PD. It is intentionally set to 100 GHz in order to remove any bandwidth limiting effect of a receiver on transmitter performance. The optical signal is detected and analyzed with a virtual oscilloscope.

The electrical simulation results shown in Fig. 5.23 are imported to the pho-

tonic circuit and the simulated optical eye diagrams are shown in Fig. 5.25. Pronounced improvement in eye quality is observed at all 3 data rates for both linear and limiting equalization. More detail is shown in Table 5.4. As expected, the optical modulation amplitude is reduced. For instance, at 60 Gp/s, the optical modulation amplitude is reduced by 2.6% in linear equalization and 1.3% in limiting equalization, which leads to a small drop in ER (0.16 dB in linear equalization and 0.11 dB in limiting equalization). However, thanks to the distributed nature of the TWMZM, this is much smaller compared to an amplitude drop in electrical eyes (Fig. 5.23). Despite of amplitude drop, the optical eye height is increased with equalization. For instance, at the same data rate, the eye height is increased from 0.6 mW to 1.2 and 1.15 mV with linear and limiting equalization, respectively. The RMS noise decreases by 19% with linear equalization and 16% with limiting equalization, owing to faster rising and falling edges. The BER is estimated from the eye diagrams based on a Gaussian model and is shown in Table 5.4. From the estimated BER, we can see the TWMZM without equalization can only achieve error free operation at 40 Gb/s ($BER < 1 \times 10^{-15}$). In contrast, with equalization, error free operation is achieved even at 60 Gb/s ($BER < 1 \times 10^{-15}$) with wide open eye. The improvement in transmitter performance is evident with the feedback equalization.

5.4 Discussion

The feedback equalized TWMZM can be implemented monolithically, wherein Si photonic and electronic devices are fabricated on the same chip in a single process. A 130 nm SiGe process is compatible with photonic devices and the SiGe layer can be used to fabricate Ge waveguide PDs. For instance, monolithic integration of photonic and electronic devices has been demonstrated in [38, 99]. Meanwhile, they can also be fabricated separately using a Si photonic process

Parameters	40 G			50 G			60 G		
	No Eq	Linear Eq	Limiting Eq	No Eq	Linear Eq	Limiting Eq	No Eq	Linear Eq	Limiting Eq
BER	$< 1 \times 10^{-15}$	$< 1 \times 10^{-15}$	$< 1 \times 10^{-15}$	1.8×10^{-10}	$< 1 \times 10^{-15}$	$< 1 \times 10^{-15}$	3.25×10^{-7}	$< 1 \times 10^{-15}$	$< 1 \times 10^{-15}$
Height (mW)	1.13	1.21	1.21	0.86	1.19	1.17	0.6	1.2	1.15
Amplitude (mW)	1.73	1.62	1.6	1.62	1.56	1.56	1.51	1.47	1.49
ER (dB)	4.886	4.461	4.45	4.545	4.305	4.365	4.199	4.038	4.087
Opening factor	0.8841	0.9155	0.919	0.8427	0.921	0.9154	0.799	0.938	0.9234
Width (ps)	20.47	23.24	23.5	15.42	16.74	17.8	11.72	12.8	12.69
R_{rms}^* (ps)	0.7798	0.3084	0.3174	0.7959	0.5352	0.341	0.8101	0.6537	0.6804
D_{pp}^{**} (ps)	4.506	1.873	2.346	4.889	3.243	2.201	4.808	3.182	3.055
rise time (ps)	15.03	12.23	12.14	12.29	11.11	10.58	10.49	9.92	9.409
fall time (ps)	11.63	11.34	11.07	10.02	6.124	5.967	8.725	5.52	8.845

Table 5.4: Optical eye quality comparison between non equalized, linear and limiting equalized TWMZM at 40, 50 and 60 Gb/s data rates. *: RMS jitter; **: peak to peak jitter.

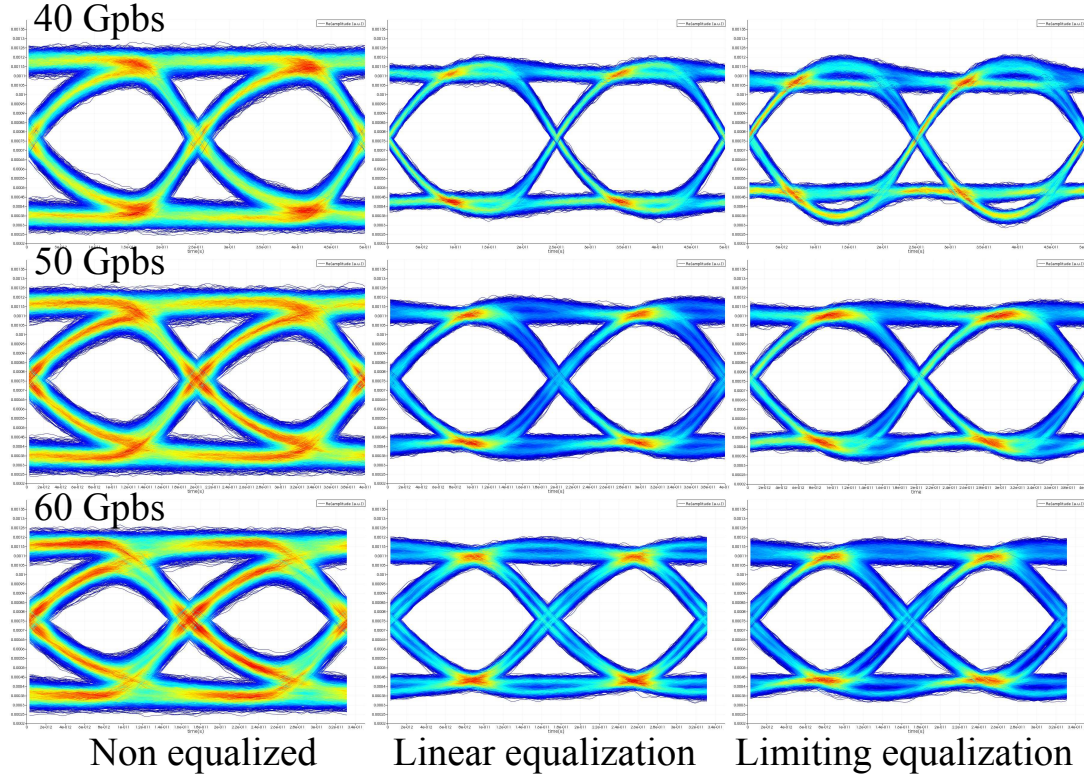


Figure 5.25: Simulated optical eye diagrams at the TWMZM output at 3 data rates: 40, 50, 60 Gb/s for non-equalized (left), linear equalized (middle) and limiting equalized (right) TWMZM.

and a BiCMOS process as exemplified in this work. Then the two chips can be flip-chip bonded together. Thanks to advances in packaging techniques, such as copper pillar bumps [124], the two chips can be bonded together with very small parasitics and pitch size.

For the TWE design, a meandering waveguide and transmission line layout is utilized to reduce the length of signal lines in the feedback circuit. One issue to be considered is the extra loss at turning sections, for both electrodes and waveguides. Using a small radius of curvature saves both Si photonic chip area as well as BiCMOS chip area, but it results in more losses. It should be noted that the optical waveguide loss is relatively small for strip waveguide because of the well-confined optical mode. Waveguide bends with a few micro meter

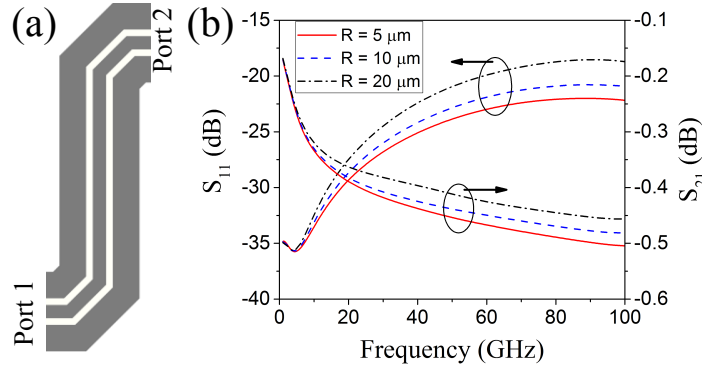


Figure 5.26: EM simulation of the S-shaped transmission line. (a) Image of the simulated S-shaped transmission line. (b) Simulated S-parameter of the transmission line at different radii of curvature.

radius of curvature has demonstrated negligible optical loss. In contrast, for the transmission line electrode, the loss may be large if the radius of curvature is small.

To investigate how the electrode radius of curvature affects the RF loss, the S-parameters of an S-shaped transmission line is simulated and plotted in Fig. 5.26. The transmission line is built with the same metal layer in the same technology as the TWE design. The impedance is matched to 50Ω . For a 3-stage equalized TWMZM, there will be 6 sections of these meandering lines in each arm. Three different radii of curvature are compared: 5, 10, and $20 \mu\text{m}$. As shown in the figure, the S_{21} drops with decreasing radius of curvature. At 30 GHz, the S_{21} is -0.38 dB for $20 \mu\text{m}$ radius and decreases to -0.41 dB at $10 \mu\text{m}$ and to -0.42 dB at $5 \mu\text{m}$. This value will have to be multiplied by 6 in a 3-stage equalized TWMZM. That is 0.18 dB increase in RF loss at 30 GHz if the radius decreases from 20 to $10 \mu\text{m}$, and 0.24 dB increase in RF loss if decreased to $5 \mu\text{m}$. The reflection (S_{11}) increases with decreasing radius of curvature. At 30 GHz, S_{11} increases by almost 3 dB when the radius of curvature decreases from 20 to $5 \mu\text{m}$. From the chip area perspective, if the radius decreases from 20 to $10 \mu\text{m}$, the area saved is almost 50%.

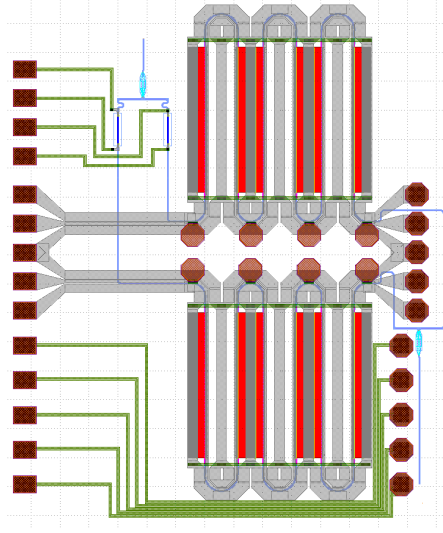


Figure 5.27: Layout of the TWMZM. The TWE and the waveguide underneath are in meandering shape to minimize the signal path in electrical circuit.

5.5 Prototype Fabrication

The TWMZM is fabricated using IMEC 50G Si photonic technology. The layout is shown in Fig. 5.27. The TWE and the waveguide underneath are in meandering shape. To minimize the optical loss, the waveguide turning sections use strip waveguide. Therefore, no doping is implemented in the turning sections. The electrical chip will be flip-chip bonded with the photonic chip. The octagonal pads are flip-chip bondpads. The total chip area is $1.5 \times 1.8 \text{ mm}^2$.

Chapter 6

Discussions and Conclusions

In this thesis, the design of high performance EPICs for optical interconnect applications has been discussed. Firstly, we discussed the design methodology: a multi-physics EPIC design flow. Then two EPIC designs have been presented with focusing on electro-optical interface: a multi-functional EPIC and a feedback equalized TWMZM. In this Chapter, possible improvements to the current designs are discussed and the thesis is summarized.

6.1 Discussions

First of all, in our multi function EPIC design, the on-chip delay line can only generate fixed delay. Although, active transmission line [125] has been proposed but the design is very complicated and it is not compatible with silicon photonic process. Moreover, the additional on-chip delay line causes extra signal loss which is not controlled. Conversely, a delay line implemented using active circuits in a CMOS or SiGe process can achieve controlled, tunable time delay and better control of signal loss. Variety of digital and analog blocks can be used to implement time delay [77–80]. Meanwhile, the attenuation can be easily achieved by a resistor network on a CMOS or SiGe chip with tunable attenuation and large bandwidth. The dc voltage shift by the resistive network is not a concern here if the signal is

AC-coupled to photonic devices. To minimize the delay and loss from packaging, flip-chip bonding can be used. Secondly, as mentioned previously, the nonlinearity of phase shifter $V_\pi L_\pi$ vs. V_R curve causes performance penalty, like inducing frequency spurs when it is used as a frequency doubler. The root cause of this nonlinearity is that in a traditional design, as the bias voltage increases, the boundary of depletion region moves further away from the center of waveguide. Then the interaction between optical mode and free carriers is weakened. To illustrate this point, we use a simplified 1D model (Fig. 6.1-a), assuming the electrical field intensity $|E(W)|^2$ follows a Gaussian distribution with 0 mean. With reverse bias voltage changes ΔV , the changes in depletion region is ΔW and the modulation efficiency is proportional to $\Delta W \cdot |E(W)|^2$, where $W \approx \sqrt{V_R + V_B}$ with V_R being the reverse bias voltage and V_B being the built-in potential of this junction. As we can see, when V_R increases, both $|E(W)|^2$ and ΔW decrease, causing the modulation efficiency to drop (Fig. 6.1-a). To solve this problem, we can move the junction to the left and leave the entire rib region p-type doped. The n-type region can be doped much heavier than p-type region so that most of the depletion region resides inside the rib (Fig. 6.1-b). Therefore, as V_R increases, instead of moving away from the center, the boundary moves towards the center of waveguide and the increase in $|E(W)|^2$ will counteract to the decrease in ΔW , leading to more or less constant $\Delta W \cdot |E(W)|^2$ and thus constant modulation efficiency. The phase shifter with constant-modulation efficiency will find a lot of applications in addition to our multi function EPIC. One example is signal generation and frequency multiplication in microwave photonics.

For our equalized TWMZM design, a few improvements can also be made. Firstly, we can make the equalization adaptive by using tunable feedback gain. In the linear equalizer, the gain is proportional to the ratio of characteristic impedance and degeneration resistor. By adding a tunable resistor in parallel with the degeneration resistor, the gain can be changed. To maintain the same bias

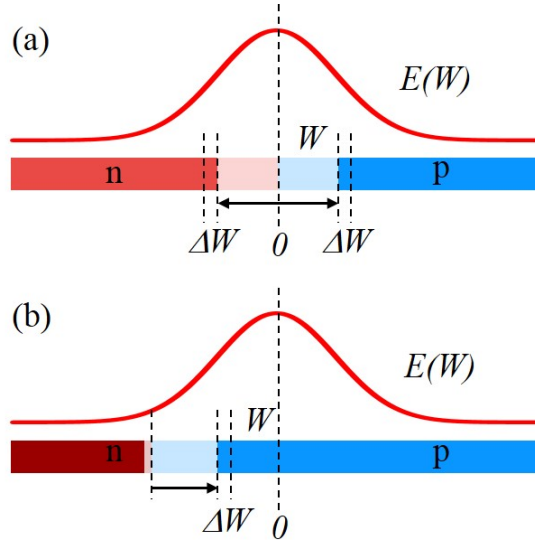


Figure 6.1: (a) Simplified 1D mode of a rib waveguide embedded with PN junction. The Gaussian curve shows the electrical field intensity distribution within the waveguide. The light color region at the middle is depletion region. As V_R increases, the boundary of depletion region moves away from center indicated by the arrow. (b) The junction was moved to the left. As V_R increases, the boundary of depletion region moves into the center indicated by the arrow.

condition, the parallel resistor must not drain any current. On the other hand, in the limiting equalizer, the gain is controlled by the tail current source. Therefore, the gain can be directly tuned by varying the tail current. However, this will also cause the GD to change. To solve this problem, we can keep the source current intact, but bifurcate the current flow in the pull-up circuit by adding a tunable current source in parallel with the load (Fig. 6.2-a). By varying the current of this tunable current source, we can adjust the feedback gain. Secondly, other than the cascade topology with one feedback path, we can use another topology that has multiple feedback paths as shown in Fig. 6.2-b. In the time domain, it works like to a DFE wherein each feedback path is used to cancel one post-cursor ISI. At the same time, in the frequency domain, multiple resonances could occur to form a frequency response that is exactly opposite to the frequency roll-off of a loaded transmission line and result in a flat response over a large frequency range.

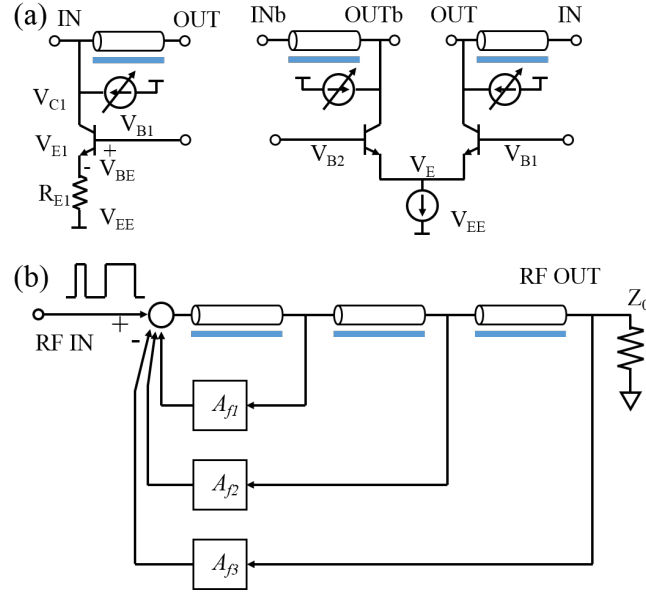


Figure 6.2: (a) Adding a tunable current source to the pull-up in linear (left) and limiting (right) feedback circuits to adjust the feedback gain. (b) New topology of feedback equalization using one stage with multiple feedback paths.

The benefit of this design is that it is more adaptive than the one-feedback path design and it is potentially capable of exact canceling of high frequency roll-off in a loaded transmission line.

6.2 Conclusions

To summarize, in this thesis, we firstly present an integrated, multi-physics design flow for EPIC in Chapter 3, which seamlessly interfaces electronics and photonics. The loading effect of photonic devices on the electrical driving signal and the effect of attenuated driving signal on the optical signal have been taken care of in this design flow. The flow is illustrated in the example design of a TWMZM based on a reverse biased PN junction. The design of each individual block has been discussed. Compact circuit modes are extracted for all the photonic and electronic components. System level simulations using both electrical circuit

and photonic circuit simulators are demonstrated.

With this design flow as the design methodology, a new multi function EPIC has been designed in Chapter 4, which exploits the new operation mode of a MZM in the time domain. The EPIC is capable of incorporating pre-emphasis function, obviating an electrical pre-emphasis driver. In addition, it can work as an optical pulse generator with frequency multiplication. The simulated EO performance verifies the functionality of the proposed circuit and demonstrates its capability of operating at a data rate beyond the modulator and detector limit. What's more, we have demonstrated its function as an optical pulse generator and its application in data communication using RZ data format. Finally, we propose an all passive electrical circuit design for complete on chip implementation. A prototype EPIC is fabricated using IMEC 50G silicon photonic technology. With its simplicity and versatility, the proposed circuit will play an important role in future EPIC system design.

As an example of electronic-photonic co-design, a high speed TWMZM with built-in feedback equalization is designed in Chapter 5. The new equalization techniques take advantage of the natural time delay in a TWE to form a built-in IIR filter. Compared to the FIR equalization, it boosts the high frequency component instead of just suppressing the low frequency component. Therefore, it produces a much higher ER. In addition, it equalizes the GD vs. frequency curve and leads to GDV reduction. Moreover, it allows cascaded design so that the required feedback gain is reduced, which not only benefits the ER, but also minimizes the effect of feedback on reflection and stability. Two feedback circuits (linear and limiting) are proposed and their performances are analyzed in terms of gain, bandwidth, noise, and stability. A 3-stage cascaded TWMZM with feedback equalization is designed. The photonic circuit is designed based on IMEC 50G Si photonic process and electrical circuit is designed based on GlobalFoundries 0.13 μm SiGe process. The design is modeled using our EPIC design flow and

system level simulation results show that the linear equalized TWE has a 3-dB bandwidth more than 2 times as high as that of the original TWE. The EO 3-dB bandwidth is improved by 65%. Transient simulation results show the equalization has improved the data rate from 40 Gb/s to 60 Gb/s. The simulated total power consumption is 190 mW for linear equalization and 171 mW for limiting equalization, in a 3-stage cascaded transmitter. The prototype photonic chip is fabricated using IMEC 50G Si photonic technology.

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Chapter A

Silicon Photodetectors

A.1 Background and Review

The photodetector (PD) is one of the key component in many photonic systems, such as optical interconnects and image sensors. It converts the optical power into electrical current through the photo-generation of free electron hole pairs (EHPs). The absorption coefficient α of the PD material is one of the most important parameters. The absorption coefficients as a function of the wavelength for some commonly used materials are plotted in Fig. A.1. This coefficient is directly related to the bandgap of the semiconductor material in that the energy of the photons incident on the material must be higher than its bandgap energy. As a consequence, many materials exhibits a cut-off wavelength in their absorption coefficient versus wavelength curve, beyond which the absorption coefficient drops significantly. For example, the absorption coefficient of GaAs and InP have sharp drops right after 960 and 850 nm due to their bandgap being around 1.34 and 1.43 eV, respectively (Fig. A.1). This cut-off is not that sharp for semiconductor with indirect bandgap, like silicon and germanium, because the phonon is usually involved in the photon absorption to conserve the momentum and the probability of the photon absorption is associated with the availability of a phonon. As shown in Fig. A.1, silicon has very low absorption coefficient for any wavelength

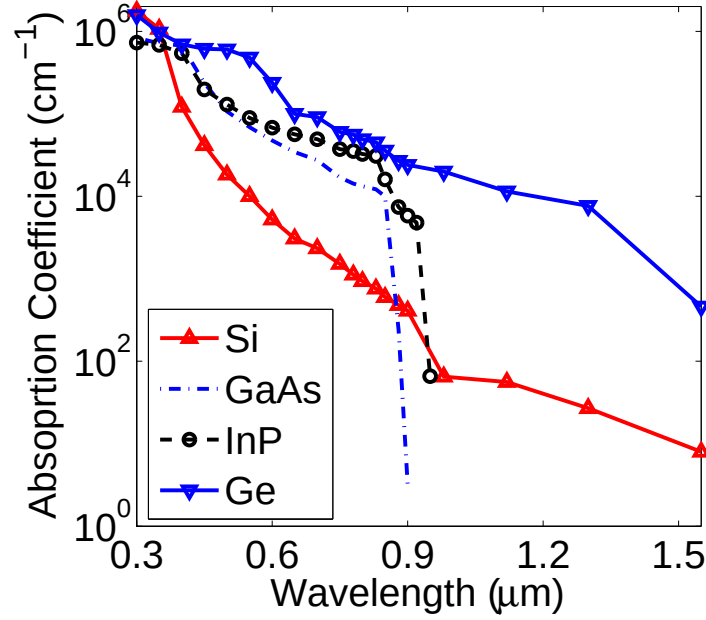


Figure A.1: Optical absorption coefficients of silicon, GaAs, InP and germanium with respect to wavelength.

beyond $1.11 \mu\text{m}$. Therefore, it cannot be used for photodetection in telecommunication wavelengths, which are around $1.3 \mu\text{m}$ and $1.5 \mu\text{m}$. Other materials (*e.g.*, germanium) are normally combined with silicon to increase the absorption. Nevertheless, the silicon has reasonably good absorption coefficient around visible range (400-700 nm) and near infrared light (700-1000 nm), which makes it good candidate for photodetection in imaging applications.

The light penetrated into the material has its intensity decrease exponentially with the penetration distance (x):

$$I = I_0 \exp(-\alpha x) \quad (\text{A.1})$$

where I_0 is the light intensity at $x = 0$. Most of absorption occurs over a distance $1/\alpha$, where intensity drops to $1/e$. This is called penetration depth δ . For silicon at 400 and 850-nm, the penetration depth is 0.1 and $16.5 \mu\text{m}$. So most of blue light will be absorbed by a few hundreds of nanometer silicon layer while infrared

light needs 10s of micrometer to have most of the power absorbed. The amount of absorption leads to another important concept in PD, quantum efficiency (QE), which is defined as the ratio between the number of photo-generated EHPs to the number of incident photons. Since one photon will only generated one EHP, the QE is directly related to how many photons being absorbed. An analytic expression for QE in terms of measurable quantities is given by:

$$\eta = \frac{I_{ph}/q}{P_0/hv} \quad (\text{A.2})$$

where I_{ph} is the photocurrent measured by an external circuit, P_0 is incident power, q is the charge of an electron, h is plank constant and v is the frequency of the incident light. The ratio of I_{ph}/P_0 gives rise to another important parameter, responsivity R , which is the primary parameter characterizing a PD. It is used to find out how much photo-current can be produced for a specified power at certain wavelength. The responsivity related to the QE by: $R = \eta \frac{q\lambda}{hc}$, where λ is the wavelength and c is the speed of light. The QE can be very high depends on the absorption coefficient and the thickness of the absorbing material. Fig. A.2-a shows the QE of silicon as a function of the wavelength at two different thickness values. It shows that the QE of a well designed silicon PD in the wavelength range of 400-600 nm can be close to 100%. For $\eta = 1$, the responsivity R is directly proportional to λ . However, the QE will finally drop because of the absorption decreases with increasing wavelength. Therefore, it is expected that the responsivity will increase with the wavelength initially and then drop as the QE decreases. Fig. A.2-b shows the responsivity as a function of wavelength for two different thickness values. For thicker silicon, the QE and responsivity are generally higher.

Another important parameter of the material is the carrier mobility, which determines how fast charge carriers can travel. Photo-generated charge carriers will travel inside the semiconductor material to the collector and be detected by

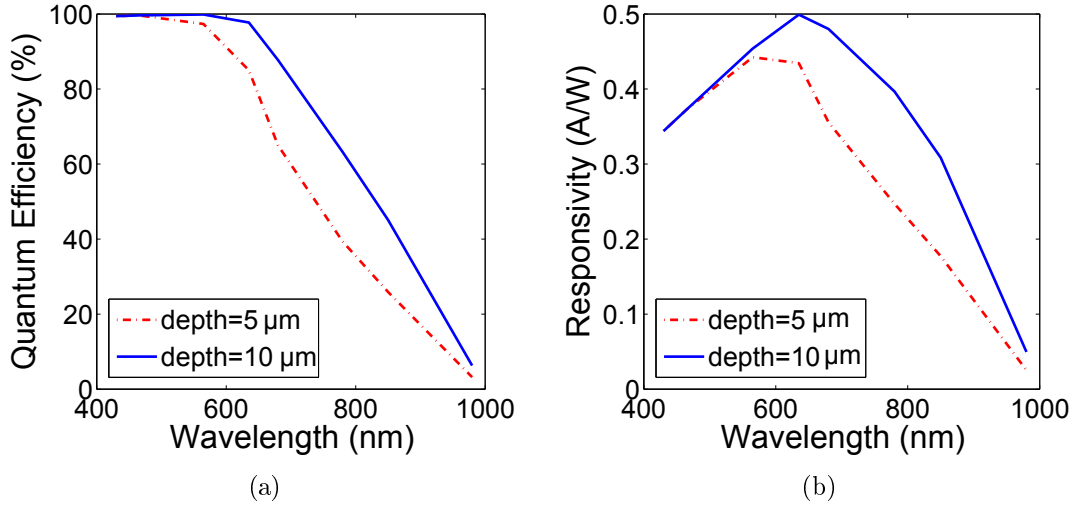


Figure A.2: (a) Maximum achievable QE and (b) responsivity for 5- and 10- μm thick silicon with respect to wavelength.

an external circuit. Without electrical field, charge carriers will just diffuse slowly in all directions, but collectively they diffuse from the regions of high carrier concentration to the regions of low carrier concentration. A diffusion length is the mean distance a carrier travels from point of generation until it recombines. It is an inverse function of the temperature and doping concentration. On the other hand, at the presence of electrical field, the velocities of the electron and hole are characterized as: $\mathbf{v}_e = \mu_e \mathbf{E}$ and $\mathbf{v}_h = \mu_h \mathbf{E}$, respectively, where μ_e and μ_h are the mobilities of electrons and holes in the semiconductor material, and $\mathbf{E}$ is the electrical field. The drift velocity is much faster than the diffuse velocity. Under high electrical field, the velocities for both electrons and holes saturate. The saturation velocities for both electrons and holes in silicon are around $1 \times 10^5 \text{ m/s}$ at electrical field intensity of $2 \times 10^6 \text{ V/m}$ and $2 \times 10^7 \text{ V/m}$, respectively. The difference is due to relatively low mobility for holes. The carrier transit time is the time taken for the photo-generated carriers to travel to collectors. This time is directly proportional to the thickness of the absorbing layer and inversely proportional to the drift velocity. For fast response, this time must be minimized

by reducing the thickness of the absorbing layer and maximizing the drift velocity. It is worth to mention that the operation speed of a PD is also strongly affected by the load R_L and the parasitic capacitance of the PD C_{PD} , where R_L is the input impedance of the amplifier right after the PD,. It imposes an extrinsic RC time constant on the frequency response of the devices.

A.1.1 Common Photodetector Structures

Many types of PDs have been developed. Most commonly used PDs utilize p-n junction or other variances (*e.g.* p-i-n junction). Internal gain can be achieved through engineering of junction doping profile. This type of devices with internal gain is called avalanche PD (APD). Each of these devices has different design parameters and performance, which make them suitable for a variety of applications.

A PN junction forms the simplest structure of a PD as shown in Fig. A.3-a. There is a depletion region inside the junction which is devoid of free carriers. As being discussed in previous chapter, the width of the depletion region ($w_n + w_p$) is a function of the reverse bias voltage plus the built-in potential $V_R + V_B$. It has an electrical field opposing the diffusing of electrons (holes) from the n-side (p-side) to the p-side (n-side). The electrical field intensity can be obtained by integrating the net space charge across this region. Under this electrical field, photo-generated EHPs can be separated and drift towards the electrodes on the two sides and be collected. This design has many drawbacks. First of all, as we can see from Fig. A.3-a, the electrical field is not uniform inside the depletion region, resulting in long transit time for the photo-generated carriers. Second, the depletion region width is small. As a consequence, the majority of photons are absorbed outside the depletion region where there is no electrical to separate the EHPs and drift them. Most of these free charge carriers recombine with each other and doesn't contributed to the photo-current and the rest diffuses slowly into the electrode and slows down the operation speed of the devices. Lastly, its junction

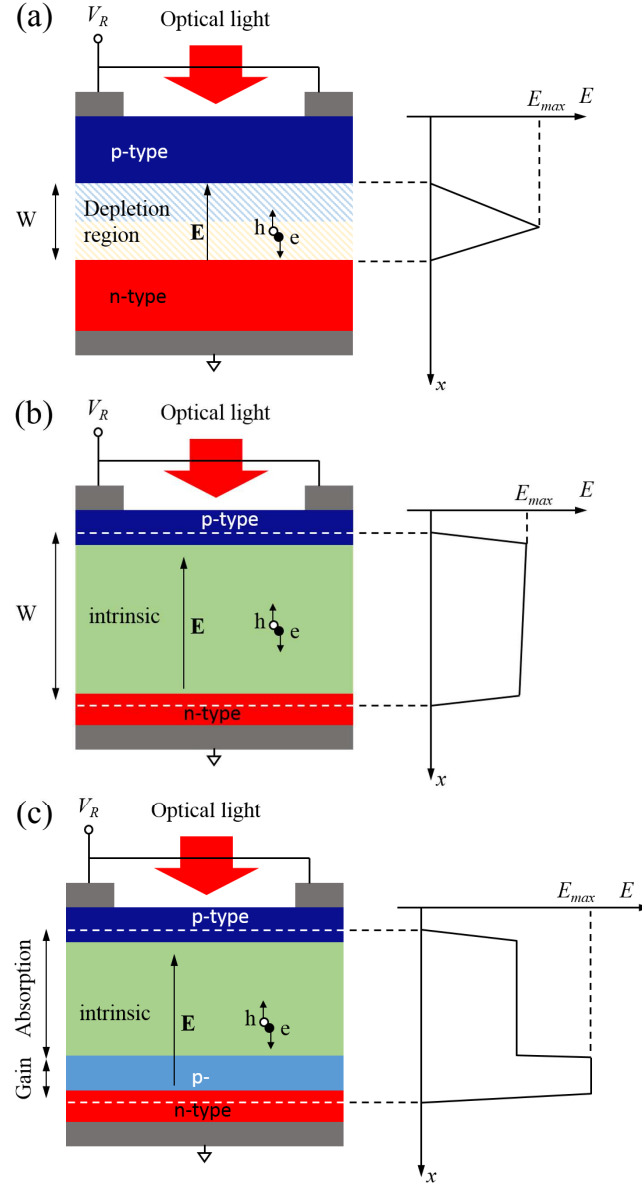


Figure A.3: Structures and electrical fields of (a) a p-n PD, (b) a p-i-n PD, and (c) an APD. V_R is the reverse bias voltage. W is the width of the depletion region. Photogenerated electrons and holes are represented with open and close circles. Arrows indicate drift/diffuse directions.

capacitance is large, which results in a large RC time constant that limits the operation speed. These problems can be substantially reduced in p-i-n junction devices, wherein an intrinsic layer is inserted into the heavily doped p and n region (Fig. A.3-b). To avoid the minority carrier recombination, the p and n region are usually made very thin. The whole intrinsic region can be depleted and a strong and uniform electrical field is present in the intrinsic region. Therefore, most of the photons can be absorbed in the depletion region and be sequentially separated and drifted quickly by the electrical field. In addition, the junction capacitance is much smaller than that of a PN junction. The capacitance can be approximated by a parallel plate capacitor:

$$C_{pin} = \frac{\epsilon_0 \epsilon_r A}{w_i} \quad (\text{A.3})$$

where w_i is the width of the intrinsic layer and A is the cross-sectional area. The width of intrinsic layer is an important design parameter which determines the operation speed and the QE of the devices. Large w_i is favored in terms of parasitic capacitance and QE, but it leads to long drift time and large parasitic series resistance. It should be noted that a practical device has lightly doped intrinsic region. Therefore, the electrical field is not exactly uniform, but has a small slope as shown in Fig. A.3-b.

The maximum responsivity of a silicon PIN PD at 850 nm is about 0.68 A/W, given 100% quantum efficiency. However, practical devices can never reach this value because of the large penetration depth of silicon (16.5 μm) at this wavelength. An APD solves this problem by providing an internal gain through an avalanche of impact ionization process, wherein the primary free carriers are accelerated by the electrical field to such a high velocity that they can impact-ionize some of the silicon valance bonds and generate secondary EHPs. These EHPs can also be accelerated and generate more EHPs. As a result, one free charge carrier can generate a large number of EHPs, all of which contribute to the photocurrent. The structure of the APD is shown in Fig. A.3-c. A moderate-heavily

p-type doped layer is insert between the n-type and the intrinsic region. This p-type layer together with heavily doped n-type layer produces very high electrical field under high reverse bias voltage and free carriers drifted into this high field region gain enough kinetic energy to generate secondary EHPs. Both electrons and holes is capable of generating secondary EHPs and the generation rate is governed the impact ionization coefficient, α_h and α_e , for electrons and holes respectively. In silicon, impact ionization coefficients are about 10^4 cm^{-1} at an electrical field intensity around $2 \times 10^5 \text{ V/cm}$ for electrons and more than $5 \times 10^5 \text{ V/m}$ for holes [101].

A.1.2 Photodetectors on CMOS

PDs have been integrated on CMOS technology since early 1970s for imaging applications. CMOS imager based on PDs on CMOS has the advantages of seamless integration of photo sensing devices to electrical circuits. In an active pixel sensor, for instance, the read-out and reset circuits are implemented side-by-side to PDs, which enables the charge accumulated by a PD to be converted into voltage inside the pixel. Each of the pixels can be directly addressed by two crossing signal lines (x, y), like in the memory chip. This significantly eases the imager design and improves the performance. Different from the conventional imaging applications, the emerging field of 3D imaging requires high speed and high sensitivity optical pulse detection for large range and high resolution distance measurement. For example, for a direct TOF measurement with optical pulse, the single shot accuracy is related to the receiver bandwidth by [126]:

$$\sigma_R \approx \frac{0.35 \times c}{2 \times f_{-3dB} \times SNR} \quad (\text{A.4})$$

where f_{-3dB} is the 3-dB bandwidth of the receiver, SNR is the peak signal voltage to rms noise voltage. Two ways to improve the accuracy are: increasing the

bandwidth, and the sensitivity (responsivity) of the PD (since most of the noise comes from electrical circuits).

Thanks to its internal gain, APDs can be used to detect weak light signal which is highly desirable for the 3D image sensing application. However, the standard CMOS process is optimized for microelectronics. It has very limited number of doping layers can be used for PD fabrication. The complex doping profile in conventional APD (Fig. A.3-c) is not available in a standard CMOS process. In a typical p-type substrate, two types of structures are typically used to make APDs in CMOS: p+/NW/p-sub and n+/PW/DNW/p-sub. These two structures take advantages of high electrical field at either p+/NW or n+/PW junction to achieve avalanche impact ionization. The schematics of these two structures are shown in Fig. A.4. In the first type of APD structure, the NW is almost completely depleted at high bias voltage. The photo-generated electrons are swept towards the n+ contact and collected. Meanwhile, the photo-generated holes drift towards the p+ contact, going through the p+/NW junction where they will trigger avalanche impact ionization due to the strong electrical field in this region. Since the avalanche impact ionization is caused by the injected holes only, this type of APD is called hole-injection type. The output photocurrent (I_{ph1}) is taken from the p+ contact inside the NW. So only the holes collected at this p+ contact counts. In the NW, besides the electrons from the p+/NW junction region, there are also electrons coming from the NW/p-type junction or even the p-type substrate. Consequently, the photocurrent at the n+ contact (I_{ph2}) is larger than the photocurrent at the p+ contact inside the NW ($I_{ph2} > I_{ph1}$). This difference is canceled out by the photocurrent collected at the p+ contact in the substrate (I_{ph3}) which arises from the photogenerated holes from the NW/p-sub junction and the substrate. So we have $I_{ph2} = I_{ph1} + I_{ph3}$. As the Fig. A.4 shows, for this devices, I_{ph1} is free from slow diffusing charge carriers from the substrate because the NW/p-sub junction has a built-in electrical field pointing downwards and blocks

the photo-generated holes in the substrate. Therefore, this structure ensures high speed operation. One variance of this structure is $p^+/PW/DNW/p\text{-sub}$, wherein the PW/DNW junction provides the high field for impact ionization (Fig. A.4-b). The $p^+/PW/DNW/p\text{-sub}$ has relatively thicker multiplication layers, hence larger gain. Due to low doping concentration, this device has to be biased at very high voltage to achieve large electrical field. Inversely, in $n^+/PW/DNW/p\text{-sub}$ structure, n^+/PW junction provides the high electrical field necessary for the avalanche impact ionization. The photo-generated holes are swept towards p^+ contact inside the PW , while the photo-generated electrons go through the high electrical field region and trigger the avalanche impact ionization. This type of APD is called electron-injection type because the injected electrons are responsible for the avalanche process. The photocurrent is taken from the n^+ contact in the PW and it is also free from slow diffusing electrons due to the built-in potential of the PW/DNW , thus the device can be operated at high speed. Noted that in all three structures, shallow trench isolations (STIs) are used to prevent pre-mature breakdown.

As a result of the thin n^+/p^+ layer, the multiplication layers for both $p^+/NW/p\text{-sub}$ and $n^+/PW/DNW/p\text{-sub}$ are usually very thin, which is to its advantage in that there will be only one type of charge carrier triggering the impact ionization, thus leads to good noise performance [127]. The corners of the heavily doped region usually have highest electrical field which can lead to avalanche breakdown at relatively low bias voltage. The avalanche in these corners doesn't multiply the photocurrent and should be avoid. To prevent this kind of pre-mature breakdown at the corner of the n^+/p^+ region, a guard ring structure is normally implemented in an APD, using a moderate-low doping n/p region [128]. In some process technologies, a STI [129] can also be used to effectively prevent early breakdown (Fig. A.4).

Interdigitated figure structure is employed in many designs to reduce the tran-

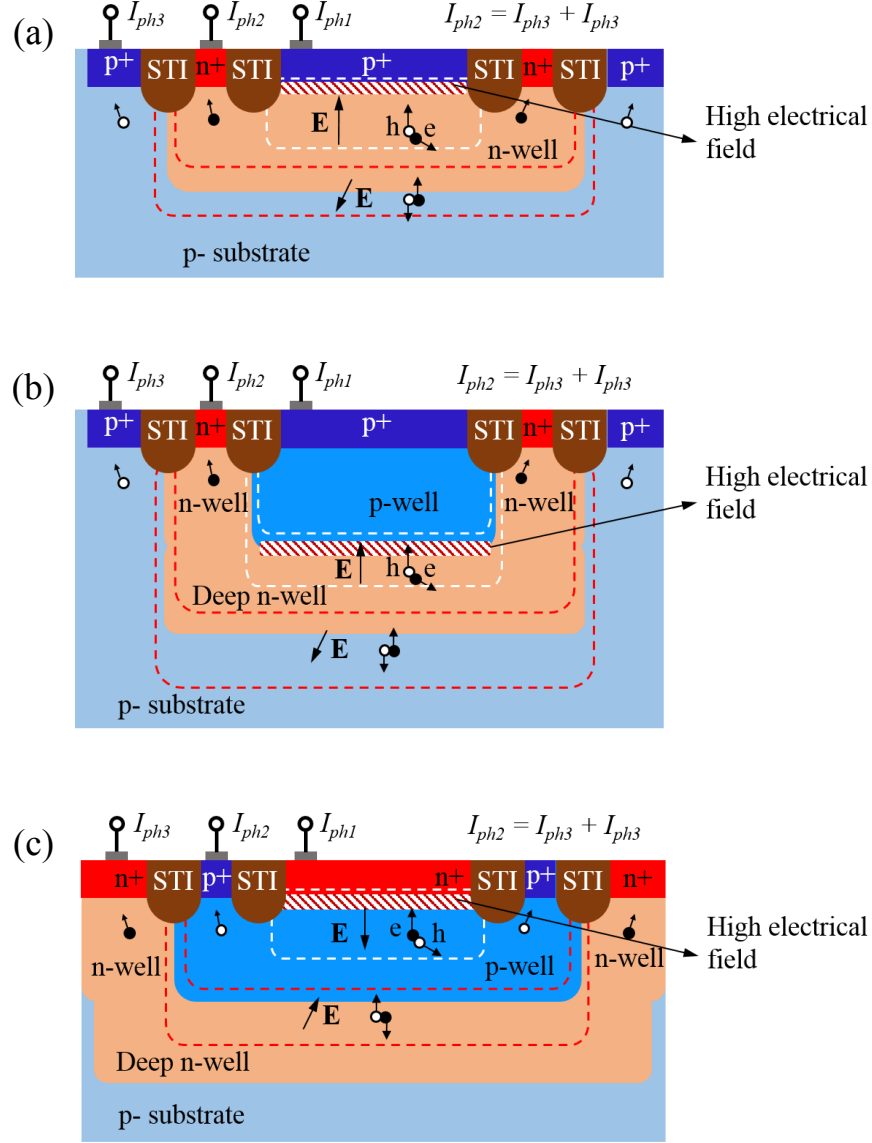


Figure A.4: Structures for APD on CMOS. (a) p+/NW/p-sub, (b) p+/PW/DNW/p-sub, and (c) n+/PW/DNW/p-sub. The regions with high electrical field are marked out. The white dashed lines indicate active depletion regions and the red dashed lines indicate depletion region for diffusing carriers shielding. STI is used to prevent early breakdown at the edges of n+/p+ regions.

sit time for the EHPs but it compromises the filling factor and increases the parasitic capacitance. The spacing between the figures is an important parameter in this type of design which is basically a tradeoff between parasitic capacitance and the transit time. During the past decades, we have seen significant progress on the performance CMOS APDs. For example, in [130], both type of APDs were fabricated by using a standard 0.18- μm CMOS process, achieving 90 GHz and 180 GHz GBP for hole-injection type and electron-injection type APDs, respectively at 850-nm wavelength. In another, Lee *et al* [131] reported a record high GBP of 1 THz using the n+/PW/DNW/p-sub structure at 850-nm wavelength. Such high GBP is attributed to the high frequency peaking observed at avalanche region. The maximum bandwidth of 3.2 GHz was obtained. CMOS APDs on advanced technology node have been reported in [132]. Two devices based on a PW/DNW/p-sub and a p+/NW/p-sub structures was fabricated on a 40-nm standard CMOS technology but only the device based on p+/NW/p-sub structure works in avalanche mode due to pre-mature breakdown in the other device. A high responsivity CMOS APD was reported in [133] to have $2.7 \times 10^4 \text{ A/W}$ under 5-nW illumination and 670-nm wavelength using n+/PW/DNW/p-epi structure. This device also achieve high gain-responsivity product of 17.4-GHz·A/W at 850-nm. Low noise operation which is essential for detecting weak light was also achieved by using CMOS APD. In [134], Pancheri *et al.* has demonstrated a low noise APD using p+/PW/DNW/p-sub. This structure utilized an undoped p-sub region as a guardring to prevent pre-mature breakdown. A recorded low excessive noise factor of $F = 6$ was reported for this devices at $M = 50$ and $\lambda = 380 \text{ nm}$ and $F = 12$ for $M = 50$ and $\lambda = 850 \text{ nm}$. The methods for optimization of the device design were also developed. Shimotori *et al.* [135] compared the bandwidth of the CMOS APDs with different interdigitated figure spacing and demonstrated that an optimized figure spacing exists for maximum GBP. In [136], Lee *et al.* studied the power dependence of the gain, noise and bandwidth characteristics of

a CMOS APD based on a p+/NW/p-sub structure under 850-nm illumination. It showed that when optical power decreases, the excessive noise factor increases and the bandwidth decreases. In another work by Chou *et al.* [137], the biasing of the DNW was shown to improve the performance of a n+/PW/DNW/p-sub structure based CMOS APD. The devices achieved highest bandwidth of 8.7 GHz and a large gain-bandwidth product of 542 GHz under 850-nm illumination.

A.2 High Performance CMOS Avalanche Photodiode

APDs fabricated in CMOS technologies are seeing continued progress, owing to their unique advantages of high sensitivity, low cost, compatibility with silicon photonics and electronics. In this chapter, I will discuss the design and implementation of a high performance CMOS APD by using a standard 130 nm CMOS technology without any process modification. The prototype device achieves maximum avalanche gain of more than 10^4 and maximum 3-dB bandwidth about 2.4 GHz. The maximum GBP is 176 GHz. In addition, we demonstrated through the detailed simulation that by controlling the NW/p-sub junction bias, further gain-bandwidth improvement can be achieved.

A.2.1 Photodiode Design

The schematic in Fig. A.5 shows one side of the APD. It utilized the p+/NW/p-sub structure because of its simplicity and reliability. Interdigitized finger structure is employed to reduce the transit time of the free charge carriers and improve the bandwidth. The NW contact is the anode and the p+ in the NW is the cathode. STI is utilized to avoid pre-mature breakdown at the corners of p+ and n+ regions [129].

The devices are simulated using a finite-element simulator, Sentaurus Device

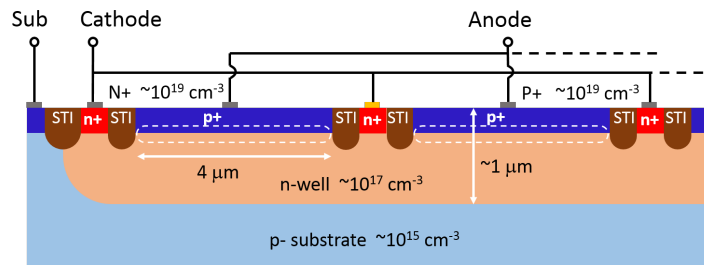


Figure A.5: Cross-section view of the APD (not to scale) showing one side of the device. The region enclosed by dashed line is avalanche region. Doping concentrations are estimated from the sheet resistance of each region. STI is utilized to prevent early break down at the edges of highly doped regions.

for design parameter optimization and better understanding of device operation mechanism. The electric field, electron/hole current, and small signal frequency response under 850 nm illumination are simulated for reverse bias voltages up to 12 V. The total thickness of the device simulated is 36 μm , which is more than twice of the penetration depth (16.5 μm) of silicon at 850 nm wavelength. In the small signal frequency response simulation, 50 Ω resistive load is connected to the cathode of the device to emulate the 50 Ω characteristic impedance of RF measurement instruments. The breakdown voltage of the simulated device is 11 V and the simulated responsivity is 37 mA/W at 0.1 V reverse bias voltage. Fig. A.6 shows the electric field distribution in the simulated device at 10 V reverse bias voltage ($V_R = -10\text{V}$). The photocurrent is taken from the anode and both cathode and the substrate contacts are connected to ground. As we can see, the strongest electric field is found at the p+/NW junction where the avalanche impact ionization process occurs. And in the NW, the electric field is very high so that the photo-generated EHPs can be quickly swept towards the anode or cathode. In addition, there is another region of high electric field is located around the NW/p-sub junction. As we discussed previously, the electric field in this region has opposite direction (pointing downwards) to the electric field above thus it can block the slow diffusing holes from the substrate. As we move further into the

substrate, the electric field almost vanished, so the photo-generated EHPs in this region will diffuse slowly in all directions. The strong electric field at the corner of the p+ region, which often causes the early break down, is effectively eliminated by the use of the SiO₂ STI.

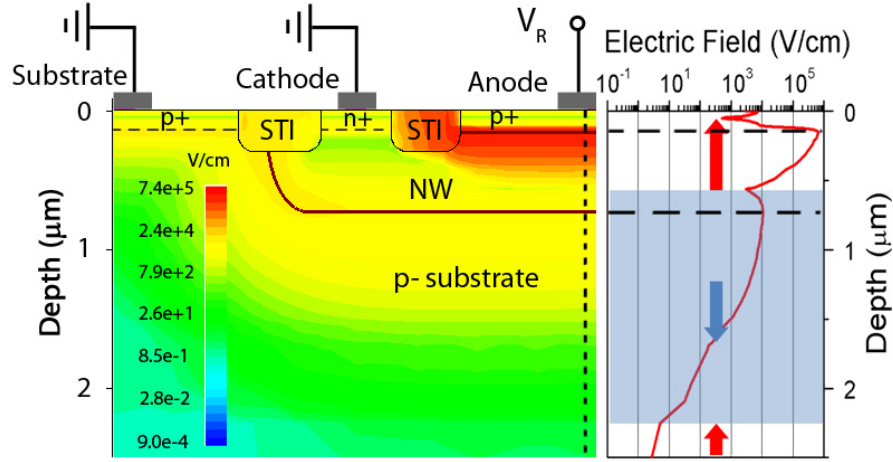


Figure A.6: Simulated electric field (intensity) distribution inside the device. The curve on the right shows the intensity of electric field along the dotted line. The arrows show the direction of electric field, and the region with electric field pointing downwards is marked with shallow.

A.2.2 Prototype Characterization

The prototype device is fabricated using IBM 8hp 130 nm BiCMOS technology without using any bipolar device related layers. There is not any process modification or optical window. As a result, there is metal fill generated automatically by the process over the active region for the fulfillment of the metal density rules. In addition, there are approximately 20 μm dielectric layers above the devices. Therefore, we expected a large portion of the incident light being reflected back by the metal fill pieces and the top dielectric layers. Nevertheless, the avalanche gain and small signal frequency response will not be affected. To facilitate the dark current measurement, a dummy replica with its active region covered with

metal layer is fabricated alongside the real device. The active area of the APD is $70 \times 70 \mu\text{m}^2$ and the best performance one reported here has a finger spacing of $4 \mu\text{m}$, a good trade-off between parasitic capacitance and carrier drift time.

For CMOS APD characterization, a 5 Gb/s, 850 nm VCSEL (Finisar V850-2173-001) is used as an optical source. In all the measurements, the cathode and substrate contacts are grounded and a reverse bias voltage V_R is applied at the anode.

In the DC measurement, the VCSEL is biased at 0.85 mW output. The I-V characteristic curves of the fabricated APDs both with and without illumination are shown in Fig. A.7-a. Well defined breakdown voltage at about 10.8 V (when the dark current reaches $10 \mu\text{A}$) is observed. The dark current is below 1 nA up to 9 V, which is equivalent to $20 \mu\text{A}/\text{cm}^2$. The avalanche gain is determined as the ratio of photocurrent between a given bias voltage and 0.1 V where the avalanche process is insignificant. As shown in Fig. 3b, the gain is almost constant at low bias and increases with the reverse bias exponentially in the avalanche region up to a maximum value of 10^4 at the bias voltage of 10.97 V. This is the highest gain reported at 850 nm wavelength for CMOS APDs and it is 5 times higher than that in [131]. Note that the responsivity of the APDs is not very high at low reverse bias due to reflection from the metal fill and dielectric layers. With a little process modification (remove the metal fill and dielectric layer in the active region), our devices will achieve even higher responsivity.

In the small signal frequency response measurement, the same 850-nm VCSEL is modulated by a 50-GHz vector network analyzer (VNA) and the output of the APD is traced back to the VNA through a bias tee. Prior to the measurement, the whole link is calibrated by using a commercial 12.5 Gb/s InGaAs/InP PIN photodiode (Cosemi LX3052) upto 4 GHz. The measured frequency response of the APD from 45 MHz to 4 GHz at various reverse bias voltages are shown in Fig. A.8a. Interestingly, at bias voltage close to breakdown, a RF peak is

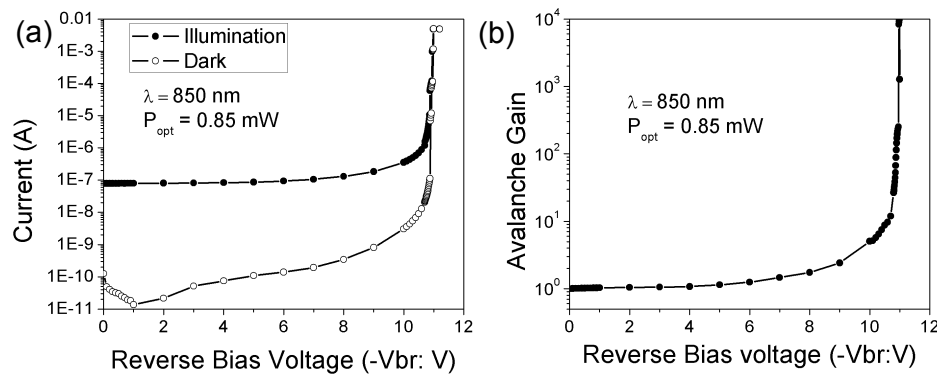


Figure A.7: (a) Measured current-voltage characteristics of the CMOS APD with and without illumination at 850 nm wavelength. (b) Measured avalanche gain of the CMOS APD as a function of reverse bias voltage.

observed on the frequency response curve which enhanced the 3-dB bandwidth of the device. Therefore, the 3-dB bandwidth increases slightly with the bias voltage at the beginning. The maximum bandwidth of 2.4-GHz is achieved at the reverse bias of 10.8 V. The RF peaking at high frequency is widely reported for APD and it is attributed to the oscillation of space charge wave induced by the finite buildup time of the avalanche current [138–140]. However, the peaking gradually disappears as we further increase the reverse bias voltage. The low frequency response surges very quickly as the reverse bias voltage increases, which causes the 3-dB bandwidth to drop rapidly. Similar behavior is also observed on the simulation results, but the breakdown voltage and peaking are slightly different. Fig. A.9 shows the measured and simulated 3-dB bandwidth of the APD as a function of the measured avalanche gain. The gain is obtained as the ratio of the signal power at certain reverse bias voltage to that at 0.1 V and at 45 MHz. According to [141], the GBP reaches a constant value for gain larger than the ratio of ionization coefficient as being reported in other literature [130, 133]. In this figure, the measured data exhibits a region of constant GBP of 176 GHz. However, with further increase of the gain, an abrupt drop of the bandwidth makes it deviate from the constant GBP line. The highest GBP of 176 GHz we measured

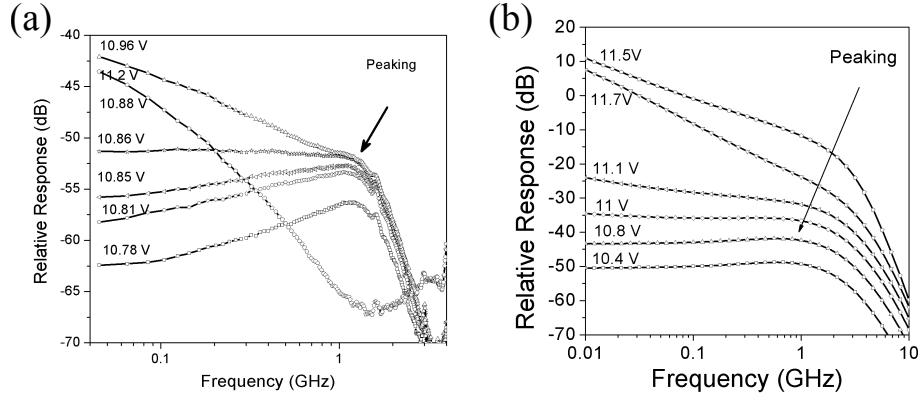


Figure A.8: (a) Measured frequency response of the CMOS APD at 850 nm wavelength at different reverse bias voltages. (b) Simulated frequency response of the CMOS APD at 850 nm wavelength at different reverse bias voltages.

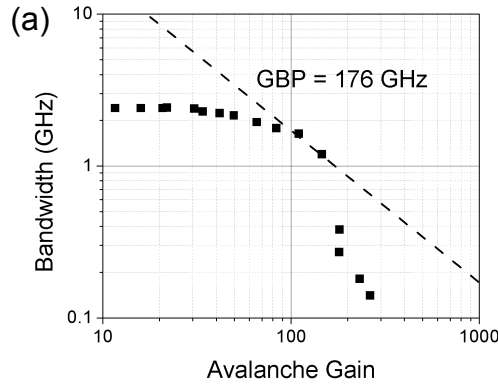


Figure A.9: Measured gain-bandwidth characteristics of the CMOS APD.

is comparable to the GBP achieved by an electron-injection type APD reported by Iiyama *et al.* [130] and much higher than other CMOS APDs reported in the literature [142, 143].

To investigate the peaking in the avalanche region, the input impedance of the CMOS APD is characterized by S_{11} measurement using the 50-GHz VNA. The calibration of the VNA is done to push the reference plane to the tip of the RF probe. The measured S_{11} from 45 MHz to 10 GHz at various bias conditions are plotted in Smith chart as shown in Fig. A.10. As can be seen, at bias voltage below 10.78 V, the impedance of the APD stays purely capacitive as expected.

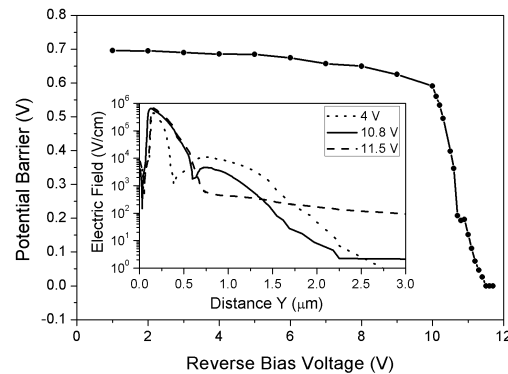


Figure A.11: Simulated potential barrier height for holes in substrate as a function of reverse bias voltage in the simulated device. The inset shows the simulated electric field profiles along the dotted line at three different bias voltages.

the holes in the substrates experiences an abruptly drop after reverse bias of 10 V (from 0.7 V to almost 0 V), which can be attributed to the vanishing barrier region under high reverse bias. This can also be observed from the electric field distribution in the inset of Fig. A.11, which clearly shows both the lowering of the electric field intensity and shrink of the barrier region from reverse bias of 4 V to 10.8 V and the barrier is almost flattened out as the reverse bias increases upto 11.5 V. Meanwhile, it has been noticed that the electric field in the substrate region increases by several orders of magnitude. Therefore, plenty of the photogenerated holes in the substrate overcome the barrier and drift towards the active region. Since the electrical field here is still much smaller than that in the active region, and the carriers have to travel long distance to reach the active region. It leads to large transit time and thus slow response. The hole-current density distributions (intensity) in the simulated device at bias voltage of 10.9 V and 11.1 V (Fig. A.12) clearly demonstrate the increases of the holecurrent. As can be observed, direct channel between substrate and p+ region is formed at bias of 11.1 V. As a result, the photocurrent increases significantly as well as the avalanche gain, as being observed in the measured results.

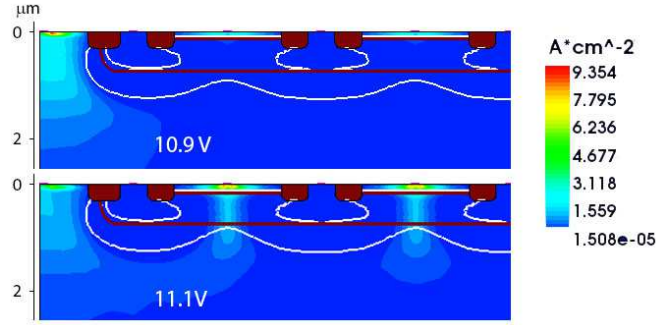


Figure A.12: Simulated hole current density at two different bias voltages.

Therefore, one way to improve the high speed performance of the APD is to control the potential barrier height by applying a reverse bias V_{sub} to the substrate contact. This makes the NW/p-sub junction reverse biased and increases the electrical field inside the junction. To verify this concept, different V_{sub} values are applied on the simulated devices. The 3-dB bandwidth versus avalanche gain curves of the simulated devices at different V_{sub} 's are plotted in Fig. A.13. When $V_{sub} = 0$ V, there is an abrupt drop in the gain bandwidth curve which matches the measured results in Fig. A.9. On the other hand, when the V_{sub} increases, the abrupt drop in the bandwidth can be avoided and GBP can be further improved. The simulation results here show 4 times increase in maximum GBP with a substrate bias voltage of $V_{sub} = -0.4$ V. The circled data points in Fig. A.13 have the same p+/NW junction reverse bias voltage. Note that the increased barrier height results in about 10 times less photocurrent due to the shielding of the holes in the substrate, but more than 100 times improvement in the 3-dB bandwidth. Therefore, the simulation results demonstrate the effectiveness of the V_{sub} in improving the high speed performance of the CMOS APD. Notice that changing V_{sub} in real integrated devices is not practical because substrate has always to be grounded. Another way of controlling of the potential barrier could be apply a bias voltage V_{NW} on the NW, which will have the same effect as reverse bias V_{sub} . Unfortunately, the fabricated devices have substrate and NW contacts

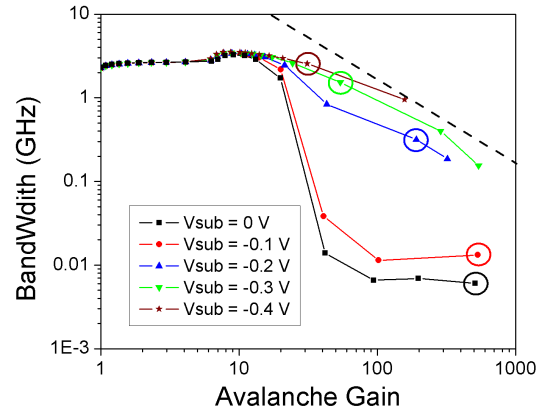


Figure A.13: Simulated gain-bandwidth characteristics of the device under different V_{sub} 's. Circled data points have the same $V_R = -11$ V. The dash line indicates constant GBP.

always connected. We cannot test this concept in the fabricated devices. A new batch of CMOS APDs will be designed and fabricated to verify this concept.